
IT-University of Copenhagen

MSc Thesis

KISPECI1SE — 30 ECTS

Emergent Taxation and Economic Behavior in a Multi-Region Environment

A Deep Reinforcement Learning Study

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Date

May 31, 2026

Keywords: Multi-agent reinforcement learning; Deep reinforcement learning; AI Economist; Emergent tax policy; Multi-region economy; Labor mobility; Market integration; Redistribution; Inequality; Social welfare; Agent-based economic simulation.

Abstract

Traditional economic models often study taxation, redistribution, and mobility through analytical or numerical frameworks in which the behavioral structure is specified in advance. However, international economies introduce additional complexity, as they are shaped by interactions across regions: goods move through trade, agents may relocate, and policy decisions in one region can affect outcomes elsewhere. These interactions are especially difficult to capture when agents and policymakers adapt to each other over time. This thesis studies these mechanisms in a multi-agent reinforcement learning environment.

Building on an existing one-region economic framework where agents and planners adapt through deep reinforcement learning, this model is extended into a two-region economy with separate social planners, market integration, and labor mobility. Agents choose how to move, gather resources, trade, build, and relocate, while regional planners learn separate income-tax schedules. The analysis compares three experimental settings: a closed-economy baseline, a trade-integrated economy, and an economy allowing labor mobility in which agents can move between regions.

The results show that inter-regional interaction changes both agent behavior and planner policy, but not through a simple tax-competition mechanism. Market integration changes the conditions under which agents trade and produce, while labor mobility allows agents to change the tax, market, and spatial environment they face. Relocation is associated with skill, period income, tax exposure, and environmental factors, rather than tax differences alone. Even when one region is assigned a persistently lower tax schedule, agents do not simply concentrate there.

Planner behavior also changes under mobility. Average tax rates are slightly lower when travel is enabled, but the learned tax schedules remain heterogeneous across brackets, regions, and trained runs. This suggests that planner actions are shaped by the combined regional welfare outcome rather than by mobility or tax competition alone.

Overall, the thesis shows how taxation, trade, and mobility can interact endogenously through learned behavior in a multi-region economy. The results should not be interpreted as policy prescriptions or fully converged equilibrium outcomes, but as evidence of learned behavioral mechanisms in a controlled deep reinforcement learning environment.

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1 Introduction

Taxation is central to how economic policy shapes behavior, welfare, and the distribution of resources. Theoretical and empirical work has long studied how income and wealth distributions emerge, how they affect the macroeconomic state, and how they evolve over time [1, 2]. Measures of equality and social welfare are therefore well-established tools for evaluating economic systems, while the design of redistributive tax policy remains central to understanding how policy affects individual incentives, welfare, and economic outcomes [3–5].

Economic models have traditionally studied these questions through analytical frameworks or numerical solution methods in which the behavioral structure is specified in advance. Such approaches have generated important insights into taxation, redistribution, and welfare, but they can become difficult to apply when environments involve many interacting agents, high-dimensional state spaces, and adaptive behavior. Some computational approaches, including heterogeneous-agent macroeconomic models and deep-learning-based solution methods, address these difficulties by allowing aggregate outcomes to depend on heterogeneous agents or by using neural networks to approximate high-dimensional dynamic models [1, 6]. Related applications of deep learning and reinforcement learning in macroeconomics show how neural-network-based agents can approximate policy functions and equilibrium behavior in complex dynamic settings [7–9]. However, these approaches often focus on equilibrium computation or policy-function approximation rather than on learned interaction between adaptive agents and an endogenous policymaker. Reinforcement learning extends this computational tradition by allowing agents to learn decision-making policies through repeated interaction with an environment [10].

Deep reinforcement learning (DRL) is therefore particularly relevant for economic settings characterized by sequential decision-making, interaction among learning agents, and behavioral adaptation. In such settings, agents and planners make decisions over time while learning from outcomes that depend on earlier actions and later responses from other learners. This is useful in economic environments where policy changes affect individual incentives. Agents adapt their behavior in response to these changes, and the resulting economic outcomes reshape the policy problem. The economic and computational foundations used in this thesis are introduced in Section 2, as well as the original AI Economist framework developed by Zheng et al. on which the thesis builds [11, 12].

The contribution of this thesis is not to introduce a new economic modeling paradigm, but to adapt an established reinforcement-learning environment to include additional economic structure. Existing implementations of the AI Economist consider a single, closed economy governed by a single social planner [11, 12]. Other computational economic models, including heterogeneous-agent macroeconomic models and deep-learning-based solution methods, provide important tools for studying inequality, policy functions, and equilibrium behavior in high-dimensional settings [1, 6–8]. However, these approaches typically do not focus on learned interaction between separate regional planners, adaptive agents, cross-regional trade, and labor mobility. These settings provide useful benchmarks, but they exclude a central feature of many real-world economic systems: interaction between regions.

In practice, economic outcomes are shaped not only by within-region policy, but also by cross-regional dynamics such as trade, labor mobility, and regional policy-making. Trade theory emphasizes how cross-regional exchange can shape economic outcomes even between similar economies [13], while tax-competition focused literature shows how economic agents and regional tax policy can create strategic interactions between governments [14–16].

Migration can affect policy-making and taxation by creating competition between regions. Work on tax competition and migration shows that mobility can change the incentives faced by regional policymakers, but not always in a simple race-to-the-bottom direction. Razin and Sadka [17] show that migration can generate fiscal externalities, so that tax competition may lead to outcomes that differ from coordinated policy arrangements. Empirical work reviewed by Kleven et al. [18] further shows that some individuals, especially high-income earners and workers whose human capital is less tied to a specific location, can respond to tax differences when making location decisions.

The European Union provides a useful real-world motivation for this type of setting. Modern European economies combine market integration with the legal principle of free movement. EU citizens have the right to move, live, and work in any member state, creating an institutional setting in which relocation is formally unrestricted within the union [19]. This institutional structure is reflected in observed mobility patterns: in 2023, approximately 1.5 million people moved from one EU country to another, and 10% of the EU population resides in a country different from their country of citizenship [19]. Mobility is therefore not only a legal principle, but also an economically relevant way for individuals to respond to differences between regions. At the same time, economic conditions across EU member states remain heterogeneous. Tax regimes differ substantially despite the existence of a common market, with top marginal income tax rates varying across countries due to differences in national policy priorities and fiscal structures [20]. Countries also differ in production environments, industrial composition, and resource availability. The model developed in this thesis does not attempt to reproduce these structural differences directly. Instead, it uses a simplified two-region environment in which the regions are comparable in aggregate resource availability, allowing the analysis to focus on market integration, mobility, and taxation rather than on exogenously imposed regional inequality. Differences are also evident in income distributions. Both within-country inequality and across countries are part of the European economic landscape, commonly measured through indicators such as the Gini coefficient [21]. Taken together, variation in taxation, economic opportunities, mobility, and inequality creates an environment in which agents may respond to incentives not only through labor and production decisions, but also through relocation. While the model abstracts from many institutional details, this international context motivates the study of taxation in a setting where agents and policymakers interact across regional boundaries.

Introducing these mechanisms changes the environment in which taxation operates. From the perspective of agents, taxation becomes only a single component of a broader decision problem that also includes production, trade, market access, congestion, and relocation costs. From the perspective of policymakers, taxation must be chosen in anticipation of behavioral responses that may extend beyond regional borders. This creates a feedback loop: policy influences behavior, behavior reshapes the regional economy, and the resulting economic outcomes affect future policy incentives.

This feedback loop motivates the central research question of the thesis:

How does the introduction of inter-regional interaction through labor mobility and market integration affect the behavior of deep reinforcement learning agents and social planners in a multi-region economy?

Figure 1 summarizes the motivation, research question, methodological approach, and main findings of the thesis.

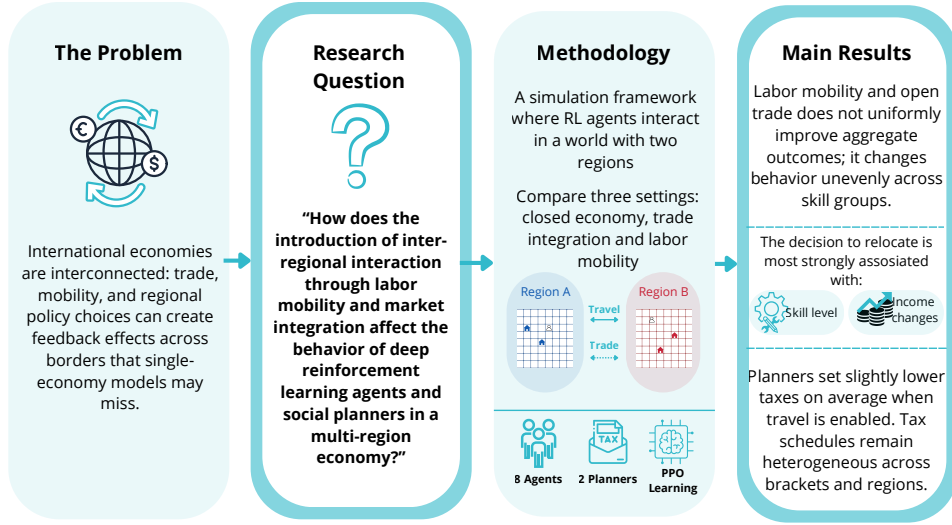


Figure 1: **Conceptual overview of the thesis.** The figure summarizes the motivation, research question, methodological design, and main findings of the thesis. The analysis studies inter-regional interaction in a two-region deep reinforcement learning economy with separate planners, market integration, and labor mobility.

To answer this question, the thesis extends the AI Economist into a two-region economy with separate regional planners, cross-regional market integration, and labor mobility. The model is specified in Section 3, and the main economic mechanisms are summarized visually in Figure 3.

The training procedure and experimental design are described in Section 4. The analysis compares three main settings: a closed-economy baseline, a trade-integrated economy, and a labor mobility economy in which agents can relocate between regions. The closed baseline establishes how agents and planners behave when the two regions are economically isolated. The trade-integrated setting introduces cross-regional trade while keeping agents in their assigned region. The mobility setting then allows agents to change region, making taxation part of a broader decision problem involving market access, spatial conditions, accumulated resources, and travel costs. This design allows the analysis to separate the effects of market integration from the effects of labor mobility.

The results, presented in Section 5, show that inter-regional interaction changes both agent behavior and planner policy, through several interacting mechanisms, including taxation, trade, production, and relocation. Market integration alters the conditions under which agents trade and produce, while mobility introduces a broader response channel through which agents can change the tax, market, and spatial environment they face. However, relocation is not explained by tax differences alone. Agents instead base their decision to relocate on many different factors such as income and economic opportunities. Similarly, planner behavior changes under mobility: Average tax rates are slightly lower, a results consistent with a limited tax-competition interpretation. However, this should not be interpreted as direct evidence that planners intentionally lower taxes to retain agents. The learned tax schedules also reflect changes in the combined welfare from many different economic components. The results therefore suggest that mobility may introduce tax-competition-like pressure, but only as part of a broader feedback system in which planner policy and agent behavior adapt to each other over time.

These findings should be interpreted within the controlled simulation setting used in this thesis. The model does not provide direct policy prescriptions, but it helps illustrate how taxation, mobility, production, and learned behavior can interact endogenously in a simplified multi-agent economy. The implications and limitations of the findings are discussed in Section 7.

2 Framework Overview

This section introduces the economic and computational foundations of the model. It first outlines the economic framework of agent behavior and taxation from the original AI economist setup, before describing how these concepts are used in a multi-agent reinforcement learning setting.

2.1 The Original AI Economist Environment

The AI Economist, introduced by [11] and extended in [12], provides a reinforcement learning-based framework for studying economic policy in simulated environments. The model represents an economy in which heterogeneous agents and a single social planner are trained jointly. Agents make decisions regarding labor and production, while the planner sets tax policies, and both adapt to each other over time. This structure enables the analysis of how policy and behavior interact in a closed economy.

A key feature of the framework is that it allows economic outcomes to emerge from learning and interactions rather than being specified in advance. This offers a flexible environment for studying taxation and redistribution in settings where behavioral responses are difficult to model explicitly.

The environment is a stylized, grid-based world designed to capture essential economic interactions such as production, trade, and taxation. It consists of a two-dimensional grid composed primarily of grass tiles on which the agent can move around, and water tiles that act as natural barriers. Within this environment, agents can interact with resources distributed across the map. See the map presented in figure 5 for a visual representation.

2.1.1 Game Objective, Resources, and Trade

Each agent seeks to maximize their income while minimizing labor, as described in the agent utility specification in Section 3.8. Coin is earned through two main channels, building houses and selling resources, both of which depend on resources. Agents can use resources to build houses, or they can sell resources to other agents through the market. Resources can be obtained either by moving around the map and gathering them from resource tiles, or by buying them from other agents using coin.

The two resources available are wood and stone. These resources spawn stochastically on designated resource tiles. Once harvested, a tile becomes temporarily empty until the resources regenerate. Agents can move through the environment to locate and collect available resources.

Agents can combine one unit of wood and one unit of stone to construct a house at their current location. Houses generate income for the agent only once when it is built, with the amount determined by the agent’s building skill level defined in Section 2.1.3. In addition to production, agents participate in a market for trading resources. The market operates as a continuous double auction, where agents submit bids (offers to buy) and asks (offers to sell) for each resource. These trades allow agents to acquire missing inputs for production or to earn coin by selling excess resources.

Moving, gathering, trading, and building all require some amount of labor. The specific labor costs used in the simulation are reported in Table 4 in the appendix. All agents face identical labor costs, ensuring that differences in outcomes arise from differences in building skill, individual observations, market conditions, and taxation rather than from different action costs. This economic loop is illustrated in Figure 2.

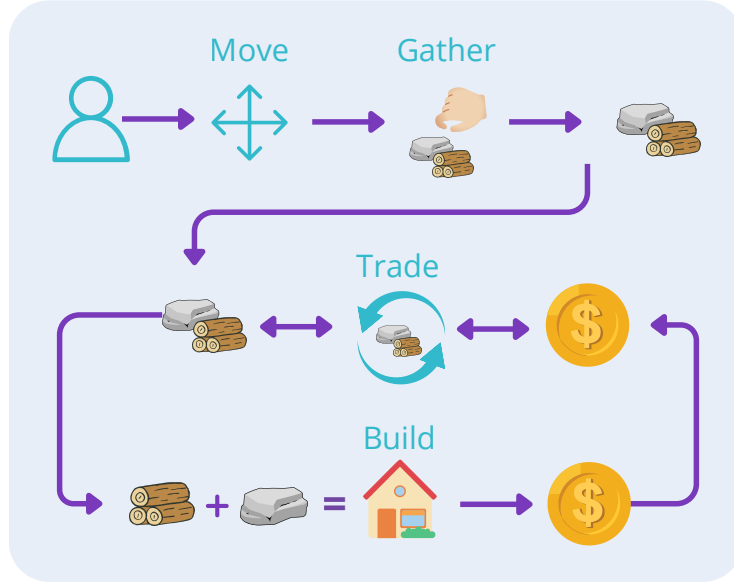


Figure 2: **Overview of the core Gather-Trade-Build loop.** Agents move through the environment to gather wood and stone, which can either be traded in the market or combined to build houses. Building generates coin income, while market exchange allows agents to buy missing resources or sell excess resources. Each action is associated with a labor cost, creating a trade-off between income generation and labor disutility.

2.1.2 Taxation

A social planner operates alongside the agents and sets a bracketed income tax policy. At the beginning of each tax period, the planner selects marginal tax rates for a fixed set of income brackets (see the specific tax brackets in the Appendix table 5). These rates remain constant throughout the period. Taxes are collected at the end of each tax period and redistributed uniformly to all agents. Each simulation has 10 tax periods, each consisting of 100 time steps. For further specification see section 3.9.

Taxation is based on income generated within the current period, and the resulting revenue is pooled and redistributed as equal lump-sum transfers. This implies that higher-income agents contribute more in taxes than they receive, while lower-income agents receive net transfers, thereby reducing inequality.

Because agents adapt their behavior in response to taxation, and the planner adapts taxation in response to observed outcomes, the tax policy is jointly optimized with agent behavior, rather than treated as an exogenous and fixed part of the environment.

2.1.3 Skill

Agents are heterogeneous in skill. Each agent is assigned a fixed skill parameter that determines its productivity. The skill level is predetermined based on the original AI economist setup, and is held fixed for all experiments [11]. Skill levels directly affects the income generated from building houses. As a result, higher-skill agents earn more from building. Lower-skill agents may therefore rely relatively more on gathering, trade, or redistribution, not because they are more productive at gathering, but because building provides them with lower income compared to high skill agents, who might be more willing to pay higher prices for resources on the market. This dynamics make skill levels the main driver of economic trading cycles and

the foundation for agent specialization.

2.2 Two-Level Multi-Agent Reinforcement Learning

The multi-region economy is modeled as a partially observable multi-agent decision process, in which mobile agents and regional planners interact within the same simulated environment but receive different observations and optimize different objectives [22]. The notation used in this section is summarized in Appendix A.1. The detailed observation and action spaces are described in Section 3.6 and 3.7.

Formally, the environment can be described by the tuple

$$(\mathcal{S}, \mathcal{A}, \mathcal{T}, r, \gamma, \mathcal{I}),$$

where $\mathcal{S}$ denotes the state space of the economy, $\mathcal{A}$ the joint action space, $\mathcal{T}$ the transition function, r the collection of reward functions, $\gamma \in (0, 1)$ the discount factor, and $\mathcal{I}$ the set of learning agents. In this thesis, $\mathcal{I}$ consists of eight mobile economic agents and two regional planners, one for the top region and one for the bottom region.

At each timestep t , the state $s_t \in \mathcal{S}$ contains the full economic and spatial configuration of the environment, including agent locations, resource availability, inventories, houses, market orders, tax schedules, and accumulated income and labor. However, this full state is not observed directly by every policy. Each mobile agent receives an individual observation $o_{i,t}$, which includes local spatial information, its own inventory and labor state, market information, and tax information. The regional planners receive planner-specific observations, including regional spatial information, their own tax schedule, and market information. These observation structures are specified in more detail in Sections 3.6 and 3.7.

The mobile agents and planners act on different time scales. Mobile agents choose actions at every environment timestep. The planners choose bracketed marginal income tax schedules only at evenly spaced out taxation timesteps. Each episode contains $H = 1000$ timesteps, divided into 10 tax periods of 100 timesteps each. Taxes are collected and redistributed at the end of each tax period, as described in Section 3.9.

The transition function

$$s_{t+1} \sim \mathcal{T}(\cdot \mid s_t, \mathbf{a}_t)$$

maps the current state and joint action vector $\mathbf{a}_t$ into the next state. In this environment, variation across rollouts arises from both stochastic policy actions and stochastic environment processes. Agents and planners sample actions from learned probability distributions, so the same observation can lead to different actions across rollouts. Conditional on the chosen action, some state updates are rule-based, such as movement, building, taxation, and relocation, while other processes, such as resource spawning and market tie-breaking, introduce additional stochasticity.

The mobile agents share one policy function,

$$\pi_\theta(a_{i,t} \mid o_{i,t}, h_{i,t}),$$

where $h_{i,t}$ is the recurrent hidden state and θ denotes the shared agent-policy parameters. Parameter sharing means that all mobile agents use the same neural network, while heterogeneous behavior can still emerge because agents differ in their observations, locations, inventories, skills, and regional tax conditions.

The two planners are represented by separate policy networks. The top planner controls the tax schedule for the top region, while the bottom planner controls the tax schedule for the bottom region. Their policies can be written as

$$\pi_{\phi_r}(\tau_{r,t} \mid o_{r,t}^P, h_{r,t}^P),$$

where $r \in \{\text{top}, \text{bottom}\}$, $\tau_{r,t}$ is the tax schedule chosen by planner r , $o_{r,t}^P$ is the planner observation, $h_{r,t}^P$ is the planner hidden state, and ϕ_r denotes the planner-policy parameters.

Because multiple agents learn simultaneously, the environment is non-stationary. Agents adapt to market conditions, relocation opportunities and planner tax schedules, while planners adapt to the income distributions, production levels, and welfare outcomes generated by agents. This feedback loop is central to the model: the policies learned by agents affect the learning problem faced by planners, and planner tax policies affect the learning problem faced by agents. This makes stable training difficult and motivates the use of PPO, described in Section 2.4.

2.3 Policy Representation and Neural Network Architecture

The policy and value functions required for learning are approximated using neural networks, allowing the model to operate in high-dimensional and partially observable environments.

Deep reinforcement learning relies on neural networks to approximate policies and value functions in high-dimensional environments [23]. In the AI Economist framework used in this thesis, agents and planners operate in a spatially structured and partially observable environment, requiring architectures that can process both spatial and temporal information.

Multi-layer perceptrons (MLPs). Multi-layer perceptrons are fully connected feedforward networks that map input vectors to outputs through a series of nonlinear transformations. In reinforcement learning, MLPs are commonly used when the observation space can be represented as a flat feature vector. They serve as general-purpose function approximators for both policy and value functions. While expressive, MLPs do not use structure in the data, such as spatial location or temporal dependencies, and are therefore less efficient for structured environments.

Convolutional neural networks (CNNs). In this setting, agents and planners observe the environment as a grid-based spatial map containing information about resources, agent locations, and ownership. Convolutional neural networks are used to process these spatial observations [24]. CNNs apply shared filters across the input grid, allowing the network to detect local patterns such as resource clusters or nearby agents while recognizing similar patterns in different locations. This significantly reduces the number of parameters compared to fully connected architectures and improves generalization by using the spatial structure of the environment.

Recurrent neural networks and LSTM. The environment is partially observable, meaning that the current observation does not fully capture the underlying state. To address this, recurrent neural networks are used to incorporate temporal information. In particular, Long Short-Term Memory (LSTM) networks extend standard recurrent architectures by maintaining an internal memory state that is updated over time [25]. This allows the model to retain relevant information from past observations, such as previous tax policies or agent interactions, and use it to inform current decisions. LSTM units mitigate issues such as vanishing gradients, enabling learning over longer time horizons.

Combined convolutional-recurrent architectures. In practice, the models used in this thesis combine convolutional and recurrent components. Spatial observations are first processed by convolutional layers to extract feature representations of the environment. These spatial features are then concatenated with non-spatial features, passed through fully connected layers, normalized, and processed by an LSTM layer, which integrates information over time. The LSTM output is projected to policy logits and to a scalar value estimate. This hybrid architecture allows the model to jointly capture spatial dependencies and temporal dynamics. See Figure 4 for a graphical representation of the multi-region setup.

Both the economic agents and the social planners are modeled as deep reinforcement learning agents. At each timestep t , an agent or planner receives a structured observation containing available information such as spatial map features, economic state variables, market conditions, and tax-related information.

The final output of the policy branch is a set of action logits. RLlib interprets these logits as categorical distributions over the valid choices in each action component, after applying the relevant action masks. Actions are sampled from these distributions, enabling stochastic policies that support exploration during learning.

The action spaces are multi-discrete, meaning that several action components are selected jointly at each decision point. For mobile agents, this allows the policy to choose across different behavioral components in the same timestep, such as building, buying, selling, gathering, and travel. Each component includes a zero or no-op choice, so the agent can leave individual behavioral components inactive. For planners, the multi-discrete action space represents tax-policy choices. Conceptually, each planner chooses one marginal tax rate for each tax bracket, allowing a complete tax schedule to be constructed in a single forward pass. In the implementation, planner actions are also subject to action masks, so tax schedules can only be changed at valid tax-decision times.

The learning structure enables agents and planners to learn complex, state-dependent behavior in a high-dimensional and partially observed environment. The architecture combines spatial, non-spatial, and temporal information within each policy network, allowing decisions to condition on both current observations and recent history.

2.4 Proximal Policy Optimization

The original AI Economist framework trains both agents and planners using Proximal Policy Optimization (PPO) [26]. PPO is well suited to this environment because learning takes place in a high-dimensional, partially observed, and non-stationary setting. Small changes in either agent behavior or planner policy can affect market outcomes, tax revenue, redistribution, and future observations. As a result, abrupt policy updates may destabilize training.

The economy is treated as a multi-agent Markov decision process with eight mobile agents and two regional planners. The eight mobile agents share one policy network, denoted a , while the two planners have separate policy networks, denoted p_{top} and p_{bottom} . Each policy is represented by an actor-critic neural network: the actor outputs masked action logits, which define a probability distribution over valid actions, while the critic estimates the value of the current observation. See section A.3 in the appendix for more detail.

PPO is a policy-gradient method. During training, the current policies generate rollout trajectories containing observations, sampled actions, rewards, and value estimates. These rollouts are used to estimate discounted returns and advantages. The advantage A_t measures whether an action performed better or worse than expected relative to the critic’s value estimate. In the basic policy-gradient formulation, the update direction is

$$\nabla_{\theta} J(\theta) \approx \mathbb{E}_t [\nabla_{\theta} \log \pi_{\theta}(a_t | o_t) A_t].$$

Thus, actions with positive advantage become more likely, while actions with negative advantage become less likely. PPO modifies this update using the probability ratio

$$r_t(\theta) = \frac{\pi_{\theta}(a_t | o_t)}{\pi_{\theta_{\text{old}}}(a_t | o_t)},$$

where $\pi_{\theta_{\text{old}}}$ is the policy that generated the rollout data. The clipped PPO objective is

$$L^{\text{PPO}}(\theta) = \mathbb{E}_t [\min(r_t(\theta) A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t)].$$

The policy and value losses are optimized by backpropagation through the actor-critic networks. The value branch is trained to predict discounted returns, while entropy regularization is used to encourage exploration for selected policies. Rollout data are collected from parallel environment replicas, split into minibatches, and reused for several stochastic-gradient updates. The main hyperparameters, including rollout length, batch size, minibatch size, number of SGD passes, learning rates, entropy coefficients, and model architecture, are reported in Appendix Table 3.

The stability of PPO is particularly important in the AI Economist setting because agents and planners learn simultaneously. Agents must adapt to changing tax policies, while the planner must adapt to changing agent behavior. PPO does not remove this non-stationarity, but it provides a practical training method for managing it in complex multi-agent environments.

3 Multi-Region Model Specification

This work extends the AI Economist framework to a setting with multiple interacting economies. The model consists of a system of two separate regions, each governed by separate social planners and populated by mobile agents. The main extension is the introduction of interaction between regions through agent mobility, enabling agents to relocate between regions in response to differing economic conditions.

The extension also introduces multiple regional planners. These planners do not interact directly, but their policies are linked indirectly through agent behavior: agents optimize individual utility and may respond to differences in regional tax schedules, market conditions, and mobility opportunities.

Restricting the model to two economies provides a minimal yet expressive setting that captures essential strategic interactions, such as competition like behavior between tax policies and the movement of labor across regions. This approach mirrors standard economic modeling, where pairwise interactions are often used to study more general multi-agent dynamics while maintaining computational tractability.

The main mechanisms are visualized in Figure 3

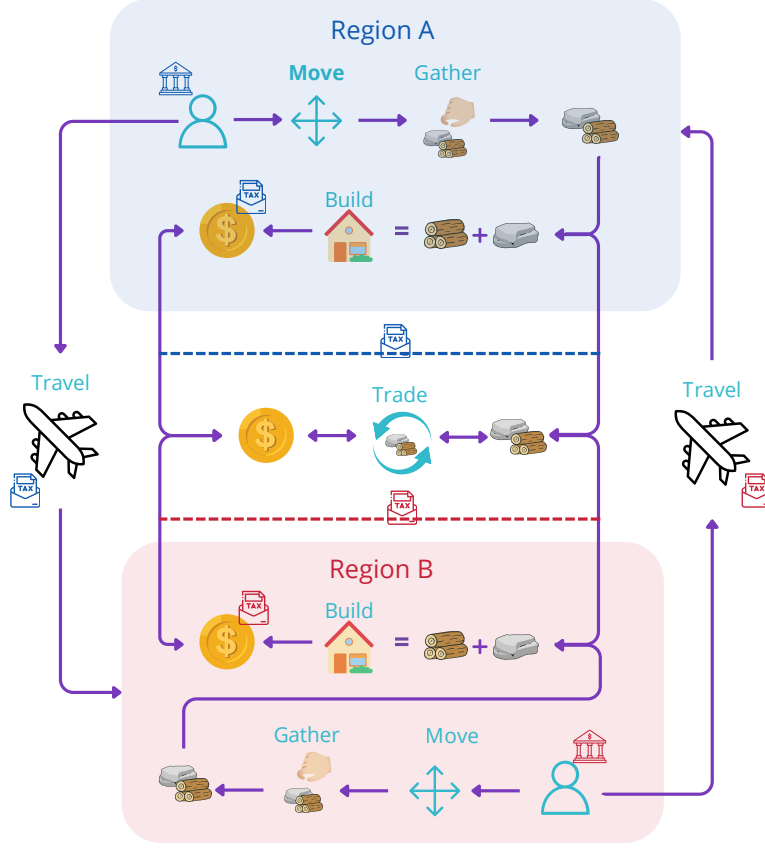


Figure 3: **Overview of the main economic mechanisms in the multi-region environment.** Turquoise labels indicate agent actions: movement, gathering, building, trade, and travel - all associated with a labor cost. Agents gather wood and stone, use them to build houses, and receive coin income from production. Trade allows resources to be exchanged across agents and, when cross-regional, is subject to import taxation. Tax icons indicate which regional planner collects revenue, while subsidy icons indicate redistribution to agents currently located in that region.

3.1 Overview

The environment is implemented as a two-dimensional grid world divided into two regions by a water boundary. These regions represent separate economies, referred to as the top region and bottom region. Each region replicates the core mechanics of the original environment, including resource generation, production, and market interactions (introduced in section 2).

Each region is governed by its own social planner. The planner individually sets a marginal tax on income in each predetermined bracket (see section 3.9 and table 5), affecting only the agents currently located within that region. This creates a setting where different regional tax schedules can emerge simultaneously.

Markets are region-specific in the closed-economy baseline. When cross-regional trade is enabled, agents may trade with orders from the other region, but imported goods are subject to a 10% import tax. Thus, market integration expands the set of possible trading partners while preserving a distinction between local and non-local exchange.

Figure 4 summarizes how the multi-region economy is represented as a two-level reinforcement learning problem, where mobile agents learn economic actions at the lower level and

regional planners learn tax schedules at the upper level.

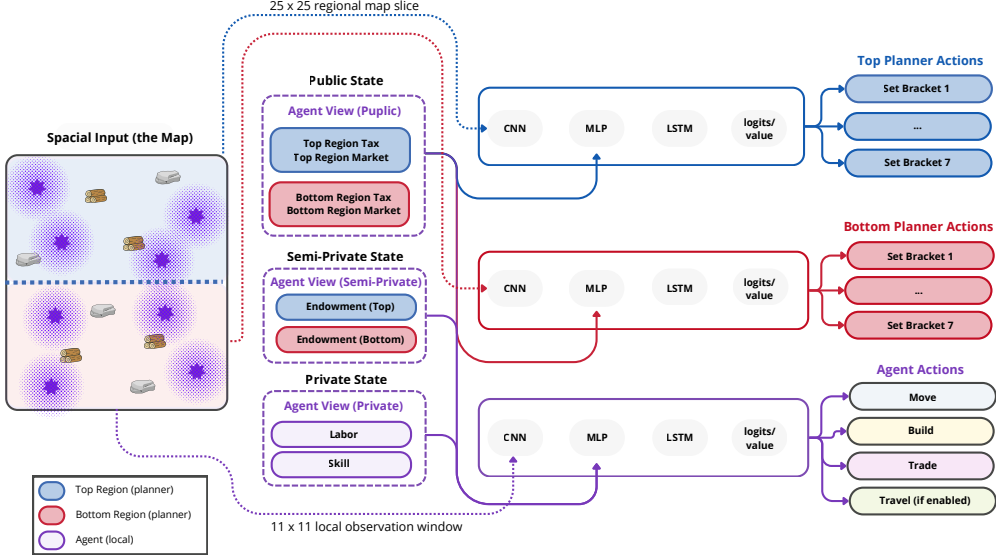


Figure 4: **Schematic overview of the general network architecture in the multi-region economy.** Spatial observations are processed by a stack of two convolutional layers (CNNs) for each planner and for the shared agent policy, and are then flattened into a fixed-length feature vector. Each planner observes only the regional part of the map corresponding to the region it governs. This spatial feature vector is concatenated with the remaining observation inputs and processed by a stack of two fully connected layers (MLP). The output is then used to update the hidden state of an LSTM, after which action logits are computed through a linear projection of the updated hidden state. Finally, the network computes a softmax probability distribution for each action head. For the agent policy, there is a single action space and action head. For the planner policies, there is a separate action head for each tax rate controlled by the planner.

Figure 4 also illustrates the main extension relative to the original AI Economist framework. Instead of a single centralized planner, the multi-region economy contains two regional planners, each associated with one region. The planners receive region-specific spatial information and choose separate tax schedules, while agents observe the full tax and market information. This allows the model to study how separate regional tax schedules emerge in an environment where agents and planners interact across regions.

The model is characterized by a feedback loop between agent behavior and planner policy. Agents respond to the incentives created by taxes, market conditions, and mobility costs, while planners adapt their tax schedules in response to the regional outcomes generated by agents. This interaction is central to the analysis, since the effects of taxation depend not only on the tax rates chosen by planners, but also on how agents learn to adjust their economic behavior across regions.

3.2 Regional Environment and Spatial Structure

The spatial environment is based on the original AI Economist open-quadrant layout [11]. In this thesis, the layout is duplicated vertically and separated by an impassable water boundary, creating two regions: a top region and a bottom region. This design preserves the core structure of the original environment while allowing the model to introduce regional taxation, market interaction, and labor mobility.

The two regions are intentionally constructed to be comparable in aggregate resource availability. This ensures that differences in agent behavior and planner policy can be interpreted primarily as consequences of trade, taxation, mobility, and learning dynamics, rather than as outcomes driven cross-regional inequality built into the map.

At the same time, the map retains local differences in resource access within each region. Following the open-quadrant structure, resource tiles are unevenly distributed across space: some areas contain both wood and stone, some are dominated by one resource type, and others contain no resource tiles. This spatial variation matters because agents must physically move to access resources, and because houses can only be built when agents acquire both wood and stone. As a result, an agent’s initial location affects its early production opportunities, trading incentives, and potential gains from relocation.

Figure 5 shows the initial spatial layout of the environment.



Figure 5: **Initial map layout of the two-region environment.** Green tiles represent wood, white tiles represent stone, blue tiles represent water, circles represent agents, and squares the same color as circles represent houses.

The initial locations are held fixed across all subsequent experimental settings. Agents in the lowest skill group start in the quadrant containing both resource types, agents in the second-lowest skill group start in the stone-only quadrant, agents in the second-highest skill group start in the wood-only quadrant, and agents in the highest skill group start in the quadrant without resource tiles.

The map layout is important for the interpretation of the results. This is because, when agents specialize, trade, or relocate, the behaviors arise from their individual observation and resources access, shaping the simulation outcomes through their action choices.

3.3 Agents, Skills, and Resource Collection

Agents are heterogeneous in their initial locations and skill levels. Building skill determines the income an agent receives from building a house. These differences allow unequal outcomes

and specialization patterns to emerge endogenously, even though all agents follow the same shared policy network.

Movement and gathering. Agents may move across the grid by selecting one of four movement actions: up, down, left, or right. When an agent moves onto a resource tile containing a resource, they may use a gathering action to collect either wood or stone, depending on the tile type. Once harvested, the tile becomes temporarily depleted and later respawns stochastically. Resource collection therefore depends on agents’ positions on the map, requiring them to move through the environment in order to acquire the resources needed for production and trade.

Building. Agents may convert one unit of wood and one unit of stone into a house at their current location, provided that the tile is not a resource tile or already occupied by another house. Building generates coin income for the agent at the timestep where the house is placed on the map, with the amount determined by the agent’s building skill. This makes building the main production activity in the environment and creates incentives for agents to acquire both resource types either through gathering or trade. A house can only be passed through by the agent who built it, which can lead to spacial congestion on the map.

Trading. Agents can submit buy and sell orders for wood and stone through an auction-style market. When submitting an order the agent can choose between 11 price levels $\{0, \dots, 10\}$, two directions (bid or ask), and two resources (wood or stone). Each trading action submits a single order that remains open for up to 50 timesteps unless it is matched or canceled. Agents may have at most five open orders per resource and cannot place orders that exceed their current coin or resource holdings.

A trade occurs when a bid price equals or exceeds an ask price for the same resource, taking any applicable taxes into account. Matching prioritizes the best price, then the earliest order, and finally random selection in the case of ties. Executed trades transfer one unit of the relevant resource at the matched price.

3.4 Markets and Cross-Regional Trade

The market determines whether resource exchange is restricted within regions or allowed across regions. In the closed-economy baseline, markets are region-specific: agents can only trade with orders submitted by agents located in the same region. This creates two separate local markets, allowing each region to develop its own price dynamics and resource constraints.

When cross-regional trade is enabled, agents may also trade with orders from the other region. Market integration therefore expands the set of possible trading partners and allows resources to move across the regional water boundary without requiring agents themselves to relocate. This separates the movement of goods from the movement of labor, allowing the effects of trade and mobility to be studied separately.

Imported goods are subject to a fixed import tax equal to 10% of the transaction price. The tax is paid by the buyer, and the resulting revenue is redistributed uniformly within the buyer’s region. Import taxation therefore creates an additional source of regional revenue that depends on trade flows, linking local welfare not only to domestic production but also to cross-regional exchange.

In the simulation, the import tax should be interpreted as a stylized trade friction rather than as a literal representation of internal EU trade policy, because there are no tariffs between EU member states [27]. Without any cost attached to cross-regional trade, the two regional markets would effectively collapse into a single integrated market, weakening the distinction between local and non-local exchange. The import tax preserves a mild incentive to trade locally when local supply is sufficient, while still allowing cross-regional trade to be beneficial when price differences or resource shortages are large enough. It can also be interpreted as a representation of transportation costs, coordination costs, or other frictions associated with trading across regions.

The impact of cross-regional trade on inequality is ambiguous and depends on market structure. On the one hand, trade may reduce inequality by alleviating local supply constraints and enabling mutually beneficial exchange. On the other hand, trade may amplify inequality if one region becomes more productive or better positioned to dominate market exchange. As a result, the distributional effect of market integration is not imposed directly, but emerges from the interaction between supply, demand, taxation, and learned agent behavior.

3.5 Labor Mobility and Travel

Labor mobility is the second channel of interaction between regions. When travel is enabled, agents can choose a cross-water travel action that relocates them from their current region to a designated location in the other region. This action changes the regional environment in which the agent operates and makes the agent subject to the tax policy of the destination region. (see section 4.3.1 for travel destination specification)

Upon relocation, the agent retains its accumulated coin holdings and resource inventory. However, any houses previously built by the agent remain fixed in their original locations, and still act as obstacles for other agents, but have no other effect since build-income is only collected once. This means that relocation does not transfer all productive assets across regions. Instead, travel changes the agent’s future spatial, market, and tax environment while preserving the effects of earlier building decisions.

Like any other active action, travel includes a labor cost but also a separate monetary cost. These costs discourage frequent or unnecessary relocation and make mobility a strategic decision rather than a costless adjustment. The monetary travel cost is redistributed within the region the agent leaves, reflecting the idea that out-migration affects the origin region while providing a residual benefit to the agents who remain.

When an agent relocates, any outstanding market orders submitted in the previous region are canceled.

The mobility mechanism alters the learning problem for both agents and planners. From the agent’s perspective, taxation becomes only one component of a broader relocation decision that also depends on resources, market conditions, accumulated inventory, skill, congestion, and travel costs. From the planner’s perspective, taxation may affect not only labor, production, and trade, but also the composition of the regional population.

As a result, the model captures a feedback loop between policy and mobility. Planners influence agent behavior through regional tax schedules, while agents influence planner outcomes through production, trade, and relocation. This allows the environment to represent tax competition indirectly, through the effect of tax policy on agent movement and regional economic activity.

3.6 Agent Observations and Policy

Agent behavior is modeled by a deep neural network policy. At each timestep t , agent i receives a structured observation consisting of spatial information, agent-specific state variables, market information, tax information, and a recurrent hidden state. The action space is multi-discrete, so $\mathbf{a}_{i,t}$ is a vector of action components (this is explained further in the appendix section A.7). Based on this information, the agent samples its actions according to.

$$\mathbf{a}_{i,t} \sim \pi_{\theta}(o_{i,t}^{\text{world}}, o_{i,t}^{\text{idx}}, o_{i,t}^{\text{agent}}, o_{i,t}^{\text{market}}, o_{i,t}^{\text{tax}}, m_{i,t}),$$

where:

- $o_{i,t}^{\text{world}}$ denotes spatial observations of the nearby environment, represented as a local 11×11 grid centered on the agent. This provides only partial information about the global state;
- $o_{i,t}^{\text{idx}}$ denotes the local spatial index map, including information such as agent-location markers and ownership indices within the visible 11×11 area;

- $o_{i,t}^{\text{agent}}$ denotes agent-specific non-spatial state, including resource endowments, coins, building skill, accumulated labor, time, and other internal variables relevant for decision-making;
- $o_{i,t}^{\text{market}}$ denotes market information, including price histories, bids, asks, and the agent’s own outstanding orders. Depending on the trade setting, this may include local-region and other-region market information;
- $o_{i,t}^{\text{tax}}$ denotes tax observations, including the agent’s own-region tax schedule and marginal tax rate, and information about the other region’s tax schedule where provided by the environment;
- $m_{i,t}$ denotes the agent action mask, which restricts the policy to valid actions at the current timestep;

The policy network also uses a recurrent hidden state $h_{i,t-1}$. The recurrent hidden state is not part of the environment observation space. It is an internal state of the LSTM policy network that is carried forward across timesteps and used together with the current observation when computing the policy.

The observation is partially observable: agents do not observe the full global state of the environment, but instead condition their decisions on local spatial information and selected public economic variables. The spatial component $o_{i,t}^{\text{world}}$ is centered on the agent and contains information about nearby terrain, resources, houses, and other agents. As a result, agents do not have direct access to the full spatial distribution of resources, houses, or agents across both regions.

A central modification in the multi-region environment is that policy and market information are encoded relative to the agent’s current region. Instead of observing unlabeled regional tax schedules, agents observe the tax policy in their own region and the tax policy in the other region. This distinction reduces representational ambiguity and allows the shared agent policy to compare the fiscal conditions associated with staying and relocating.

The same principle is applied to market information. Agents observe region-specific market conditions, distinguishing between local and non-local bids, asks, and trading opportunities. This allows agents to compare the economic conditions in their current region with those in the alternative region.

This observation design is particularly important in the mobility setting. When agents can relocate, their decision problem includes not only whether to gather, trade, or build, but also whether the expected benefit of another region outweighs the cost of travel. By explicitly labeling tax and market information as belonging to the current or alternative region, the observation space gives agents the information needed for the relocation decision they face.

Agents do not observe the income distribution or complete economic state of the other region. They may observe the other region’s tax policy and selected market information, but their knowledge of non-local economic outcomes is incomplete. Relocation decisions are therefore made under imperfect information: agents can compare some policy and market incentives across regions, but they cannot perfectly forecast all consequences of moving.

All mobile agents share the same policy parameters θ . Parameter sharing improves sample efficiency and encourages the policy to learn general behavioral rules, while differences in behavior arises from differences in location, inventory, skill, labor history, market exposure, and regional tax conditions. (this is further discussed in section A.9)

3.7 Planner Observations and Policy

Each planner receives a regional spatial observation corresponding to the side of the map it governs. This means that planners do not observe the full global map through their spatial observation. Since the regional map includes visible agent-location markers, each planner get indirect information about the number and location of agents currently present in its region.

Each planner also observes the state of its own tax component. This includes whether the current timestep is a tax day, whether it is the first day of a tax period, the current regional tax phase, the regional income information from the previous tax period, and the planner’s current marginal tax rates.

The planner policy for region $r \in \{\text{top}, \text{bottom}\}$ can be written as

$$\tau_{r,t} \sim \pi_{\phi_r}(o_{r,t}^{\text{world}}, o_{r,t}^{\text{idx}}, o_{r,t}^{\text{tax}}, o_{r,t}^{\text{market}}, o_{r,t}^{\text{agents}}, m_{r,t}),$$

where $\tau_{r,t}$ denotes the tax schedule chosen by planner r , ϕ_r denotes the parameters of the regional planner policy, and $h_{r,t-1}^P$ denotes the planner’s recurrent hidden state.

The planner observation consists of the following components:

- $o_{r,t}^{\text{world}}$ denotes the regional spatial map observed by planner r .
- $o_{r,t}^{\text{idx}}$ denotes the regional index map. This contains region-specific spatial index information, including visible agent-location markers;
- $o_{r,t}^{\text{tax}}$ denotes the planner’s own regional tax state. This includes whether the current timestep is a tax day, whether it is the first day of a tax period, the current tax phase, income information from the previous tax period, and the current marginal tax rates.
- $o_{r,t}^{\text{market}}$ denotes the planner’s local market state for wood and stone. For each resource, this includes the market rate, price history, full asks, and full bids.
- $o_{r,t}^{\text{agents}}$ denotes per-agent information for the agents located within region r at time t .
- $m_{r,t}$ denotes the planner action mask. The action mask restricts which tax actions are available at a given timestep, ensuring that bracket updates are only active at valid tax-decision times and that unavailable tax-rate choices are masked out.

The recurrent hidden state $h_{r,t-1}^P$ is not an external observation from the environment, but an internal state of the LSTM policy network that summarizes information from previous timesteps.

The planner’s action is a bracketed marginal income tax schedule. Each tax schedule consists of seven income brackets, and the planner chooses one marginal tax rate for each bracket. The planner action can therefore be represented as a vector

$$\tau_{r,t} = (\tau_{r,t}^1, \tau_{r,t}^2, \dots, \tau_{r,t}^7),$$

where $r \in \{\text{top}, \text{bottom}\}$ indexes the region and $\tau_{r,t}^b$ denotes the marginal tax rate chosen for bracket b . For each active bracket decision, the planner selects a rate from the discrete set

$$\{0, 0.05, 0.10, \dots, 1.0\}.$$

In the implemented multi-discrete action space, each bracket component also contains a no-op action used when the bracket is not being updated. Conceptually, each regional planner chooses seven tax-bracket decisions, one for each bracket. The specific income bracket cutoffs used in the experiments are reported in Table 5. The selected tax rates remain fixed for the duration of the tax period. Action masking ensures that planner actions are only applied at the beginning of each tax period, so that tax schedules are temporally consistent within a period.

The planner does not observe separate components of its reward in the form of equality and productivity. Instead, the planner learns from the current regional social welfare change. This means that a planner cannot directly observe whether an increase in reward was caused by higher productivity, greater equality, or a combination of both. It can only learn from the overall welfare signal produced by the environment.

The planner reward is based on the social welfare of the agents currently located within its region at the time outcomes are evaluated. Since agents may relocate during a tax period, regional membership is determined by current location rather than by initial location or by

where income was earned. This also determines which tax schedule applies at tax collection. If an agent earns income in both regions during the same tax period, the income is not split by region. Instead, the agent’s net income over the full tax period is taxed according to the tax schedule of the region in which the agent is located at the time of tax collection. Thus, an agent located in the top region at tax collection is taxed by the top planner’s schedule, while an agent located in the bottom region is taxed by the bottom planner’s schedule.

This timing rule makes mobility directly relevant for both agent incentives and planner learning. From the agent’s perspective, relocation can change the tax schedule applied to the entire period’s net income. From the planner’s perspective, mobility changes the set of agents whose income, productivity, and welfare enter the regional outcome. A planner that gains or loses agents therefore faces a different regional income distribution and welfare objective, which can affect future tax policy learning.

3.8 Utility, Welfare, and Rewards

This section introduces the utility and reward structures that shape the behavior of agents and planners in the model. Figure 6 provides a high-level visual summary of these mechanisms.

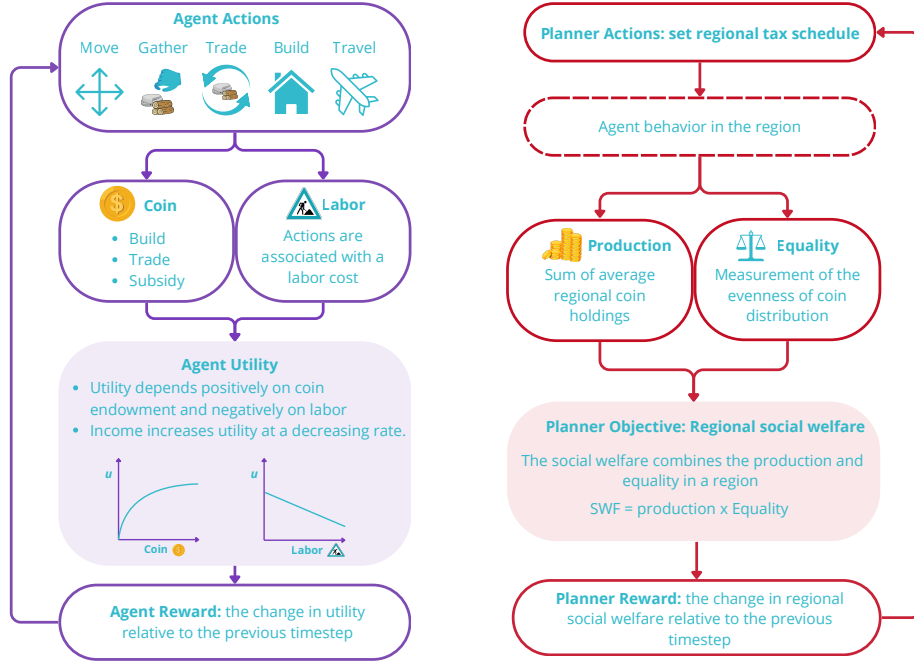


Figure 6: **Overview of the reward structure for agents and planners.** Agent actions may affect coin holdings and labor, which jointly determine agent utility. Agent rewards are given by the change in utility relative to the previous timestep. Planner actions set regional tax schedules, which influence agent behavior and regional outcomes. Planner welfare combines regional production and equality, and planner rewards are given by changes in regional social welfare over time.

Agents get utility from consumption and disutility from labor. In the model, consumption is given by the agent’s coin endowment $x_{i,t}^c$, and preferences are represented using a constant

relative risk aversion (CRRA) utility function:

$$\text{CRRA}(x_{i,t}^c) = \begin{cases} \frac{(x_{i,t}^c)^{1-\eta} - 1}{1-\eta}, & \eta \neq 1, \\ \log(x_{i,t}^c), & \eta = 1. \end{cases}$$

where $\eta > 0$ governs the degree of diminishing marginal utility [28].

Labor enters as a linear cost. Let $l_{i,t}$ denote cumulative labor performed up to time t . Agent i 's individual utility is then given by

$$u_i(x_{i,t}^c, l_{i,t}) = \text{CRRA}(x_{i,t}^c) - \kappa \cdot l_{i,t},$$

where $\kappa > 0$ determines the scale of disutility from labor (in the current setup the parameter values are $\eta = 0.23$ and $\kappa = 0.21$). Although utility depends on cumulative labor, the reward structure defined in equation (1) implies that agents face marginal labor costs at each time step.

The agents therefore each face a trade-off between income and effort. Taxation affects this decision by altering the effective return of labor.

A higher tax rate can reduce the private gain from earning additional coin, while redistribution can increase the coin endowment of lower-income agents. Planner tax policy therefore enters the agent decision problem by changing the after-tax return from building and trading, while redistribution changes agents' coin endowments at the end of each tax period. At the same time, agent behavior determines the income distribution, production level, and tax base observed by the planner. The planner's problem is therefore linked to agent behavior through a feedback loop: taxes influence agent incentives, while agent actions determine the welfare outcomes used to evaluate planner policy.

The planner policies are evaluated using a social welfare function that captures both efficiency and distributional outcomes.

Let $x_t^c = (x_{1,t}^c, \dots, x_{N,t}^c)$ denote the distribution of coin endowments across all agents at time t . In the multi-region setting, welfare is evaluated at the regional level. Let $\mathcal{I}_{r,t}$ denote the set of agents located in region $r \in \{\text{top, bottom}\}$ at time t , and let

$$x_{r,t}^c = (x_{i,t}^c)_{i \in \mathcal{I}_{r,t}}$$

denote the coin endowments of agents currently located in region r . Regional production is then defined as

$$\text{prod}_r(x_{r,t}^c) = \frac{1}{|\mathcal{I}_{r,t}|} \sum_{i \in \mathcal{I}_{r,t}} x_{i,t}^c, \quad \text{if } |\mathcal{I}_{r,t}| > 0.$$

Inequality is measured using the Gini coefficient [29]. For the full economy, the Gini coefficient is denoted by $\text{gini}(x_t^c)$, where x_t^c is the distribution of coin endowments across all agents. The coefficient takes values between 0, representing perfect equality, and 1, representing maximal inequality. Regional equality is then defined as

$$\text{eq}_r(x_{r,t}^c) = 1 - \text{gini}(x_{r,t}^c).$$

Social welfare combines equality and production. For planner r , welfare is computed over the agents currently located in that planner's region:

$$\text{swf}_{r,t}(x_{r,t}^c) = \text{eq}_r(x_{r,t}^c) \cdot \text{prod}_r(x_{r,t}^c).$$

Thus, regional welfare is high only when both total regional production and equality among agents in that region are high.

The reinforcement learning problem is defined in terms of changes in these objective functions over time.

Agent rewards are given by changes in individual utility. To avoid confusion with the region index r , rewards are denoted by ρ . For agent i , the reward at timestep t is

$$\rho_{i,t} = u_i(x_{i,t}^c, l_{i,t}) - u_i(x_{i,t-1}^c, l_{i,t-1}). \quad (1)$$

Each regional planner receives its own reward, defined as the change in social welfare for the agents currently located in its region. The reward for planner r is

$$\rho_{r,t}^P = \text{swf}_{r,t}(x_{r,t}^c) - \text{swf}_{r,t-1}(x_{r,t-1}^c).$$

This makes planner rewards region-specific: the top planner is rewarded according to welfare changes among agents currently located in the top region, while the bottom planner is rewarded according to the corresponding welfare changes in the bottom region.

This formulation ensures that both agents and the planner optimize marginal improvements rather than absolute levels. As a result, agent decisions and policy adjustments respond directly to changes in economic conditions within the simulation.

3.9 Tax Collection and Redistribution

At the end of each tax period of length 100 timesteps, each agent i realizes period income as the change in coin holdings since the beginning of the period:

$$z_{i,t} = x_{i,t}^c - \bar{x}_{i,t}^c,$$

where $x_{i,t}^c$ is the agent's current coin endowment and $\bar{x}_{i,t}^c$ is its coin snapshot from the start of the tax period. Since agents may lose coin during a period, taxable income is defined as

$$z_{i,t}^+ = \max\{z_{i,t}, 0\}.$$

Taxes are computed separately within each region using the fixed income bracket cutoffs reported in Table 5. Let $m = (m_1, \dots, m_{B+1})$ denote the common bracket cutoff vector, with $m_{B+1} = \infty$. For taxable income $z_{i,t}^+ \geq 0$, the tax liability under region r 's schedule is

$$T_r(z_{i,t}^+) = \sum_{b=1}^B \tau_{r,t}^b [(m_{b+1} - m_b) \mathbf{1}[z_{i,t}^+ > m_{b+1}] + (z_{i,t}^+ - m_b) \mathbf{1}[m_b < z_{i,t}^+ \leq m_{b+1}]].$$

Here, $\mathbf{1}[\cdot]$ is an indicator function equal to one when the condition inside the brackets is true and zero otherwise. The first term applies when taxable income fully exceeds bracket b , so the entire bracket width $m_{b+1} - m_b$ is taxed at the marginal rate $\tau_{r,t}^b$. The second term applies when taxable income falls inside bracket b , so only the portion of income above the lower cutoff, $z_{i,t}^+ - m_b$, is taxed at the same marginal rate. Thus, $T_r(z_{i,t}^+)$ gives the total tax liability under the tax schedule chosen by planner r for the current tax period.

Let $r(i, t)$ denote the region in which agent i is located at the time of tax collection. When mobility is enabled, this is the region that determines the applicable tax schedule, regardless of where the income was earned during the tax period. To avoid taxing coins that are already committed to pending market orders, tax payments are limited to the coins currently available in the agent's inventory:

$$\hat{T}_{i,t} = \min\{T_{r(i,t)}(z_{i,t}^+), x_{i,t}^{c,\text{inv}}\}.$$

Regional tax revenue is given by the sum of tax payments from agents currently located in that region:

$$R_{r,t}^{\text{tax}} = \sum_{i \in \mathcal{I}_{r,t}} \hat{T}_{i,t}^{\text{tax}}.$$

The region may also receive revenue from travel payments and trade tariffs. Let $R_{r,t}^{\text{travel}}$ denote revenue from agents leaving region r , and let $R_{r,t}^{\text{tariff}}$ denote tariff revenue collected from regional trade. Total regional revenue is therefore

$$R_{r,t} = R_{r,t}^{\text{tax}} + R_{r,t}^{\text{travel}} + R_{r,t}^{\text{tariff}}.$$

Budget balance is achieved through per-capita lump-sum redistribution within each region. The transfer in region r is

$$g_{r,t} = \frac{R_{r,t}^{\text{tax}} + R_{r,t}^{\text{travel}} + R_{r,t}^{\text{tariff}}}{|\mathcal{I}_{r,t}|}.$$

The transfer is then credited to inventory coins. After taxes and transfers are applied, the system updates the coin snapshot $\bar{x}_{i,t}^c \leftarrow x_{i,t}^c$ for the next tax period.

4 Training and Experimental Design

4.1 Phased Learning

Policies are trained within a structured multi-phase training procedure designed to make learning more stable. Learning proceeds in four phases, each introducing or holding fixed a specific component of the system:

Phase 1: Agent Pretraining. (500 iterations). Mobile agents are trained while planner policies remain fixed. This allows agents to develop baseline economic behaviors without the influence of taxation and redistribution. During this phase, agents learn to navigate toward resource tiles, collect resources, and use the auction market to buy missing inputs or sell excess inventory. The resulting policies provide a stable foundation for the later and more complicated learning phases.

Phase 2: Taxation Learning. (1500 iterations). Planners are introduced in phase 2 and begin to optimize tax policies. To avoid instability from abrupt policy changes, taxation is gradually incorporated through an annealing schedule in which the maximum allowable marginal tax rate increases linearly from 10% to 100% over the course of 400 iterations. This enables agents to adapt smoothly while allowing planners to learn responses to changing agent behavior.

Phase 3a: Agent Adaptation with Fixed Planners. (500 iterations). Cross-regional mobility is introduced by enabling the travel action for agents, while planner policies are held fixed. This isolates the effect of mobility on agent behavior in a stable policy environment.

Phase 3b: Planner Adaptation to Mobile Agents. (1500 iterations). Finally, planner policies are unfrozen and trained jointly with agents with mobility enabled. This phase captures the fully joint adaptation between agents and planners. When agents are able to relocate planners must then account for the risk of losing productive agents to the other region, while agents continue to adjust their behavior in response to changing tax policies. The resulting dynamics generate a feedback loop between migration decisions and taxation policies.

4.2 Training Scale and Parallel PPO Updates

The simulations are trained using the multi-agent PPO setup described in Section 2.4. Training is organized into four phases with a total of 4,000 training iterations. Each iteration collects a batch of 800 environment timesteps before updating the policy. Across the full training process, this corresponds to approximately 3.2 million environment timesteps. Since each episode lasts 1,000 timesteps, the total amount of training is equivalent to roughly 3,200 simulated episodes.

Training data are collected using parallel rollout workers. In practice, several independent replicas of the same economic environment are simulated at the same time. Each replica contains its own agents, planners, resources, markets, and stochastic events, but all replicas use the current policy parameters when generating trajectories. These trajectories contain the observations, sampled actions, rewards, value estimates, recurrent hidden states, and next observations experienced by agents and planners during rollout.

Once enough experience has been collected, trajectories from the parallel environments are combined into a single training batch. The batch is then divided into minibatches and reused for several stochastic-gradient updates of the corresponding actor-critic networks. The same PPO update rule is used for both mobile agents and regional planners, but the policies are trained from their own experience streams. The eight mobile agents are mapped to the shared agent policy a , while the top and bottom planners are mapped to the separate policies p_{top} and p_{bottom} . Thus, agent learning updates a shared behavioral policy, while planner learning updates the two regional tax policies independently.

Parallel rollout collection is used for computational efficiency. Running several environment replicas at once increases the amount of experience collected per training iteration and reduces the wall-clock time required to train the policies. A more detailed explanation of the parallel training configuration, including workers, environment replicas, rollout fragments, batch sizes, and minibatch optimization, is provided in Appendix A.10. The full interaction loop is summarized in Algorithm A.3, and the main training hyperparameters are reported in Appendix Table 3.

4.3 Observation Space Consistency

The observation space is held fixed across all experimental conditions. Even in settings where cross-regional trade or labor mobility is disabled, agents continue to receive the same types of local and non-local tax and market observations. This ensures that differences between experiments are not driven by differences in what agents or planners can observe.

The same is true for the multi-phase training structure. Planner policies remain fixed during Phase 1 and 3a even when mobility is disabled, before being reintroduced in phase 2 and the final phase 3b.

This design ensures that behavioral changes across experimental conditions are driven by changes in the rules of the environment, such as whether trade or travel is allowed, rather than by changes in what agents are able to observe. For example, when cross-regional trade is disabled, agents may still observe non-local market information, but they cannot act directly on it through trade. Similarly, when travel is disabled, agents may observe tax differences across regions, but they cannot respond by relocating.

Keeping the observation space fixed therefore preserves comparability across the closed-economy, trade, and mobility conditions. It allows the analysis to isolate the effect of enabling new actions or interactions, rather than mixing those effects with changes in information availability.

4.3.1 Relocation and Spatial Generalization

To limit unnecessary learning complexity arising from spatial variation, relocation destinations are defined relative to agents’ initial positions rather than assigned randomly. When agents move across regions, they are placed in locations that are structurally comparable to their starting positions.

This design choice is particularly important given that agents observe only a limited 11×11 window of the environment. Without such control, spatial heterogeneity could introduce additional variation unrelated to the economic mechanisms of interest. By preserving local spatial similarity, behavioral differences can be more cleanly attributed to changes in taxation, market conditions, and agent interactions.

4.4 Runs, Seeds, and Evaluation Rollouts

Due to computational constraints, the main analysis is based on three independently trained runs for each experimental setting. Each setting is therefore trained three times from different random seeds, and each completed training run is evaluated using 20 rollout episodes after training is finished. The results should still be interpreted as exploratory evidence of learned dynamics in the implemented environment, rather than as statistically definitive estimates,

but the use of multiple independently trained runs helps reduce the risk that the findings are driven by a single path-dependent training trajectory.

The evaluation rollouts use the final trained policies. Variation across the 20 rollout episodes captures stochasticity during evaluation, including action sampling, resource spawning, and market matching. Variation across the three trained runs captures a broader source of uncertainty, since it also reflects differences in initialization, training path dependence, and how policies adapted to each other during training.

The results are therefore interpreted primarily through patterns that appear consistently across trained seeds, evaluation rollouts, and related figures. Isolated differences in a single metric are treated cautiously, especially when variation across trained runs or rollout episodes is large.

5 Results

5.1 Results Overview and Interpretation of Figures

This section presents the simulation results from the three main experimental settings: the closed-economy baseline, the trade-integrated economy, and the labor mobility setting where agents can relocate between regions. The results are organized to build up the mechanisms gradually. The baseline first establishes how agents and planners behave when the two regions are economically isolated. The trade setting then introduces cross-regional market access while keeping agents fixed in their original regions. Finally, the labor mobility setting allows agents to relocate, making taxation part of a broader decision problem involving market access, spatial conditions, accumulated resources, and travel costs.

The main purpose of the results section is to identify how inter-regional interaction changes the learned behavior of the system. The central question is whether market integration and labor mobility alter agent behavior, regional outcomes, and learned tax policy. The analysis focuses on the mechanisms behind these changes: how agents use trade, individual relocation incentives, how planner tax schedules respond to learned regional outcomes, and whether tax differences can explain movement between regions.

The following sections therefore build the argument in stages: first by identifying the baseline mechanisms of specialization and taxation, then by introducing market integration, and finally by examining whether labor mobility is best understood as tax minimization or as a broader learned economic response.

5.2 Baseline: Closed Economy

The baseline setting considers two closed regions, meaning that agents cannot travel between regions, and markets strictly local. Each region therefore evolves as a local economy with its own market, agent population, and regional planner. This setting provides a benchmark for understanding how agents allocate labor, how local markets develop, and how planners learn tax schedules in response to agent behavior without the possibility of relocation.

The baseline shows that even without inter-regional interaction, heterogeneous outcomes emerge from the interaction between skill, spatial position, resource access, market activity, and planner taxation. It further shows that taxation itself is volatile and dependent on agent behavior even when regional agent composition remain constant.

5.2.1 Agent Actions and Spatial Context

The first part of the baseline analysis examines how individual agents move and act within the grid environment. Although agents are trained using shared neural network parameters θ , their realized behavior can differ substantially because each agent conditions its actions on its own observation, hidden state, location, inventory, skill level, and local market conditions. Shared policy parameters therefore do not imply identical behavior across agents.

Spatial position is important, since resource availability is not uniform across the map (see Figure 5). Agents observe only a local window of the environment, so their initial location affects which resources, agents and houses are visible. Agents starting near resource tiles may gather more frequently, while agents located further away from resource tiles may have stronger incentives to trade.

Agent behavior changes over the course of an episode as the economy develops. Early actions are shaped by nearby resources. Later behavior reflects accumulated economic state, including market prices, previous trades, taxation, redistribution, and earlier house construction. Houses are particularly important because they alter the spatial structure of the environment: since a house can only be passed through by the agent who built it, previous building decisions can affect later movement paths and access to resources.

Although the baseline regions are initialized symmetrically, their outcomes need not be identical. Stochasticity in resource spawning, action sampling, market matching, and policy learning can lead to different early trajectories, which may then be amplified through agent and planner responses. The baseline therefore provides a useful setting for examining how heterogeneous behavior and unequal outcomes emerge before inter-regional trade or mobility is introduced.

Figures 7 and 8 provide a special view of one closed-economy rollout, and show where in the different agents move and build. These single-run plots are not interpreted as population-level averages, but as evidence of the mechanisms generated by the learned policies.

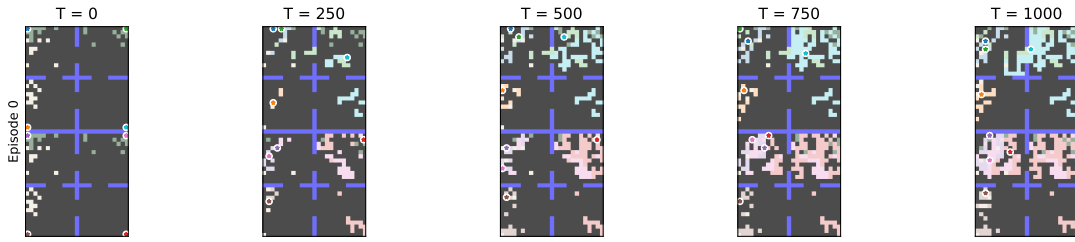


Figure 7: **Spatial development of the closed economy over one rollout.** The panels show the environment at selected timesteps from $t = 0$ to $t = 1000$. The figure illustrates how initially empty building areas become increasingly occupied as agents gather resources, trade, and construct houses. Each colored square indicates a house built by the same colored agents.

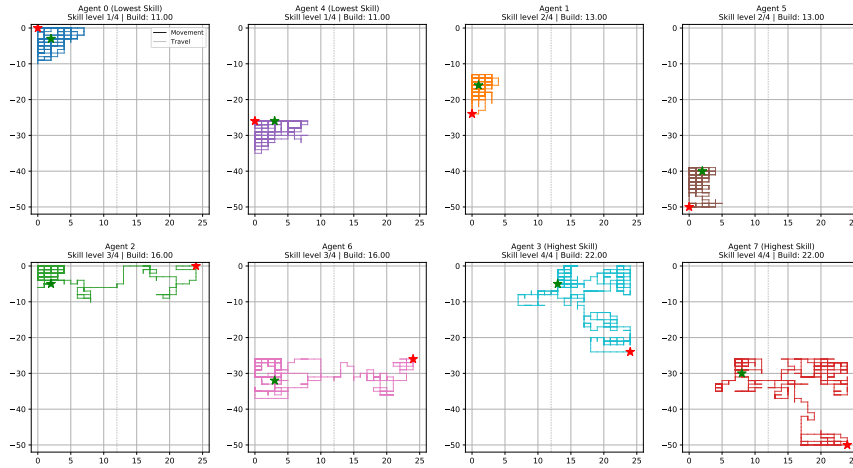


Figure 8: **Individual movement paths in the closed economy.** Agents with identical skill levels are shown in pairs next to each other. The figure shows that each agent’s movement trajectory, with start and end positions marked. It illustrates that agents with the same skill levels have similar but not identical movement paths and that movement path across skill levels are different.

Figures 7 and 8 show that heterogeneous behavior emerges even though agents share the same policy parameters. Figure 7 shows how the initially undeveloped map becomes shaped by movement and building decisions over the rollout. Many agents remain relatively close to their initial quadrant, but some move farther in response to local resource constraints and building opportunities.

Figure 8 makes this heterogeneity clearer. Some agents move substantially away from their starting locations, and this is most pronounced for the highest-skill agents. This suggests that high-skill agents actively search for resources that are not available near their initial positions, likely because access to both wood and stone is necessary in order to benefit from their higher building productivity.

Agents starting in the wood-dominated quadrant (agent 2 and 6) also appear to explore the environment outside of their starting quadrant in this rollout. This movement pattern indicates either a special constraint from house building or that stone is less available through the auction market at a price that can compete with seeking the stone themselves. This behavior therefore suggests that spatial constraints and market access are part of shaping realized behavior.

To examine what patterns are reflected in broader outcomes, Figure 9 reports a set of average baseline metrics grouped by building skill across all 3 tested seeds. These metrics summarize the main economic outcomes of the closed-economy setting and provide additional context for interpreting the results of the baseline economy.

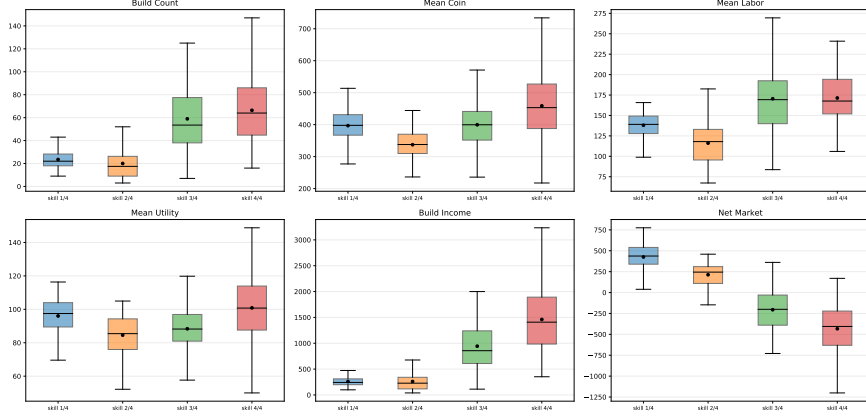


Figure 9: **Baseline agent outcomes grouped by building skill.** Boxplots summarize outcomes across dense logs for build count, mean coin, mean labor, mean utility, build income, and net market position. The figure shows that higher-skill agents generally build more and earn more build income, while lower-skill agents rely more on positive net market income.

In figure 9 the build-count and build-income panels show a clear differences across skill levels. The highest-skill agents build the most on average, followed by the second-highest skill group. This confirms that building skill strongly affects production outcomes, since higher-skill agents receive more income from each house and therefore have stronger incentives to build when resources are available. Interestingly enough the lowest skill agents build on average more than the second lowest, a probable cause of this result is because of their advantageous starting location with accesses to both types of resources (explained in section 3.2).

The mean coin panel shows that the highest-skill agents also hold the most coin on average. This is consistent with their higher build income and larger number of completed houses. However, the lowest-skill group has the second-highest average coin holdings, despite producing relatively little build income. This difference is explained by the market panel: the lowest-skill agents have a strongly positive net market position, indicating that they earn coin primarily by selling resources rather than by building houses. Their starting position in a quadrant with access to both wood and stone gives them a resource advantage, which allows them to participate as sellers in the market.

The second-lowest skill group has both low build income and lower net market income, resulting in the lowest mean coin and explains why they also have the lowest utility among the groups. This suggests that skill alone does not determine outcomes; spatial access to resources and market position also matter. The two highest-skill groups generate the most build income, but have average negative net market positions, indicating that they spend coin on resources in order to support production. This is consistent with the interpretation that higher-skill agents use the market to acquire missing inputs for building, while lower-skill agents are more likely to earn income through resource sales.

Overall, the results show a consistent interpretation of the closed-economy baseline. The spatial evidence suggests that agents face different opportunities depending on their starting location and access to resources. High-skill agents move farther to acquire the inputs needed for building, while agents with better local resource access rely more on gathering and market sales.

These spatial differences translate into distinct economic roles. High-skill agents build more and earn more build income, and often spend coin in the market to acquire resources. Lower-skill agents generate less build income, but some partly compensate through positive net market income from resource sales.

5.2.2 Agent Labor Allocation and Economic Activity

The agents' labor allocation across activities provides a population-level view of how the closed economy organizes production, gathering, movement, and market participation.

The balance between resource collection, building, and trading is particularly informative. A high share of labor spent gathering suggests that the resources' worth in buildings or on the market is high compared to the labor needed to acquire them. A high share spent trading indicates that agents rely on the market to acquire resources or sell excess for coin. Building activity reflects the extent to which agents chose to convert resources into income. Together, these labor patterns show the economies specialization pattern across agents.

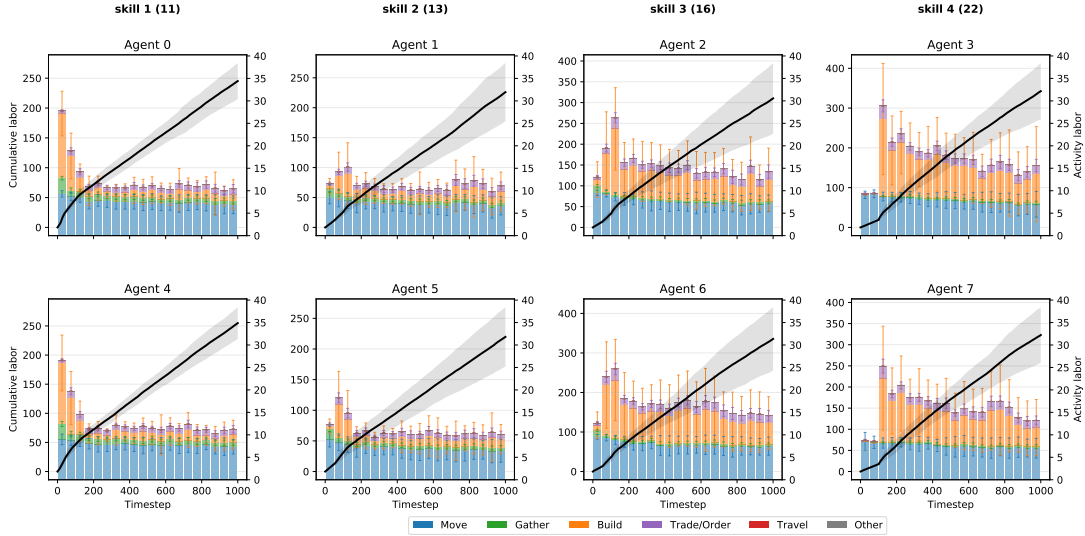


Figure 10: **Labor allocation across actions in the closed-economy baseline.** Bars show mean action-specific labor across dense logs, with error bars indicating variation, while the black line shows cumulative labor. The figure shows that movement and trade are common across agents, while building and gathering differ more across agents, reflecting differences in building skill, starting location, and access to resources.

Figure 10 shows how each agent allocates labor across actions during the episode. Movement is the most consistent behavior, both across agents and over time, indicating that agents continuously reposition themselves throughout the rollouts. Trade is also widely used across all agents, and after the initial part of the episode it becomes highly uniform, suggesting that market interaction becomes a regular part of agent behavior. The figure also shows consistent labor allocations in the different skill groups across the two regions suggesting that the skill level is somewhat reflected in the agents action chooses regardless of region.

The largest differences across skill groups appear in building activity. Higher-skilled agents generally build more, but they also show larger variation across dense logs. An explanation for this variation is that high-skilled agents that reaches a resource-rich quadrant can build more, earn more coin, buy more resources, and continue building, creating a reinforcing loop. However, If the agents spend more time in a less favorable position, the opposite can occur. Both of the high skill agents do not build at the beginning of any of the runs. This pattern occurs consistently because of the initial starting locations (see section 3.2). Agents first need to move to a different quadrant in order to gather the resources required for building a house. They also cannot buy these resources on the market at the start of the episode, since agents begin with no coin and their own starting quadrant does not contain any necessary resources.

Gathering is used more by lower-skilled agents, which suggests that they rely more on direct resource collection than on building-based income, which is to be expected based on the lower relative income.

Overall, the plot shows that movement and trade are common across agents, while building and gathering reflect differences in skill and spatial opportunity. This supports the interpretation that specialization emerges in the simulated economy.

5.2.3 Planner Tax Schedules and Welfare Components

Tax policies reflect the planners' response to agent behavior, but agent behavior is itself partly shaped by tax policy. Agents optimize utility by choosing how much to work, trade, build, and accumulate income under the tax schedule they face. The number of agents in each tax bracket can therefore be interpreted as a rough indicator of how agents respond to planner actions, subject to constraints such as skill, resources, market access, and location.

This makes it useful to ask whether planners choose tax schedules that appear responsive to where agents fall in the income distribution. Figure 11 gives a descriptive view of the average realized bracket populations and average marginal tax rates across tax days.

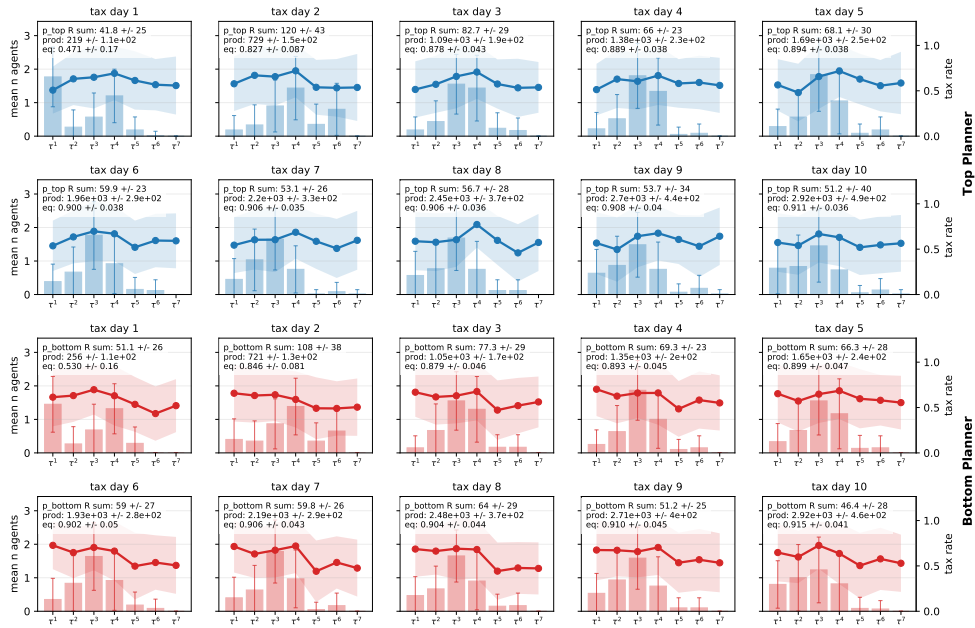


Figure 11: **Average tax-day snapshots across dense logs and seeds.** Bars show the mean number of agents in each income bracket, while lines show the mean marginal tax rate selected for each bracket. Higher production and equality indicate a higher welfare level, while planner reward reflects the change in welfare over time. The figure shows no simple one-to-one relationship between bracket population and marginal tax rates, but it suggests that welfare levels generally improve over time, especially through rising production.

The figure shows how each regional planner's marginal tax schedule relates to the realized income distribution of agents. The panel shows that agents are concentrated primarily in the lower and middle income brackets. The regional schedules differ across the top and bottom planners and vary across tax days, suggesting that planner behavior is shaped by region-specific income distributions and welfare feedback rather than converging to a single identical tax rule.

The reported production and equality values describe the level of the regional welfare components at each tax day, while the planner reward sum measures the accumulated change in welfare over the tax period. Thus, planner rewards do not necessarily increase whenever production and equality levels are higher; they are larger when welfare improves more during that period. The figure therefore suggests that planners learn from marginal improvements in regional welfare, not from tax revenue or welfare levels alone.

The descriptive panel suggests that planner tax schedules are not easily reducible to the realized bracket distribution alone. To examine this more systematically, Appendix Figures 23 and 24 report the correlation between marginal tax rates and the number of agents in each bracket across tax days. These correlations show no strong evidence that planners use bracket population as a primary basis for tax decisions.

Overall, Figure 11 suggests that planner behavior cannot be explained by a simple count-based rule. This is important because the number of agents in each bracket is itself an outcome of agent behavior. Agents move through the income distribution by choosing how much to gather, trade, build, and accumulate coin under the tax schedule they face. Planner tax decisions and agent income choices therefore evolve together. As production grows and equality remains high, positive planner rewards reflect that regional welfare continues to improve over the tax periods. The variation in schedules across regions and tax days therefore suggests that planners are adapting to the feedback between taxation, agent behavior, and regional welfare outcomes. Because of this complexity, the learned tax schedules are best interpreted as part of a feedback process rather than as direct responses to bracket counts alone.

5.3 Cross-Regional Trade and Market Integration

The second experiment enables cross-regional trade while keeping labor mobility disabled. Agents remain in their original regions, but resources can be exchanged across the regional boundary. This isolates the effect of market integration on production, prices, resource allocation, and tax revenue before introducing relocation. (see section 3.4 for the detailed setup)

Market integration can affect not only bids and asks, but also gathering, building, inventory management, and movement toward resources within the local environment. Agents may adjust collection rate, when they trade, and whether they build based on the market opportunities they have observed over time.

5.3.1 Market Integration, Cross-Regional Trade, and Redistribution

The key question in this setting is whether opening the market changes resource allocation, prices, and redistribution, or whether trade remains mostly local despite the possibility of cross-regional exchange.

The comparison between the closed and open-market settings suggests that market integration primarily changes market access and matching conditions, rather than simply increasing trade activity throughout the episode. After the initial adjustment phase, escrowed resources are lower in the open-market setting, consistent with improved matching between supply and demand in a larger market. Average transaction prices are slightly higher, indicating that cross-regional access changes effective demand as well as supply. Trade activity spikes early in both settings and then stabilizes at broadly similar levels, with the open-market setting only slightly higher for much of the episode. Appendix Figure 25 reports these supporting patterns.

Figure 12 provides the main evidence on how cross-regional trade is used. The figure separates within-region and cross-region trades for wood and stone across tax periods. The pie charts summarize the overall share of within-region and cross-region sales for each resource, while also reporting average seller prices. This makes it possible to examine whether cross-regional trade becomes an economically meaningful part of the market, and whether it differs systematically from local trade.

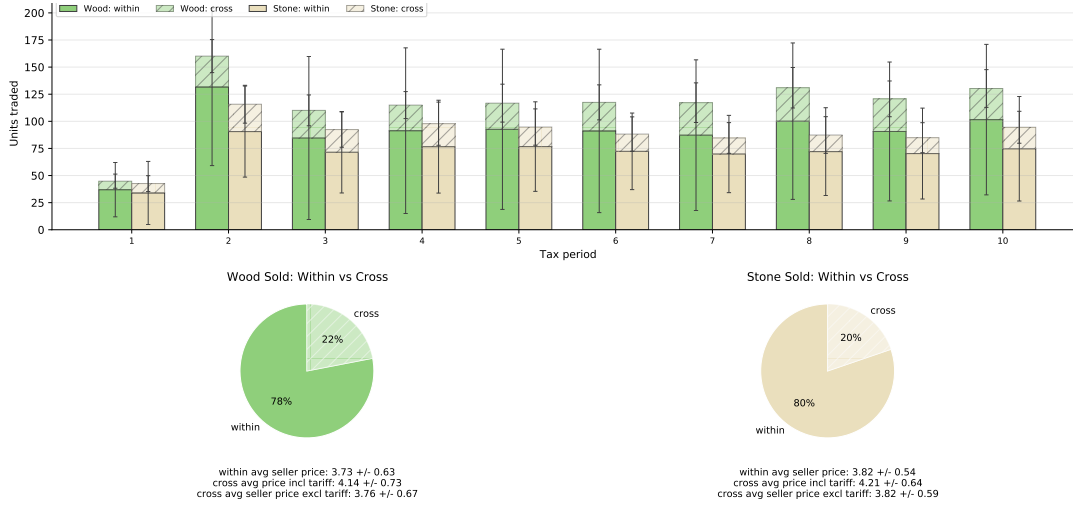


Figure 12: Cross-regional trade under market integration. The upper panel reports units traded by tax period, separated by resource and by whether the trade occurs within or across regions. The low initial trade volume followed by a spike and later stabilization reflects an early adjustment phase, as agents first need to acquire coin and resources before market activity becomes regular. The pie charts summarize trade shares and average seller prices for wood and stone, both including and excluding the 10% import tax. Cross-regional trades account for around one quarter of both wood and stone sales, showing that agents actively use the integrated market despite the trade friction.

The figure 12 shows that wood is generally traded more both within region and across region. The error bars show that the fluctuation in wood trades are larger than stone trades.

Trading activity follows a similar pattern across tax periods for both resources. Trade is initially low, then rises sharply, and finally stabilizes. This pattern appears both in the closed baseline and in the trade-enabled setting, as also shown in Appendix Figure 25. This pattern is likely related to the role of high-skill agents in the market. The previous section showed that high-skill agents buy the most resources on average and engage in higher building activity in the second tax period but non in the first, as shown in Figures 9 and 10. (The section 5.2.2 also explains why this pattern appears). The low trade volume in the first tax period can be interpreted as an initial adjustment phase. Once high-skill agents have enough coin, trading increases sharply, likely supporting the higher building activity observed in the previous section. After this point, trade stabilizes as agents continue to use the market as part of regular production rather than as a one-time adjustment.

Figure 12 also shows that cross-region trades make up a noticeable share of total resource exchange throughout the episode. For both wood and stone, cross-region sales account for around a fifth of total sales, with approximately 22% of wood sales and 20% of stone sales occurring across regions. This indicates that cross-regional trade is not a marginal feature of the simulation. Agents actively use the integrated market, even though imported goods are subject to the additional 10% tax.

The average seller prices in the pie chart summaries also suggest that cross-region and within-region trades occur at comparable prices. However for both wood and stone, the average seller price is slightly higher for cross-region trades. This supports the interpretation that agents appear to use the open market when it helps resolve resource imbalances or when non-local orders provide useful trading opportunities.

There are several mechanisms that may explain why agents use cross-regional trade even when average cross-region seller prices are slightly higher than local prices. For lower-income agents, the effective cost of buying imported resources may be partly offset through redistribution. For higher-income agents, the decision may instead reflect a trade-off between current spending and future production. Buying resources at one timestep can enable house construc-

tion at a later timestep, generating income that may outweigh the higher purchase price.

The timing of purchases and production may also matter. Spending coin on resources reduces current coin holdings, while the income from building is realized later. This can affect both the agent’s tax exposure and its utility, since the utility function gives diminishing marginal value to additional coin (see section 3.8). As a result, holding more coin is not always as valuable as converting coin into resources that enable future production. Cross-regional trade can therefore be useful even when it is more expensive on average, because it may help agents overcome local resource constraints, smooth production opportunities over time, or convert coin into inputs with higher expected future value.

The redistribution effects of the import tax are reported separately in Appendix figure 26. This shows that redistribution remains dominated by income taxation. Import-tax revenue appears after the first tax period and remains present throughout the episode, but it is much smaller than the revenue redistributed through income taxes. This suggests that import taxation creates a consistent secondary redistribution channel, but not one that drives the main redistributive structure of the economy.

This distinction is important for interpreting the trade experiment. Cross-regional trade mainly affects the economy through market access, prices, and resource allocation, while import-tax redistribution plays a smaller supporting role. The import tax matters because it preserves a friction between local and non-local exchange and creates some redistribution from cross-regional market activity. However, the main behavioral effect of market integration appears to come from agents being able to access a broader market for the resources needed in production.

Overall, the trade experiment shows that market integration changes the economy mainly through market access rather than through import-tax redistribution. These results from the mentioned figures suggest that the main effect of cross-regional trade is to expand the set of possible trading partners and make non-local resources available when local conditions are limiting.

5.4 Labor Mobility and Relocation

The final experiment enables labor mobility. Agents can now relocate between regions and thereby change the tax policy, market conditions, spatial environment, and population context they face. Mobility therefore introduces a new behavioral margin: agents no longer respond to taxation only by changing labor, trade, or production decisions, but may also respond by changing location.

The purpose of this section is to evaluate whether relocation emerges as a learned economic response and how this affects regional outcomes, as well as impact factors on the relocation decision. Because agents move with their inventories, leave houses behind, face travel costs, and enter a different spatial and market environment, relocation reflects a multi-dimensional decision problem.

This setting is more complex than the previous ones because relocation affects multiple channels at once. Moving changes the tax system applied to an agent’s income, the local resource environment, and the market tax faced by the agent. It also changes regional population composition, which in turn affects planner rewards and future tax policy. For this reason, the analysis considers both learned planner policies and fixed tax schedules. Fixed tax schedules are useful because they isolate agent responses to taxation from simultaneous planner adaptation.

A natural hypothesis is that labor mobility makes agents responsive to regional tax differences. If agents can relocate, they may be expected to move toward the region where the expected tax burden is lower, especially when they have accumulated income during the tax period. In this case, tax policy would become a direct driver of regional sorting, and the social planners might face pressure to lower tax rates in order to retain or attract agents.

However, relocation is not only a tax decision in this environment. Moving is costly, outstanding market orders are removed, houses remain fixed in the origin region, and the destination region may differ in current resource access and congestion. The following sections

therefore treat tax incentives as one possible driver of relocation, rather than assuming that tax minimization alone determines movement.

5.4.1 Introducing Labor Mobility

Before analyzing individual relocation decisions in detail, it is useful to compare the aggregate outcomes across the three main experimental settings: the closed baseline, cross-regional trade, and cross-regional trade with travel enabled. This comparison provides an initial view of how mobility changes the economy relative to the previous experiments. The purpose is not only to ask whether travel improves aggregate outcomes, but also whether it changes the distribution of behavior across skill groups and whether planners adjust their tax schedules when agents are allowed to move between regions.

Figure 13 reports summary outcomes across the three settings. Enabling travel changes the action space available to agents, but relocation is not costless. Agents spend both labor and coin when moving between regions, and additional movement may be required after relocation before they can gather resources, trade, or build effectively. Some labor that could otherwise have been used for gathering or production may therefore be redirected toward mobility. At the same time, relocation can place agents in regions with different resource access, market conditions, tax exposure, or local congestion.

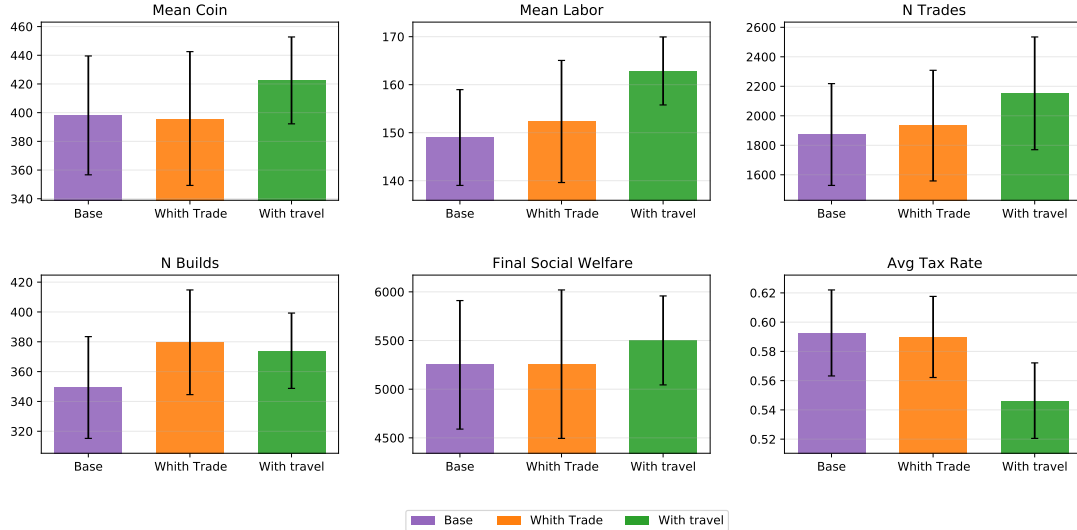


Figure 13: **Summary metrics across the baseline, cross-regional trade, and travel-enabled settings.** Bars show means across evaluation rollouts from the final trained checkpoints across seeds, with error bars indicating one standard deviation. The travel-enabled setting does not uniformly dominate the other settings. Mean coin, labor, builds, and final social welfare remain broadly comparable across settings, while the average tax rate is slightly lower when travel is enabled. Number of trades seems to be slightly higher for trade and travel. Note that the y-axes are scaled.

Figure 13 provides an aggregate comparison across settings. The purpose of the figure is to identify whether each outcome changes systematically across the baseline, trade-only, and travel-enabled settings.

A pattern in the figure is that the travel-enabled setting is associated with a higher average number of trades. Allowing travel expands the agents’ effective opportunity set. Mobility changes which resources, prices and tax schedules an agent can access, making trade more valuable as part of a broader learned strategy. Labor also slightly increases travel-enabled setting, which is to be expected with more trades, and also because the travel action itself is associate with a labor cost.

For mean coin, number of builds, and final social welfare, the differences across settings are relatively modest compared with the variation across runs. This suggests that enabling travel does not uniformly improve aggregate outcomes. Instead, mobility appears to change the composition of behavior more than it produces a clear welfare or production increase.

The average tax-rate panel shows a slightly lower average tax rate in the travel-enabled setting, at just above a 4-pct point decrease, which indicates that planner behavior changes under mobility possibly as a response to the agents being able to relocate. Lower average taxes in the travel-enabled setting compared to the two other settings is consistent across all the observed seeds (this result is shown in the Appendix figure 30). This planner response is examined more directly below.

The mobility setting also changes the problem faced by the planners. When agents can relocate, each planner no longer governs a fixed population: the set of agents entering the regional welfare calculation can change over time. A natural hypothesis is therefore that mobility may create pressure for planners to lower tax rates in order to retain or attract agents. Figure 14 examines this hypothesis by comparing the learned marginal tax schedules across the baseline, trade-only, and travel-enabled settings.

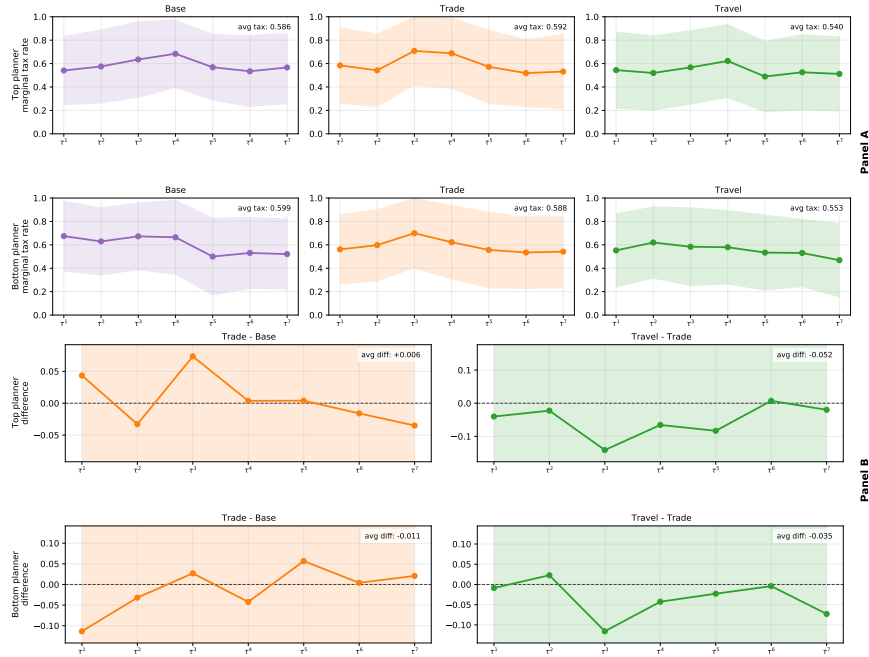


Figure 14: **Planner tax schedules across experimental settings.** Panel A shows average marginal tax schedules by bracket for the baseline, trade-only, and travel-enabled settings, separately for the top and bottom planners. Shaded bands indicate variation across runs. Panel B shows bracket-level differences in average marginal tax rates between consecutive settings. The travel-enabled setting is associated with slightly lower average tax rates for both planners compared with the trade-only setting, but the differences vary across brackets and remain small relative to the variation across runs.

Figure 14 shows that planner average tax schedules differ across experimental settings. In Panel A, average tax rates are slightly lower in the travel-enabled setting than in the trade-only setting for both planners. The average tax rate falls from 0.592 to 0.540 for the top planner and from 0.588 to 0.553 for the bottom planner. This indicates a modest downward shift in average marginal tax rates when mobility is introduced.

Panel B shows that this decrease is not uniform across brackets. For the top planner, the travel-minus-trade difference is negative for most brackets, but close to zero or slightly positive in some parts of the schedule. For the bottom planner, the decline is also uneven

across brackets and is not concentrated only in the highest income brackets. Because the tax system is marginal, lower rates across several brackets can reduce the tax burden for agents whose income reaches those brackets. The broad downward shift in the travel-enabled setting is therefore consistent with a tax-competition interpretation: when agents are able to relocate, planners may face pressure to set somewhat lower tax rates. The bracket-level pattern suggests that the downward shift under mobility is not concentrated at the very bottom of the income distribution. Since taxation is marginal, decreased rates in the middle bracket reduce tax liabilities for agents whose income reaches those brackets, including middle- and higher-income agents. This pattern is consistent with the interpretation that, when relocation is possible, planners may face stronger incentives to keep economically productive agents within their region. However, the pattern should be interpreted as a descriptive difference between learned schedules rather than causal evidence of tax competition, since the learned schedules also reflect changes in production, redistribution, regional composition, and stochastic policy learning.

To understand the aggregate results more clearly, Figure 15 separates several mechanisms by building-skill group. This is important because the baseline analysis showed that skill and spatial opportunity generate different economic roles across agents. If mobility changes the economy, it may do so unevenly across skill groups rather than through a uniform shift for all agents.

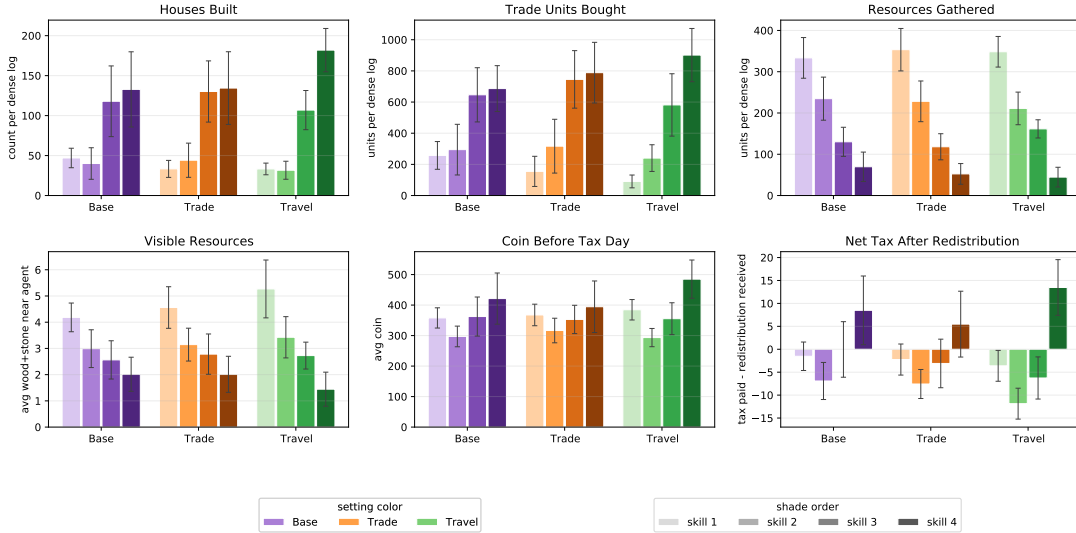


Figure 15: **Skill-group mechanisms across experimental settings.** Bars report group-level averages for houses built, trade units bought, resources gathered, visible resources, coin before tax day, and net tax after redistribution across the baseline, trade-only, and travel-enabled settings. The figure shows that skill-based specialization persists across all settings: higher-skill agents build and buy more, while lower-skill agents gather more. In the travel-enabled setting, some of these differences become more pronounced, especially for high-skill building and resource purchases.

Figure 15 shows that the skill-based specialization observed in the baseline remains present when trade and travel are introduced. Higher-skill agents build more houses and buy more resources across all three settings, while lower-skill agents gather more resources. This indicates that mobility does not remove the underlying economic roles created by the interaction between skill and spatial resource access.

The travel-enabled setting appears to strengthen some of these differences. The highest-skill group builds the most houses and buys the most resources in the travel setting, while the lowest-skill group continues to gather the most. This suggests that mobility may allow high-skill agents to access inputs, markets, or locations that support production, while lower-

skill agents remain more closely tied to gathering and resource provision. The result should be interpreted cautiously because the variation across runs is still visible, but the direction of the skill gradient is consistent across several metrics.

The tax-related panel also shows a stronger separation between skill groups when travel is enabled. The highest-skill agents receive a positive net tax after redistribution, while the lower-skill groups tend to have negative values. This does not mean that travel directly causes this redistribution pattern, but it is consistent with the interpretation that high-skill agents are more economically active and may be more likely to move into situations where they generate taxable income. The relocation-count figure in Appendix 31 provides supporting evidence that higher-skill agents travel more frequently.

Overall, the introduction of mobility does not replace the mechanisms observed in the closed baseline. Instead, it adds another adjustment margin on top of the existing interaction between skill, resource access, market participation, and taxation. The aggregate summary metrics show that travel does not lead to a uniform improvement across all outcomes, while the skill-group figure suggests that mobility affects agents differently depending on their economic role. At the same time, the planner comparison shows that tax schedules also change under mobility, with slightly lower average rates in the travel-enabled setting. However, this change is not a simple or uniform tax-cutting response. Taken together, these results suggest that mobility changes both agent behavior and planner policy, but through a broader adaptive process rather than through one direct mechanism. This motivates the next section, which examines relocation decisions directly and asks which conditions are associated with agents choosing to move between regions.

5.4.2 Relocation Decisions: Environmental and Tax Incentives

Relocation is one of the central mechanisms in the multi-region economy. Since travel is costly in both labor and coin, agents should only relocate when the expected benefit of entering the other region is large enough to offset these costs. This section tests whether relocation is mainly associated with lower tax exposure, or whether travel periods are also characterized by differences in skill, income, resources, and local spatial conditions.

To examine these mechanisms, this section compares the conditions observed in periods where agents travel with periods where they remain in their current region. The analysis combines three types of evidence. First, Figure 16 gives an overview by comparing different relevant metrics in both travel and non-travel periods, as well as report a counterfactual tax comparison. Second, Figure 17 reports regression-based evidence on which variables are conditionally associated with travel. Third, Figure 18 shows how travel probability varies across skill groups.

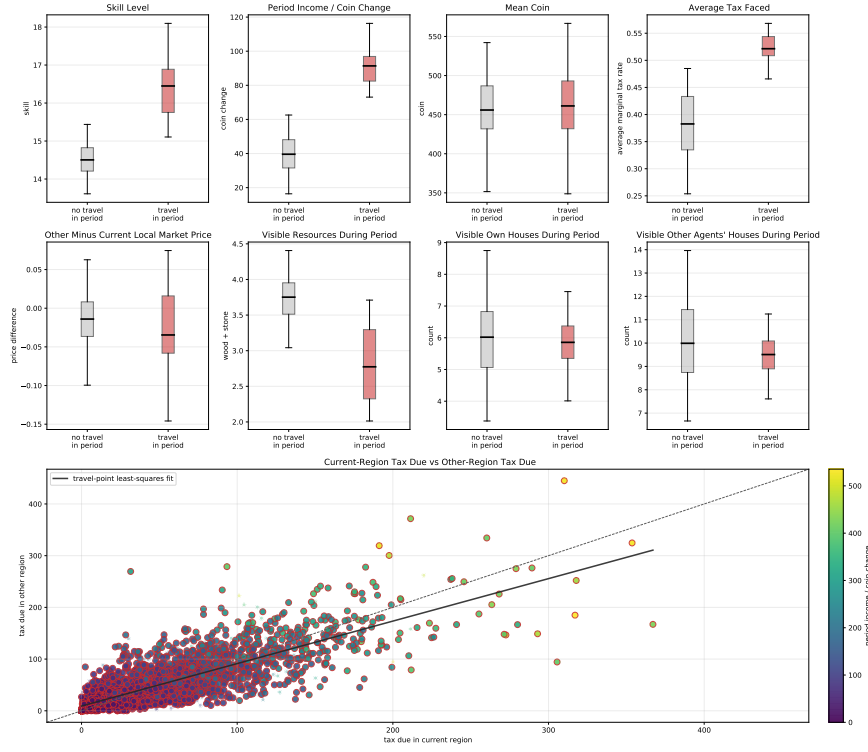


Figure 16: **Tax-period travel context and counterfactual tax.** The box plots compare travel and non-travel periods across skill, income during the period, mean coin holdings, average tax rate faced, the difference between other-region and current-region local market prices, visible resources, visible own houses, and visible other agents' houses. The scatterplot compares tax due under the current-region tax schedule with the counterfactual tax due under the other region's schedule. The figure shows that travel periods are on average associated with higher skill, higher period income, higher average tax rate exposure, and fewer visible resources. The counterfactual tax panel shows that at higher current-region tax liabilities, the fitted line lies below the diagonal, suggesting that some travelers move into a destination region with a lower tax burden for the same period income. However, several travel observations still lie above the diagonal, so lower destination-region taxation is not a necessary condition for relocation.

Figure 16 shows that travel periods differ from non-travel periods along several dimensions. Agents who travel tend to have higher building skill, higher income during the period, and higher average tax exposure. This supports the interpretation that relocation is partly associated with economic capacity: agents with higher productivity or larger recent gains are more able to absorb the cost of travel.

The counterfactual tax panel provides a more direct comparison of tax burdens across regions. Points below the diagonal indicate cases where the other region would tax the agent less than the current region, while points above the dotted diagonal indicate that the other region would tax the agent more and vice versa. The fitted line lies mostly below the diagonal at higher tax-due values, suggesting that some higher-income travel observations are associated with a lower counterfactual tax burden in the other region. However, the pattern is not universal: some agents travel even when the other region would imply a higher tax burden. The counterfactual tax comparison therefore provides evidence that tax incentives are present in some travel decisions, especially for higher-income observation periods, but the mixed placement of points around the diagonal shows that tax savings are not a universal condition for travel.

To examine relocation more systematically, I estimate logistic regressions where the dependent variable is whether an agent travels in a given period. A more detailed explanation of the

regression specification is provided in Appendix A.8. These regressions are not interpreted as causal estimates. Instead, they show which observed variables remain related to travel after accounting for the other variables included in the regression. These variables are skill, period income, mean coin, differences in market price, visible resources, and visible houses split into own house and other agents houses. Figure 17 reports logistic regression results from two specifications: one using all observations together, and one including agent fixed effects. For each model, the left panel shows odds ratios, which indicate the direction and relative size of the association with travel, while the right panel reports p-values, which indicate the strength of statistical evidence for each association.

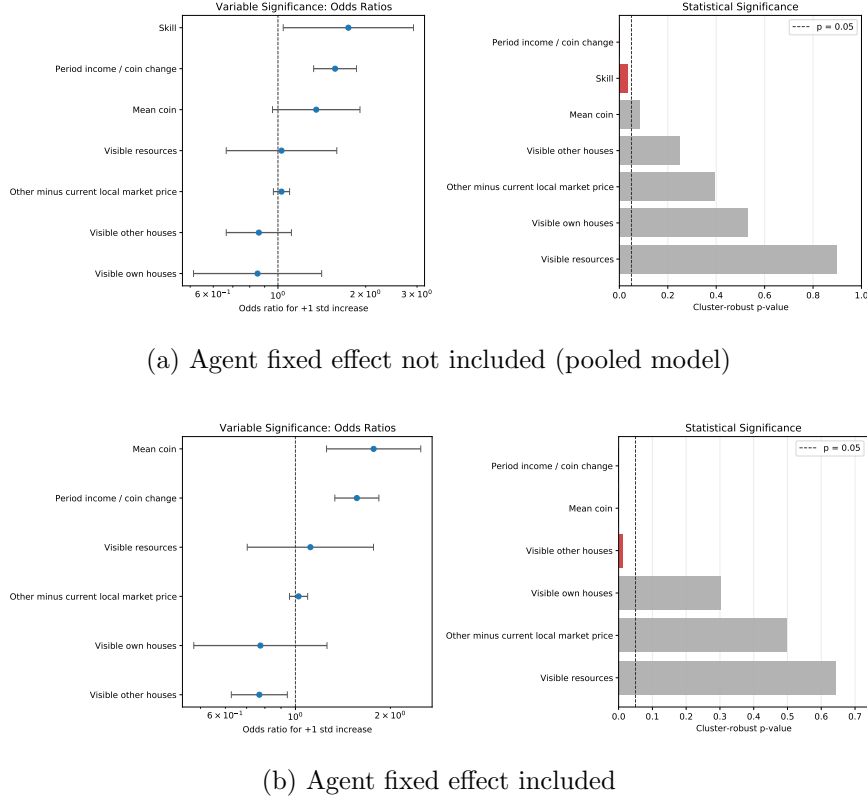


Figure 17: **Statistical evidence from logistic regressions of travel decisions.** Panel (a) reports results from the pooled model using all observations together, while Panel (b) reports results from a regression with agent fixed effects. In each panel, the left plot reports odds ratios, where values above one indicate a positive association with travel and values below one indicate a negative association. The right plot reports cluster-robust p-values, where lower p-values indicate stronger statistical evidence for a conditional association with travel. The results should be interpreted as descriptive conditional associations, not causal effects. At the 5% significance level, both regressions show statistically significant positive associations between travel period income/coin change and in the fix effect model also mean coin. The fixed-effects regressions also show a statistically significant negative association between travel and visible houses belonging to other agents. In the pooled model, skill is also positively associated with travel, but it is not estimated in the fixed-effects model because skill is constant within agents and is absorbed by the agent fixed effects.

The pooled model estimates how each variable is related to travel when all agent-period observations are considered together, while controlling for the other included variables. This model is useful as a descriptive baseline because it captures both differences across agents and changes within agents over time. The fixed-effects specification asks a stricter within-agent question. By giving each agent its own baseline tendency to travel, it controls for

stable differences between agents, such as skill level, starting position, persistent behavioral tendencies, and long-run differences in wealth or building activity.

The pooled and fixed-effects regressions are therefore complementary. The pooled model shows that building skill and period income are positively associated with travel. Higher-skill agents are more likely to travel, and travel is more likely in periods where agents experience larger positive changes in coin holdings. Skill cannot be estimated separately in the fixed-effects model because it is constant for each agent and is absorbed by the agent fixed effects.

Some relationships become clearer in the fixed-effects specification. In the pooled regression, mean coin is not clearly associated with travel. This may be because cross-agent differences muddy the relationship: some agents may generally hold more coin but rarely travel, while others may hold less coin on average but travel more because of their location, skill, or learned role. These stable differences between agents can blur the overall association. The fixed-effects model removes this stable between-agent variation and instead compares each agent to itself over time. In this specification, mean coin is positively associated with travel, suggesting that agents are more likely to relocate in periods where they have greater economic capacity than usual.

A similar logic applies to visible houses belonging to other agents. In the fixed-effects model, visible other houses are negatively associated with travel. This means that, for the same agent, periods with more nearby houses owned.

One possible interpretation is that if other agents have built houses nearby, the area may provide opportunities not fully captured by the agent's own visible resources or coin holdings. In that case, a denser built environment may reduce the incentive to move rather than increase it. These results should still be interpreted as descriptive conditional associations rather than causal effects.

The final plot examines these relationships descriptively by separating agents according to building skill. This is useful because the same environmental variable may affect agents differently depending on their productivity and economic role in the environment. Figure 18 shows binned travel probabilities across several variables, split by skill group.

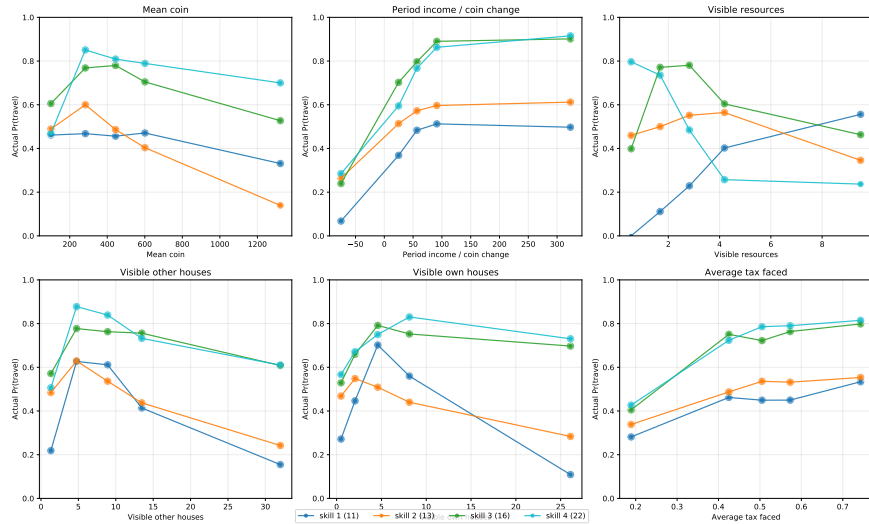


Figure 18: **Travel probability by skill group and observed conditions.** The figure shows binned descriptive relationships between travel probability and selected variables, separated by building skill. The plot is descriptive rather than regression-based, and is used to show how relocation patterns differ across agent types. The points are not evenly spaced because the x-axis shows the average value of each variable within each bin, rather than fixed bin positions. Uneven spacing therefore reflects the underlying distribution of observations.

The clearest pattern in Figure 18 is found for period income. Travel probability increases

sharply as period income moves from low or negative values to moderate positive gains, especially for the higher-skill groups. After period income reaches approximately 100 coin, the relationship appears to flatten, suggesting that additional income is associated with only marginal increases in travel probability once agents have enough coin to absorb the cost of relocation. The final bin should be interpreted cautiously, since the x-axis reports the mean value within each bin. If the highest bin contains a small number of very high-income observations, its mean can be pulled far to the right, creating the appearance of a jump even when the underlying relationship is more gradual.

Visible resources show the least uniform relationship across skill groups. For the highest-skill group, travel probability is highest when few resources are visible and declines as visible resources increase. This is consistent with the interpretation that, for highly productive builders, nearby resources reduce the need to relocate because they provide inputs required for building. For the lowest-skill group, the pattern is closer to the reverse: travel probability increases when more resources are visible. This may reflect that visible resources have a different meaning for low-skill agents. Since these agents rely more on gathering and resource sales than on building income, as shown in Figure 9, a resource-rich local environment may be associated with higher income and therefore a greater ability to afford relocation. Visible resources may therefore reduce relocation incentives for high-skill builders, while increasing relocation capacity for lower-skill gatherers.

Taken together, the figures suggest that relocation is shaped by several interacting forces. Tax incentives appear relevant, but they do not fully determine travel decisions. The descriptive evidence shows that travel periods are associated with higher skill, higher coin holdings, higher average tax exposure, fewer visible resources, and slightly fewer visible houses. This suggests that relocation is related not only to agents' economic capacity, but also to the local environment they face before moving. The regression results refine this interpretation by showing which associations remain after controlling for other observed variables. Period income and skill are positively associated with travel, while visible houses are negatively associated with travel in the conditional regressions. This is noteworthy because visible houses could plausibly reflect congestion, which might be expected to increase relocation incentives. However, the negative association suggests that visible houses may instead proxy for local attachment, accumulated capital, or valuable local economic conditions. Thus, the results do not support a simple interpretation in which agents leave built-up areas merely because they are congested. Rather, conditional on the included controls, visible houses are associated with a lower likelihood of relocation.

5.4.3 Fixed Tax Schedules and Isolated Agent Responses

To separate agent responses from simultaneous planner adaptation, the simulation is also evaluated under fixed tax schedules. In this setting, regional tax differences are held constant, allowing the analysis to focus on how agents respond to a stable fiscal incentive rather than to changing learned tax schedules. This helps distinguish whether relocation is driven primarily by tax policy itself or by other factors such as resource access, market conditions, congestion, and local population composition.

In this experiment, the top region is assigned a tax schedule that is persistently 5 percentage points lower than the bottom region across all brackets and tax periods. The purpose is to examine whether a stable low-tax region becomes more attractive to agents, whether this leads to crowding, and whether relocation differs systematically across agent types. Because the tax schedules do not change over time, any differences in movement, production, or welfare can be interpreted as agent responses to a fixed regional tax gap rather than as responses to planner adaptation.

The fixed-tax experiment provides a test of the tax-response hypothesis, because the top region is always assigned a lower tax schedule. If tax minimization were the dominant driver of relocation, agents should increasingly concentrate in the top region.

To assess whether a stable tax advantage is sufficient to attract agents to one region, Figure 19 summarizes where agents spend their time under the fixed-tax regime. The left

panel reports the average share of total residence time spent in each region, while the right panel shows the average stay length in the top and bottom regions for each agent. Together, these plots indicate both whether the lower-tax region becomes more attractive overall and how persistently agents remain in a given region once they move.

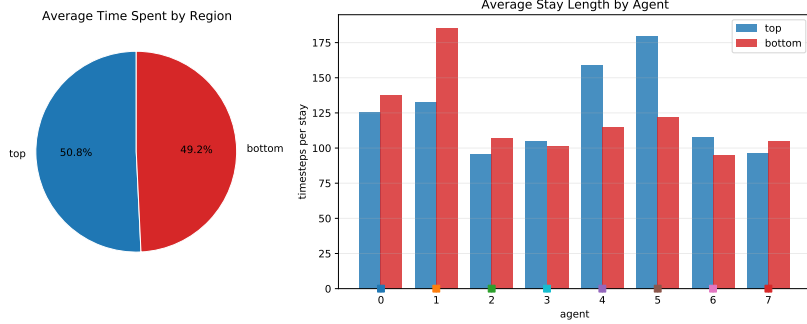


Figure 19: **Regional residence patterns under fixed tax schedules.** The left panel shows the average share of time spent in the top and bottom regions, while the right panel reports the average stay length by agent in each region. The top region, which has the persistently lower tax schedule, attracts only a slightly larger share of total residence time, and average stay lengths remain short, indicating continued frequent relocation despite the stable tax difference.

Figure 19 shows only a very small tendency for agents to spend more time in the top region, which is the region with the persistently lower tax schedule. On average, agents spend 50.8% of their time in the top region and 49.2% in the bottom region. This difference is modest and does not suggest a strong flocking response toward the lower-tax region.

The right panel further shows that agents continue to travel frequently between regions. Average stay lengths are generally short relative to the full episode, and for several of the higher-skill agents they are close to the travel cooldown of 100 timesteps. This indicates that agents often move again soon after they become eligible to travel. In other words, even when taxes are fixed throughout both training and evaluation, and agents can in principle learn that the top region always offers the lower tax burden, they do not simply settle there permanently.

This is an important result. It suggests that a stable tax advantage alone is not enough to dominate relocation behavior. Instead, travel decisions remain shaped by other factors, such as resource access, congestion, local production opportunities, and the spatial distribution of houses and agents. The fixed-tax experiment therefore supports the broader interpretation that relocation is driven by a combination of incentives rather than by taxation in isolation. While lower taxes may make a region somewhat more attractive, they do not produce a strong one-way sorting of agents into the low-tax region. This provides a direct test of the tax-minimization hypothesis: even when the top region is persistently lower-tax, agents do not simply sort into that region. Mobility in the learned economy therefore does not appear to be driven primarily by simple tax minimization.

5.4.4 Controlled Single-Agent Relocation

To evaluate whether the learned policies generalize across regional contexts, I conduct a controlled single-agent relocation experiment. This experiment differs from the endogenous mobility results above because the selected agent does not choose to travel. Instead, one agent is moved once from its current region to the other region during an otherwise standard rollout.

The experiment is designed as a counterfactual shock to the learned economy. The trained Phase 3B policies are restored from the checkpoint, and the economy is allowed to evolve normally until a predetermined timestep t^* . At $t^* = 250$, one selected agent is directly relocated to a valid empty tile in the other region. Outstanding market orders for the moved agent are cancelled before the rollout continues under the same trained policies. No further interventions

are made. Cross-regional trade remains enabled, as in the main mobility experiments. The purpose is to examine whether the moved agent continues to function in the new regional environment, and whether the relocation affects the utilities of other agents or the rewards of the regional planners.

Figure 20 compares two forced-relocation experiments. In Panel (a), agent 0, a lowest-skill agent, is moved from the top region to the bottom region. In Panel (b), agent 3, a highest-skill agent, is moved in the same direction. The upper panels show the utility of the moved agent and the average utility of other agents in the top and bottom regions. The lower panels show the rewards of the two regional planners before and after relocation.

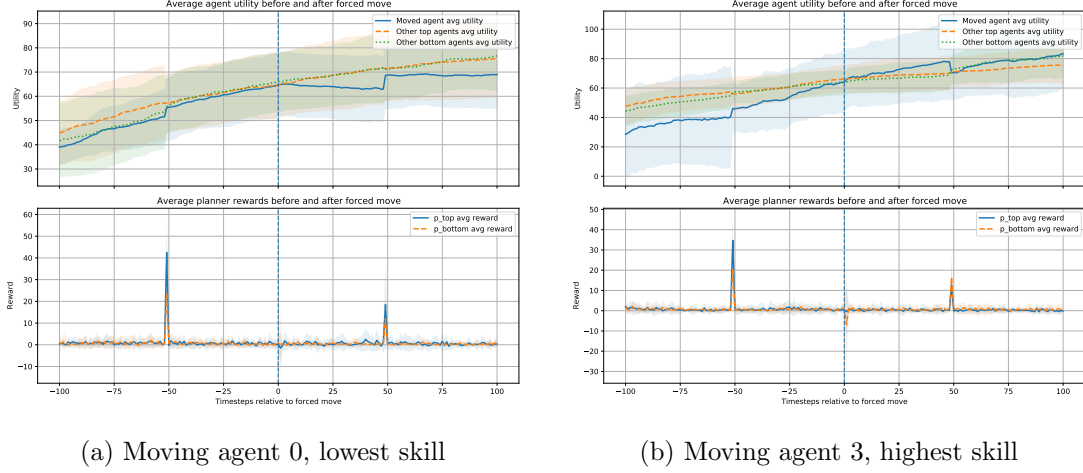


Figure 20: **Controlled single-agent relocation experiment.** Each panel shows utility and planner rewards before and after a forced move at $t^* = 250$, marked by the vertical dashed line. Panel (a) relocates a lowest-skill agent, while Panel (b) relocates a highest-skill agent. The upper plots compare the moved agent’s utility with average utility among agents in each region, and the lower plots show regional planner rewards. The tax-day spikes indicate that redistribution continues after relocation. The highest-skill agent adapts smoothly, while the lowest-skill agent’s utility stagnates after relocation. The trends in the other agents and planner rewards remain broadly stable.

Figure 20 shows that the forced relocation does not break the learned economy. The moved agents continue to accumulate utility after relocation, and the utilities of other agents continue to evolve smoothly. This suggests that the learned policies are able to operate after an unexpected regional change, even though the relocation was imposed rather than chosen by the agent.

The large reward spikes around approximately 50 timesteps before and after the relocation correspond to tax collection and redistribution events. Since the forced move occurs at $t^* = 250$, these spikes occur around the tax days at $t = 199$ and $t = 299$. The same timing is also visible in the utility plots, where redistribution changes the utility paths of both the moved agent and the other agents. This confirms that the tax and redistribution mechanisms continue to operate after relocation.

The response differs across agent types. When the highest-skill agent is moved, the moved agent’s utility continues to increase after relocation, suggesting that the agent can still use its productivity in the new region. This is consistent with the interpretation that high-skill agents can adapt relatively quickly if they already hold sufficient resources or can access the market after moving. In contrast, when the lowest-skill agent is moved, their utility adjustment appears modest, but is partly increased after the following tax redistribution. A plausible explanation is that when agents relocate their current auction submissions get canceled, so in order to get income from selling, the agent has to spend time and labor resubmitting asks to the new regional market. Other agents’ utility does not appear to be impacted by the relocation of

either high or low skill agents. The planner rewards are generally consistent before and after the relocation. Panel (b) shows that there is a decrease in the bottom planners' reward after agent 3 is transferred to the region. This is possibly a consequence of the agent on average having a high coin holding compared to the other agents resulting in higher inequality in combination with potential lower production right after the transfer. Panel (a) shows a small but noticeable increase in the spread of the bottom planners reward after agent 0 is transferred to the region. This might indicate that the agents productivity in the first couple of timestep in the new region is volatile, possibly because of the market state and the same mechanisms as mentioned above.

Overall, the controlled relocation experiment supports the interpretation that the learned economy is robust to isolated regional shocks, but that the effect of relocation depends on the type of agent being moved. High-skill agents appear better able to continue producing after relocation, while low-skill agents may create a short adjustment period in the destination region. The results reinforce that the effect of relocation depends on the agent being moved and on the state of the local economy after relocation, including skill, inventories, market access, spatial position, and tax timing.

5.4.5 Training Dynamics Across Experimental Settings

Because the results are generated by learned policies, the final outcomes must be interpreted in relation to the training process that produced them. The learning curves are therefore important for assessing whether the agents and planners develop meaningful behavior over time, and whether differences between experimental settings emerge during training rather than only in the final evaluation.

This is especially relevant because the model is trained in stages. During some phases, planners' training is frozen, and in the travel-enabled experiments the travel action is introduced only partway through training (the phased learning setup is explained in section 4.1). Each phase therefore changes the learning problem. When a new mechanism is introduced, or when previously fixed policies are unfrozen, behavior learned in earlier phases may no longer be fully adapted to the new environment.

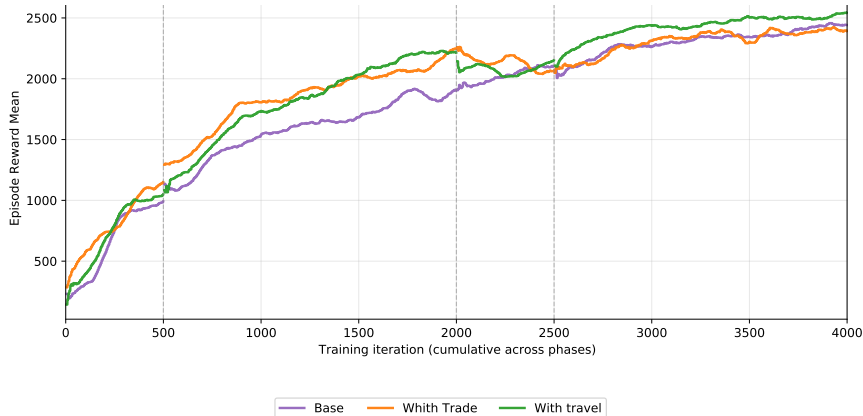


Figure 21: **Mean episode reward during training across the baseline, trade-enabled, and travel-enabled settings.** The vertical dashed lines indicate transitions between training phases. All settings show substantial reward improvement over time. Changes around phase transitions are consistent with shifts in the learning environment, such as planner freezing or unfreezing, tax-annealing, or the introduction of travel.

Figure 21 shows that all three settings improve substantially over training, indicating that the agents and planners learn increasingly effective behavior over time.

The travel-enabled setting initially follows a trajectory close to the baseline, while the trade-enabled setting tends to reach higher rewards earlier. After the later training phases, the

travel-enabled setting eventually achieve higher reward. This suggests that mobility introduces a more complex learning problem, but also one that can eventually support higher-performing behavior once agents and planners adapt. Travel expands the action space and changes the strategic environment, since agents must learn not only how to gather, trade, and build, but also when it is beneficial to change regions. Planners must also adapt to a population whose regional composition may change over time.

The phase transitions produce visible changes in reward, especially around the vertical dashed lines. These shifts should not be interpreted as ordinary smooth learning dynamics alone, since the training settings change at these points. Instead, they are consistent with changes in the optimization problem faced by the policies, such as planner freezing, planner unfreezing, changes in tax annealing, or the introduction of travel.

At the same time, the figure should not be interpreted as evidence of full convergence. The reward curves, especially in the travel-enabled setting, continue to show some movement near the end of training. This suggests that additional training could further refine both agent behavior and planner policies (this is further discussed in section 7.3).

Taken together, the training curves strengthen the interpretation of the results. They show systematic improvement over time, adaptation after phase transitions, and meaningful differences between the baseline, trade-enabled, and travel-enabled settings. For the research question, this supports the claim that inter-regional interaction changes the learned behavior of the system, while also motivating caution because additional training may still affect the final policy outcomes.

5.5 Summary of Results

Overall, the results show that inter-regional interaction affects both learned agent behavior and learned planner policy, but not through a single simple mechanism. No individual figure explains the full pattern in isolation; the interpretation comes from understanding the baseline, trade, mobility, fixed-tax, controlled relocation, and training results together.

The closed-economy baseline shows that unequal outcomes and specialization patterns already emerge before inter-regional interaction is introduced. Agents share the same policy network, but differ in skill, starting location, resource access, and market position. Higher-skill agents rely more on building and market purchases, while lower-skill agents rely more on gathering and resource sales. This establishes that agent behavior is shaped by economic context.

Market integration changes the environment by allowing resources to move across regions. Cross-regional trade becomes an active part of the economy, even with the import tax, suggesting that agents use the wider market when local resource conditions are limiting. The main effect of trade is therefore expanded market access rather than import-tax redistribution.

Labor mobility adds a relocation decision to the learned economy. Travel periods are associated with higher skill, higher period income, higher mean coin holdings, and higher average tax exposure. However, lower destination-region taxation is not a necessary condition for relocation. The regression results show that period income and skill are positively associated with travel, while visible houses owned by other agents are negatively associated with travel after controlling for other variables. This suggests that relocation depends not only on tax exposure, but also on economic capacity and the local environment agents face before moving. The fixed-tax experiment strengthens this conclusion: even when one region is persistently lower-tax, agents do not simply concentrate there. Mobility is therefore best understood as a context-dependent decision shaped by taxation, skill, income, resources, market conditions, built capital, and travel costs.

Planner tax rates are modestly lower when travel is enabled, suggesting that mobility changes planner behavior in a way that is consistent with limited tax competition. However, the decrease is uneven across brackets and planners, so the result should be interpreted as a descriptive pattern in learned tax schedules rather than causal evidence that planners simply lower taxes when agents can relocate.

Taken together, the results answer the research question by showing that inter-regional interaction links taxation to the wider economic environment. Agents respond to taxes together with resources, markets, income opportunities, and built capital, while planners adapt to the regional welfare outcomes generated by those responses. Market integration and labor mobility therefore create a co-adaptive system, rather than a single tax-driven mechanism.

6 Related Work

This section presents existing work on reinforcement learning in economic environments, computational macroeconomic modeling, and learned policy design. Its purpose is to show what academic progress has already been made in related work as well as highlight some of its current limitations.

6.1 The AI Economist Framework

The most direct point of reference is the *AI Economist* framework developed by Zheng et al. [11] and extended by the same and other authors a year later [12]. Their work introduces a reinforcement learning based framework for studying economic policy in simulated environments. At its core, the model represents an economy in which heterogeneous agents and a centralized social planner are trained jointly.

A key feature of the framework is that economic outcomes emerge from interactions among learning agents rather than being specified in advance. This makes the work by Zheng et al. particularly relevant for studying taxation and redistribution in settings where behavioral responses are difficult to model explicitly. The framework therefore provides the methodological foundation for this thesis.

An important aspect of these results is their robustness to strategic behavior. Agents in the model learn to respond to taxation, including through forms of tax avoidance, yet the learned policies remain effective. This highlights the advantage of simulation-based approaches in capturing behavioral adaptation that is difficult to incorporate in traditional models.

The approach is further validated through experiments with human participants, where AI-designed tax policies results in welfare outcomes comparable to standard frameworks. Together, these results establish the AI Economist as a useful baseline for studying adaptive taxation in simulated economies.

The thesis extends the closed economic baseline by introducing a two-region structure with separate social planners and cross-regional interaction. In relation to the original AI Economist, the contribution is therefore not a new learning algorithm, but an extension of the economic environment in which learned taxation is studied.

6.2 Other Reinforcement Learning and Computational Economic Models

Beyond the AI Economist, Leibo et al. [30] apply multi-agent reinforcement learning to economic and social dilemmas, studying how institutional settings affect collective outcomes in environments such as commons management and public goods provision. Their work provides valuable insight into strategic interaction and incentive design, but it does not incorporate the same macroeconomic structure, learned taxation policy, or endogenous redistribution studied in the AI Economist framework.

Deep reinforcement learning has also been applied to standard macroeconomic models as a numerical solution method. Atashbar and Shi [7] study a real business cycle framework in which households and firms learn policy functions through interaction with the environment. Using deep deterministic policy gradient methods, they show that deep reinforcement learning can approximate equilibrium behavior in high-dimensional continuous state spaces, including settings where households choose labor supply and firms determine production. In

this approach, however, government policy is treated as fixed, and the focus is on equilibrium approximation rather than on adaptive policy design.

Chen et al. [8] study a dynamic stochastic general equilibrium (DSGE) model in which agents are using neural networks learning intertemporal decision rules through interaction with the environment. In this framework, agents do not have full knowledge of the economic structure, and learning serves as a form of bounded rationality. Applied to monetary policy models, the framework shows that learning dynamics can converge to multiple equilibria, even in cases where standard local stability conditions would suggest otherwise. At the same time, convergence may be slow or unstable without additional restrictions.

Related work by Chen et al. [9] applies deep reinforcement learning to a monetary model with fiscal-policy interaction, showing that a neural-network-based household agent can learn the model across a wide range of policy regimes. Compared to traditional adaptive learning approaches, this method provides a more flexible way to model bounded rationality, but the focus remains on learning or recovering equilibrium behavior rather than on studying learned fiscal policy in an adaptive multi-agent environment with regional policy-making.

Beyond reinforcement learning, deep learning can be used to solve dynamic economic models by approximating decision rules with neural networks. Maliar, Maliar, and Winant [6] implement this by reformulating Bellman and Euler equations as regression problems. Their approach is especially useful for high-dimensional heterogeneous-agent models, including the type of macroeconomic heterogeneity studied by Krusell and Smith [1]. However, these methods are primarily designed for equilibrium computation, not for studying interaction between adaptive agents and policymakers.

A recent survey by Desai [31] documents the rapid expansion of machine learning in economic research, particularly in macroeconomic modeling, forecasting, and policy analysis. At the same time, it emphasizes ongoing challenges related to interpretability, stability, and the economic validity of learned policies. These challenges are also relevant here, where learned policies must be interpreted in relation to the training process that produced them.

6.3 Multi-Economy Interaction and Policy Spillovers

The economic mechanisms studied in this thesis are also related to work on international tax competition and mobility. This literature examines how regional policy-making can affect redistribution when agents or tax bases are mobile across jurisdictions. For example, Razin and Sadka [17] revisit the race-to-the-bottom hypothesis and analyze how migration can constrain redistributive taxation by giving agents an outside option. Similarly, Kleven et al. [18] review empirical evidence showing that mobility responses to taxation can be substantial for some groups, particularly high-income individuals and workers whose human capital is less tied to a specific location.

This literature is relevant because the present thesis also studies the interaction between mobility, redistribution, and decentralized policy-making. However, the approach differs in important ways. Rather than estimating mobility responses empirically or deriving tax competition results analytically, this thesis studies how related mechanisms appear in a learned multi-agent simulation environment where both agents and planners adapt through reinforcement learning.

Only a limited subset of the reinforcement learning literature considers settings with multiple interacting economies. Madhulatha and Ghorri [32], for example, apply deep learning and reinforcement learning methods to exchange-rate forecasting using time-series data and international economic variables. While this work incorporates cross-country economic information, its focus is prediction rather than adaptive taxation, redistribution, or welfare analysis.

Recent work by Tomašev et al. [33] on virtual agent economies is also relevant as a broader point of comparison. Their work highlights how populations of AI agents can form complex economic systems, creating both opportunities for coordination and risks related to instability, inequality, and emergent strategic behavior. Although their focus differs from the tax-policy environment studied here, the paper situates this thesis within a broader research agenda on AI-driven economic interaction.

Overall, this thesis contributes to the mentioned literature by combining elements that are usually studied separately: learned taxation, separate regional planners, agent mobility, and multi-region interaction. The goal is not to replace analytical or empirical models of tax competition, but to examine how similar mechanisms appear or fail to appear when policy and behavior are both produced by reinforcement learning.

7 Discussion

7.1 Interpreting Relocation in the Economy

The results suggest that mobility in the model should not be interpreted as a simple response to tax differences alone. Although agents show some tendency to travel in periods where they face higher average marginal tax rates, the comparison between current-region and alternative-region tax burdens does not show a clear one-to-one relationship. Similarly, the fixed-tax experiment does not show that agents always move toward the region with lower tax rates. The combined results instead suggest that relocation decisions are shaped by a broader set of conditions, including local resources, visible houses, wealth, period income, skill level, market access, and the agent’s previous experience in the environment. This relates to the migration and tax-competition literature by showing that mobility does not necessarily create a simple tax-minimization response. Kleven et al. [18] emphasize that high-income individuals can respond to differences in taxation across locations. The results here are consistent with the idea that taxation can matter for relocation, but they also show that tax exposure is only one part of a broader learned decision problem involving income, skill, market access, resources, and spatial conditions.

This finding is important for the research question because it shows that mobility makes regional location part of the agents’ economic decision problem.

The planner results are consistent with this interpretation. In the travel-enabled setting, average planner tax rates are slightly lower compared to the other settings. This is compatible with a tax-competition interpretation, since mobile agents may make high tax rates more costly for planners if productive agents can leave. However, the evidence should not be interpreted as proof that planners intentionally lower taxes to retain agents. The learned tax schedules also reflect production, redistribution, regional composition, stochastic learning dynamics, and the combined regional welfare objective, which are all influenced by the agents ability to relocate. The results therefore do not confirm a simple tax-competition mechanism, but they also do not rule it out. Instead, they suggest that mobility may create tax-competition-like pressures within a broader learned feedback system.

From a scientific perspective, the results illustrate the usefulness of reinforcement-learning-based economic simulations for studying mechanisms that are difficult to isolate analytically. Agents and planners in this environment respond through several connected decisions at once simultaneously. This does not mean that the model captures the full complexity of real economies, but it provides a controlled setting in which interactions between taxation, market access, and endogenous behavior can be studied jointly.

In real economies, relocation decisions are rarely driven by tax rates alone. They also depend on income opportunities, housing, local networks, transport costs, market access, and institutional constraints. This suggests that policy analysis of mobile agents should consider how taxation interacts with local production opportunities and market structure, rather than treating tax rates as an isolated driver of movement.

Overall, the results support a nuanced interpretation of the tax-competition literature: mobility introduces potential pressure on regional tax policy, but the learned outcome depends on how this pressure interacts with redistribution, production, market access, spatial constraints, and agent heterogeneity. For this thesis, the main implication is therefore that mobility creates a multi-dimensional response channel. Agents relocate based on the broader expected benefit of another region, including but not limited to taxation, while planners adapt to the regional welfare outcomes produced by those agents’ decisions.

7.2 Planner Behavior Under Mobility

A central concern in multi-region taxation is that labor mobility may generate tax competition. If agents can relocate in response to tax differences, planners may have an incentive to reduce taxes in order to retain or attract productive agents. In the strongest case, this can produce a race-to-the-bottom dynamic, where each social planner repeatedly lowers tax rates and redistribution becomes difficult to sustain. The existing literature does not give a single prediction for how mobility affects taxation. While the race-to-the-bottom argument suggests that mobility can push tax rates downward, Razin and Sadka [17] show that migration can also generate fiscal externalities that make tax competition produce higher tax rates than coordination. The relevant point for this thesis is therefore not that mobility must reduce taxes, but that mobility seems to change the tax-policy incentives faced by regional planners.

The results do not show a clear race-to-the-bottom dynamic. Although the average tax is slightly lower, the tax policy remains heterogeneous across brackets, regions, and evaluation rollouts. This suggests that mobility does not mechanically eliminate redistribution in the learned economy.

The absence of a clear race-to-the-bottom dynamic should be interpreted cautiously. The model contains only two regions, a limited number of agents, fixed travel costs, and a welfare objective in which equality directly affects planner rewards. The result should therefore not be read as a general theoretical claim about tax competition. Rather, it shows that within this environment, labor mobility does not automatically lead regional planners to abandon redistribution. In relation to the tax-competition literature, the result therefore adds nuance to existing interpretations of migration and taxation dynamics: mobility creates pressure that is consistent with tax competition, but this pressure is not strong enough in the present experiments to produce uniform tax reductions or the collapse of redistribution.

7.3 Training limitations and convergence

A limitation of the experiments is the restricted training amount. The models are trained for approximately 3.2 million environment steps (see Section 4.2), which is substantially shorter than in the original AI Economist experiments [11, 12]. The results should therefore not be interpreted as directly comparable to the performance levels reported in those papers, or as fully converged equilibrium policies.

However, direct comparison is not the purpose of this thesis. The AI Economist framework has already been developed and evaluated in prior work [11, 12]. The aim here is to use that framework as a basis for studying how the introduction of cross-regional trade, labor mobility, and regional planners changes the learned behavior of agents and planners.

The shorter training period still affects how the results should be interpreted. With fewer training iterations, agents and planners may have less time to explore the environment, refine their value estimates, and adjust to the strategies learned by other policies. This is particularly important in a multi-agent setting, where each policy is trained in an environment that changes as other agents and planners learn. As a result, learned behavior may partly reflect path dependence and early stochastic discoveries rather than full convergence to a stable policy configuration.

This limitation is especially relevant for planner learning. Planners act only on specified taxation-timesteps, meaning that they take actions far less frequently than mobile agents. Although planners observe the same episodes as the agents, the number of planner decision points is much smaller. This makes planner learning slower and more difficult to evaluate. Moreover, planner performance depends strongly on agent behavior: the social welfare function reflects both production and equality. A low or unstable planner reward can therefore reflect planner policy, agent adaptation, or the interaction between the two.

For agents, the shorter training horizon may also explain why behavior in the baseline setting does not always appear as strongly utility-optimized as in the original AI Economist experiments. Nevertheless, the agents do learn the main mechanisms including the once introduced in this thesis. In particular, they learn to use cross-regional travel, even though

travel is introduced only in the second half of the training phases. This is important because labor mobility is central to the research question. Experiments with shorter training periods showed weaker use of travel under the same configuration, suggesting that the mobility behavior observed here is learned rather than purely random.

A related limitation is the small number of independent training seeds. Each experimental setting is trained using three independent seeds and evaluated using twenty rollout trajectories. Formal multiple-comparison tests, such as ANOVA or non-parametric alternatives followed by Dunn’s pairwise tests, would therefore risk overstating statistical significance with the current data. With a sufficient number of independent training seeds, such tests could be used to assess whether differences between experimental settings are statistically distinguishable from training variability. For example, they could test whether average tax rates are systematically lower in the mobility setting than in the trade-only setting, whether travel-enabled economies produce significantly different levels of production or equality, or whether cross-regional trade changes welfare outcomes relative to the baseline. This would make it possible to distinguish more clearly between patterns that are robust across training runs and patterns that may reflect stochastic variation in learned policies. This is a common limitation in deep reinforcement learning experiments, where training is computationally expensive and outcomes are stochastic due to initialization, exploration, sampled trajectories, and stochastic-gradient optimization [34, 35].

Overall, the limited number of training iterations and independent seeds does not remove the relevance of the results, but it does place limits on their interpretation. The results are most meaningful when interpreted as descriptive patterns across trained policies and as comparisons between experimental settings, rather than as fully converged or statistically definitive outcomes. Additional training and a larger number of independent seeds could lead to more stable policies, stronger specialization, more reliable statistical comparisons, or different planner responses. These remain important directions for future work. Different approaches that might help address these challenges are discussed in Section 8.2.

7.4 Simultaneous Learning, Travel, and Implementation Sensitivity

One of the central challenges in this thesis was training and interpreting a system in which agents and planners learn simultaneously while the environment itself changes across experimental phases. This challenge becomes especially pronounced when labor mobility is enabled. Travel changes the decision problem faced by individual agents, because relocation places them in a different regional environment with different tax exposure and neighboring agents.

Mobility also changes the learning problem faced by planners. Since planner rewards are computed from the welfare of agents currently located in their region, relocation changes the population over which the planner objective is evaluated. A planner’s reward may therefore change not only because regional production changes, but also because mobility changes the number of agents included in the regional welfare calculation and the population across which redistribution is shared. This makes the two-level learning problem more difficult to interpret: agent behavior affects planner rewards, planner policies affect agent incentives, and travel changes the link between the two.

The implementation process also revealed that the learned dynamics are sensitive to relatively small changes in environment mechanics. During the thesis work, many simulations were run while correcting or refining details of the environment, such as updating the continuous double-auction mechanism so that import taxes were correctly incorporated when matching bids and asks. Such changes were necessary for the model to represent the intended economic mechanisms, but they also affected learned behavior. This is expected in a multi-agent reinforcement learning environment, especially using PPO, where small modifications to action feasibility, market clearing, tax timing, or relocation rules can alter the incentives faced by agents and planners [36].

Because agents and planners co-adapt, small changes in the mechanics can lead to different training trajectories and final policies. This sensitivity is an important limitation of the results. Changes to the specific parameter values chosen in the experiments are therefore expected to

have a significant impact on the outcomes. This sensitivity means that some patterns may depend on the particular implementation choices made in the environment, including action masks, market clearing, tax timing, and relocation rules.

7.4.1 Sensitivity of PPO and Implementation Choices

A further limitation concerns the sensitivity of PPO to implementation choices. PPO is used in this thesis firstly because this is what the original learning algorithm was built on [11, 12] and also because it provides a practical policy-gradient method for training agents and planners in high-dimensional, partially observed environments [26]. However, the policies learned with PPO depend not only on the PPO loss function, but also on code-level implementation details and how the multi-agent environment is implemented [36].

This is directly relevant for the present thesis. Action masking determines action availability at different timesteps. Small changes to these mechanisms can alter the learning signals received by both agents and planners, especially because the policies co-adapt over time.

The observed tax schedules, trade patterns, and relocation behavior may therefore depend on the PPO configuration and implementation details, making robustness checks across hyperparameters and training seeds important future work. This does not make the results uninformative, but it limits their scope. Future work could therefore test whether these patterns remain stable across alternative implementation choices, PPO hyperparameters, and independent training seeds.

7.5 Using AI and Deep Learning in Economic Modeling

The use of artificial intelligence and deep learning in economics raises questions beyond computational efficiency. Reinforcement learning is increasingly used as an alternative approach for studying dynamic economic problems, especially when traditional dynamic programming becomes difficult in high-dimensional state and action spaces [10]. This is relevant in macroeconomic and policy settings because many economic decisions are sequential: agents act today while learning from outcomes that unfold over time. Reinforcement learning is therefore useful for studying adaptive behavior, where agents learn policies through repeated interaction with an environment rather than solving a fully known model in advance [7, 9, 10].

This perspective is central to interpreting the results of this thesis. The learned tax policies should not be understood as closed-form optimal tax rules, but as outcomes of a co-adaptive learning process. Planner policies affect agent incentives, while agent production, movement, and accumulation affect the outcomes that planners observe and respond to. This two-level interaction is also central to the AI Economist framework, where agents and planners learn simultaneously in a simulated economy [11, 12]. In this sense, deep reinforcement learning is not only a method for solving the model, but also part of the economic mechanism being studied.

This also connects to broader work on virtual agent economies. Recent work by Tomašev et al. argues that autonomous AI-agent systems may form a new economic layer in which agents interact through sandbox economies, with potential benefits for coordination but also risks related to instability and amplified inequality [33]. The simplified multi-agent environment studied here can be interpreted as a small-scale laboratory for examining such interaction effects, especially how policy, mobility, production, and learned behavior interact endogenously.

The contribution of using deep reinforcement learning in this thesis is therefore not that it provides a final answer to optimal tax design. Rather, it allows taxation to be studied as an adaptive process in a dynamic multi-region economy. At the same time, learned policies depend on training time, environment design, reward specification, observation structure, and implementation choices. The results should therefore be interpreted as evidence of possible mechanisms rather than as general policy prescriptions [36].

7.6 What the Results Cannot Establish

The purpose of this thesis is not to identify an optimal tax system for an international or European-style economy. The model is too stylized for that interpretation, and the experiments do not provide enough training time or seed variation to support definitive claims about equilibrium convergence or globally optimal policies. Instead, the thesis should be understood as a mechanism-oriented study: it uses the AI Economist framework as a foundation for examining how learned agents and planners respond when inter-regional trade, labor mobility, and separate regional tax schedules are introduced.

This distinction is important. The original AI Economist work provides evidence that deep reinforcement learning can be used to learn tax schedules that perform well in simulated economies [11, 12]. This thesis does not attempt to revalidate that result or directly compare performance against the original framework. Rather, it builds on the framework to study a different question: how the interaction between agents and planners changes when agents are no longer confined to a single closed economy.

What the results can support is a more limited but still meaningful claim. They show how agents and planners respond within this simulated environment when market integration and labor mobility are introduced. In particular, the model makes it possible to examine how market integration changes trade and resource allocation, how mobility shapes relocation decisions and regional composition, and how separate planners adapt when the population entering their welfare objective can change over time. The contribution is therefore not a final policy prescription, but an analysis of learned behavioral mechanisms in a multi-region extension of the AI Economist framework.

8 Future Work

8.1 Implementation expansions

Future work can build directly on this thesis by adding more complexity to the model and making the environment closer to real-world economic settings. The most immediate extensions includes spatial structure, regional heterogeneity, economic complexity, and robustness across alternative parameter choices.

One extension would be to vary the map layout. Since spatial position and resource accessibility shape agent behavior, future work could test whether similar patterns of specialization emerge under a uniform resource distribution, where all agents have equal access to wood and stone. This would help separate behavior driven by learned economic incentives from behavior driven by local resource scarcity. One would expect such a change to increase inequality. In the current setup, low skill agents have greater access to nearby resources, while high skill agents initially face less favorable resource access. A uniform resource distribution would reduce this spatial advantage cliff. If this increases the income gap between agents, it would also affect planner rewards. As a result, changing the resource distribution could lead to different planner responses and tax schedules.

A related extension would be to introduce structural inequality across regions. In the current implementation, the two regions are designed to be comparable in aggregate resource availability. An alternative experiment would give one region systematically higher resource availability than the other. This would allow the model to study whether trade, relocation, and redistribution reduce or reinforce initial regional inequality. It would also make it possible to examine whether agents migrate toward high-resource environments, and how regional planners respond when their tax base and welfare outcomes are shaped by unequal initial conditions.

A further step would be to introduce systematic differences in agent skill across regions. Future work could initialize one region with a higher concentration of high skill agents. This would be particularly relevant for examining whether relocation leads to persistent high-skill and low-skill regions, or whether congestion, redistribution, and market conditions prevent such sorting from becoming stable.

The economic environment could also be expanded by introducing more complex production possibilities. Future versions of the model could allow agents to build different types of structures, tools, or intermediate goods. For example, some agents might be able to construct higher-paying houses from different resource combinations, or produce improved building materials that can be traded with other agents. Such extensions could generate richer forms of specialization, including specialization not only in gathering and building, but also in producing intermediate inputs for other agents.

Increasing the number of regions would also be a valuable extension. With only two regions, agents face a simple relocation choice. A model with more regions would allow agents to choose among several possible destinations with different tax policies, resource conditions, market opportunities, and congestion levels. This could create regional specialization rather than only agent-level specialization. Some regions might become primarily resource-gathering regions, while others might become big building grounds or even trading hubs trying to beat the market. Such a setting could also generate stronger congestion effects and provide a closer representation of the structure of real-world economies.

Finally, future work could perform a more systematic sensitivity analysis of the model parameters. Many mechanisms in the environment can plausibly affect the learned outcomes, including the number of agents, region sizes, building skill distribution, tariff size, travel cost, travel cooldown, and learning hyperparameters. The legacy code also contains a gathering-skill mechanism, which could be reintroduced to study whether differences in resource collection ability generate stronger specialization or sorting. Varying these parameters would help distinguish mechanisms that are robust features of the environment from outcomes that depend on a particular calibration.

8.2 Computational Improvements

A possible way to address some of the computational limitations faced in this thesis is to use GPU-accelerated multi-agent reinforcement learning frameworks. The implementation used in this thesis follows the original AI Economist/RLlib-style training setup, which supports distributed rollout collection but still makes each experiment computationally expensive. Long episodes, multiple agents, spatial observations, market mechanisms, and simultaneous planner learning all increase the cost of training.

Newer JAX-based MARL libraries, such as JaxMARL, are designed exactly around vectorized environments and hardware acceleration, allowing many environments or training runs to be executed in parallel on GPUs [37]. JaxMARL reports substantial wall-clock speedups in its benchmark environments, including faster single-run training and much larger gains when multiple runs are vectorized in parallel [38].

For an environment such as the one studied here, similar infrastructure could make it possible to increase the number of agents, enlarge the spatial world, run longer training schedules, and evaluate policies across more random seeds. However, these gains would likely require re-implementing the environment and training pipeline in a vectorized JAX-compatible form, rather than simply replacing the current RLLib trainer. Such developments would allow future work to test whether the patterns observed in this thesis remain under richer population dynamics and larger regional economies. At the same time, improved parallelization would not remove the conceptual challenges of simultaneous agent-planner learning, but it would make it easier to distinguish computational limitations from structural properties of the learned economy.

A lower-workload way to address some computational limitations would be to update the Python, Ray, and RLLib versions and migrate from the older `PPOTrainer` API to the newer RLLib PPO interface. This would not guarantee faster training, since performance may still be limited by environment stepping, memory use, or model execution. However, it could improve resource utilization in the present setup because the simulations are run on SLURM-based high-performance clusters. The implementation used in this thesis relies on an older Ray/RLLib setup, based on the legacy `PPOTrainer` API. This created practical difficulties when running the simulations on the SLURM-based high-performance cluster. Ray is designed

around a head-worker architecture, where a head node coordinates multiple worker nodes. SLURM, by contrast, allocates cluster resources through scheduled jobs [39]. If Ray is not configured explicitly within the SLURM allocation, the training script may fail to use the allocated nodes efficiently. A newer Ray/RLlib setup, combined with a proper SLURM launch configuration, could therefore improve parallel rollout collection and reduce unused cluster capacity. This would be a more incremental improvement than rewriting the environment in a JAX-based framework, while still addressing an important practical limitation of the current implementation.

Taken together, a JAX-based framework could substantially improve simulation speed in future work, but only if the environment were redesigned as a vectorized, JAX-compatible simulation. A less labor-intensive improvement would be to update the current Python, Ray, and RLlib stack and migrate from the older `PPOTrainer` API to the newer RLlib PPO interface.

8.3 Methodological Extensions

The current implementation trains agents and planners with multi-agent PPO, where the mobile-agent policy and the regional planner policies are updated across the same training process. This creates a non-stationary learning problem: as agents adapt to tax policies and local economic conditions, the planners simultaneously adapt to changing agent behavior. As a result, observed outcomes reflect both the economic incentives within the environment and the dynamics of the learning procedure itself. Methodological extensions, such as additional ablation studies, fixed-policy evaluations, or alternative multi-agent training methods, could help distinguish which patterns are robust features of the environment and which are sensitive to the training setup.

One natural extension would be to use centralized-training methods with decentralized execution. In the current setup, agents act from their own observations, while planners act from regional observations. This is appropriate for the economic interpretation of the model, since agents should not have access to the full global state when making decisions. However, during training, a centralized critic could be given access to more complete information about the joint state of the economy, while the learned actors would still execute using only their local observations. This type of centralized critic is common in multi-agent reinforcement learning because it can reduce the apparent non-stationarity faced by each learner and provide a more informative value estimate during training [40, 41].

9 Conclusion

This thesis asked how inter-regional interaction changes learned behavior in a multi-region economy with deep reinforcement learning agents, created from an existing single-region framework known as the AI Economist. The answer is that market integration and labor mobility do not simply add more activity to the environment. They change the decision problem faced by both agents and planners. Once agents can trade and relocate across regions, taxation becomes part of a larger adaptive system in which production, market access, spatial opportunity, accumulated resources, and regional policy interact.

Opening the market changes the conditions under which agents trade and produce by allowing them to access non-local resources without moving. In the trade-integrated setting, cross-regional trade becomes an active part of the market and imported resources are traded at an average higher buyer price because of the import tax. However, the introduction of cross-regional trade does not appear to produce large additional changes in aggregate welfare, suggesting that its main effect is on market access rather than on overall economic welfare.

On the agent side, the strongest result is that mobility is not equivalent to tax minimization. Agents do respond to tax exposure, but relocation is more strongly understood as a broader economic adjustment. Travel is associated with higher skill, larger period income, and local spatial conditions. The fixed-tax experiment makes this especially clear: even when one region is always assigned the lower tax schedule, agents do not simply concentrate there. This shows

that agents evaluate relocation as a change in their full economic environment, not only as a change in tax rate.

On the planner side, the results show that regional tax policy becomes more difficult to interpret under mobility, because planners no longer govern a fixed set of agents. As agents move, both the regional tax base and the welfare objective change. The learned tax schedules are slightly lower on average when travel is enabled, but they do not collapse into a uniform race-to-the-bottom pattern. Nor are they explained by a simple mechanical relationship with the number of agents in each bracket. Planner behavior is therefore best understood as a learned response to the combined regional welfare signal, where equality and production are jointly affected by agent behavior.

The main contribution of the thesis is therefore to show that the AI Economist framework can be extended from a closed economy into a multi-region setting where trade, mobility, and regional taxation interact endogenously. This setting adds a new layer of complexity to the feedback loop: agents can respond to planner actions, market conditions, and other agents by relocating between regional economies, not only by adjusting production and trade.

The thesis provides a controlled environment for studying how learned agents and planners adapt when economic behavior is no longer confined to a single regional tax and market environment. The main implication is that labor mobility in such a setting should be understood as a multidimensional response channel. Taxation matters, but only as one part of a broader learned decision problem involving economic capacity, individual abilities, spatial opportunity, and planner adaptation.

References

- [1] Per Krusell and Anthony A. Smith. “Income and Wealth Heterogeneity in the Macroeconomy”. In: *Journal of Political Economy* 106.5 (1998), pp. 867–896. DOI: 10.1086/250034. URL: <http://www.jstor.org/stable/10.1086/250034>.
- [2] Anthony B. Atkinson, Thomas Piketty, and Emmanuel Saez. “Top Incomes in the Long Run of History”. In: *Journal of Economic Literature* 49.1 (2011), pp. 3–71. DOI: 10.1257/jel.49.1.3.
- [3] Peter A. Diamond and James A. Mirrlees. “Optimal Taxation and Public Production I: Production Efficiency”. In: *American Economic Review* 61.1 (1971), pp. 8–27.
- [4] Emmanuel Saez. “Using Elasticities to Derive Optimal Income Tax Rates”. In: *Review of Economic Studies* 68.1 (2001), pp. 205–229. DOI: 10.1111/1467-937X.00166. URL: <https://eml.berkeley.edu/~saez/derive.pdf>.
- [5] Thomas Piketty and Emmanuel Saez. “A Theory of Optimal Inheritance Taxation”. In: *Econometrica* 81.5 (2013), pp. 1851–1886. DOI: 10.3982/ECTA10712. URL: <https://eml.berkeley.edu/~saez/piketty-saezECMA13.pdf>.
- [6] Lilia Maliar, Serguei Maliar, and Pablo Winant. “Deep Learning for Solving Dynamic Economic Models”. In: *Journal of Monetary Economics* 122 (2021), pp. 76–101. DOI: 10.1016/j.jmoneco.2021.07.004. URL: <https://www.sciencedirect.com/science/article/pii/S0304393221000799>.
- [7] Tohid Atashbar and Rui Aruhan Shi. *Deep Reinforcement Learning in a Real Business Cycle Model*. Tech. rep. WP/23/40. International Monetary Fund, 2023. URL: <https://www.imf.org/-/media/files/publications/wp/2023/english/wpiea2023040-print-pdf.pdf>.
- [8] Mingli Chen et al. *Deep Reinforcement Learning in a Monetary Model*. Staff Working Paper 1142. Bank of England, 2025. URL: <https://www.bankofengland.co.uk/-/media/boe/files/working-paper/2025/deep-reinforcement-learning-in-a-monetary-model.pdf>.
- [9] Mingli Chen et al. *Deep Reinforcement Learning in a Monetary Model*. Version 2, last revised 5 January 2023. 2021. DOI: 10.48550/arXiv.2104.09368. arXiv: 2104.09368 [econ.EM]. URL: <https://arxiv.org/abs/2104.09368>.
- [10] Thomas R. Cook and Nathan M. Palmer. “Reinforcement Learning in Macroeconomics”. In: *Oxford Research Encyclopedia of Economics and Finance*. Ed. by Anindya Banerjee. Oxford University Press, 2025. DOI: 10.1093/acrefore/9780190625979.013.945. eprint: https://academic.oup.com/book/0/chapter/546484272/chapter-ag-pdf/67637974/book_61801_section_546484272.ag.pdf. URL: <https://doi.org/10.1093/acrefore/9780190625979.013.945>.
- [11] Stephan Zheng et al. “The AI Economist: Improving Equality and Productivity with AI-Driven Tax Policies”. In: *arXiv preprint* (2020). arXiv:2004.13332. URL: <https://arxiv.org/abs/2004.13332>.
- [12] Stephan Zheng et al. “The AI Economist: Optimal Economic Policy Design via Two-Level Deep Reinforcement Learning”. In: *arXiv preprint* (2021). arXiv:2108.02755. URL: <https://arxiv.org/abs/2108.02755>.
- [13] Paul Krugman. “Scale Economies, Product Differentiation, and the Pattern of Trade”. In: *The American Economic Review* 70.5 (1980), pp. 950–959. URL: https://www.princeton.edu/~pkrugman/scale_econ.pdf.
- [14] John Douglas Wilson. “Theories of Tax Competition”. In: *National Tax Journal* 52.2 (1999), pp. 269–304. DOI: 10.1086/NTJ41789394. URL: <https://www.journals.uchicago.edu/doi/abs/10.1086/NTJ41789394>.

- [15] George R. Zodrow and Peter Mieszkowski. “Pigou, Tiebout, Property Taxation, and the Underprovision of Local Public Goods”. In: *Journal of Urban Economics* 19.3 (1986), pp. 356–370. DOI: 10.1016/0094-1190(86)90048-3. URL: <https://www.sciencedirect.com/science/article/pii/0094119086900483>.
- [16] Hans-Werner Sinn and Wolfgang Ochel. “Social Union, Convergence and Migration”. In: *Journal of Common Market Studies* 41.5 (2003), pp. 869–896. DOI: 10.1111/j.1468-5965.2003.00467.x. URL: <https://ideas.repec.org/a/bla/jcmkts/v41y2003i5p869-896.html>.
- [17] Assaf Razin and Efraim Sadka. *Tax Competition and Migration: The Race-to-the-Bottom Hypothesis Revisited*. Tech. rep. National Bureau of Economic Research, 2011. DOI: 10.3386/w16670.
- [18] Henrik Kleven et al. “Taxation and Migration: Evidence and Policy Implications”. In: *Journal of Economic Perspectives* 34.2 (2020), pp. 119–142. DOI: 10.1257/jep.34.2.119.
- [19] Eurostat. *Migration and Asylum in Europe – 2025 Edition: Free Movement of Persons in the EU*. 2025. URL: <https://ec.europa.eu/eurostat/web/interactive-publications/migration-2025#free-movement-of-persons-in-the-eu>.
- [20] Tax Foundation. *Top Personal Income Tax Rates in Europe, 2022*. 2022. URL: <https://taxfoundation.org/data/all/eu/top-personal-income-tax-rates-europe-2022/>.
- [21] Eurostat. *Living conditions in Europe - income distribution and income inequality*. 2025. URL: https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Living_conditions_in_Europe_-_income_distribution_and_income_inequality.
- [22] Frans A. Oliehoek and Christopher Amato. *A Concise Introduction to Decentralized POMDPs*. Springer, 2016. URL: <https://www.ccs.neu.edu/home/camato/publications/OliehoekAmato16book.pdf>.
- [23] Volodymyr Mnih, Koray Kavukcuoglu, David Silver, et al. “Human-level control through deep reinforcement learning”. In: *Nature* 518 (2015), pp. 529–533. URL: <https://doi.org/10.1038/nature14236>.
- [24] Yann LeCun et al. “Gradient-Based Learning Applied to Document Recognition”. In: *Proceedings of the IEEE* 86.11 (1998), pp. 2278–2324. DOI: 10.1109/5.726791. URL: <https://ieeexplore.ieee.org/document/726791>.
- [25] Sepp Hochreiter and Jürgen Schmidhuber. “Long Short-Term Memory”. In: *Neural Computation* 9.8 (1997), pp. 1735–1780. DOI: 10.1162/neco.1997.9.8.1735. URL: <https://www.bioinf.jku.at/publications/older/2604.pdf>.
- [26] John Schulman et al. “Proximal Policy Optimization Algorithms”. In: *arXiv preprint arXiv:1707.06347* (2017). URL: <https://arxiv.org/abs/1707.06347>.
- [27] Council of the European Union. *EU Tariffs Explained*. Council of the European Union. May 12, 2025. URL: <https://www.consilium.europa.eu/en/policies/eu-tariffs-explained/> (visited on 05/29/2026).
- [28] Richard M. H. Suen. “Bounding the CRRA Utility Functions”. In: *MPRA Paper* (2009). URL: <https://mpra.ub.uni-muenchen.de/13260/>.
- [29] Michael Catalano, Tanya Leise, and Thomas Pfaff. “Measuring Resource Inequality: The Gini Coefficient”. In: *Numeracy* 2.2 (2009). DOI: 10.5038/1936-4660.2.2.4. URL: <https://digitalcommons.usf.edu/numeracy/vol2/iss2/art4/>.
- [30] Joel Z. Leibo et al. “Multi-Agent Reinforcement Learning in Sequential Social Dilemmas”. In: *Proceedings of the 16th International Conference on Autonomous Agents and Multiagent Systems*. 2017, pp. 464–473. URL: <https://arxiv.org/abs/1702.03037>.

- [31] Ajit Desai. *Machine Learning for Economics Research: When, What and How*. Staff Analytical Note 2023-16. Bank of Canada, Oct. 2023. DOI: 10.34989/san-2023-16. URL: <https://www.bankofcanada.ca/2023/10/staff-analytical-note-2023-16/>.
- [32] T. Soni Madhulatha and Md. Atheeq Sultan Ghor. “Deep Neural Network Approach Integrated with Reinforcement Learning for Forecasting Exchange Rates Using Time Series Data and Influential Factors”. In: *Scientific Reports* 15.1 (Aug. 2025), p. 29009. DOI: 10.1038/s41598-025-12516-3. URL: <https://www.nature.com/articles/s41598-025-12516-3>.
- [33] Nenad Tomašev et al. *Virtual Agent Economies*. 2025. DOI: 10.48550/arXiv.2509.10147. arXiv: 2509.10147 [cs.AI]. URL: <https://arxiv.org/abs/2509.10147>.
- [34] Peter Henderson et al. *Deep Reinforcement Learning that Matters*. 2019. arXiv: 1709.06560 [cs.LG]. URL: <https://arxiv.org/abs/1709.06560>.
- [35] Andrew Patterson et al. *Empirical Design in Reinforcement Learning*. 2024. arXiv: 2304.01315 [cs.LG]. URL: <https://arxiv.org/abs/2304.01315>.
- [36] Logan Engstrom et al. “Implementation Matters in Deep Policy Gradients: A Case Study on PPO and TRPO”. In: *arXiv preprint arXiv:2005.12729* (2020). DOI: 10.48550/arXiv.2005.12729. URL: <https://arxiv.org/abs/2005.12729>.
- [37] FLAIR. *JaxMARL: Multi-Agent RL Environments and Algorithms in JAX*. GitHub repository. 2024. URL: <https://github.com/FLAIROx/JaxMARL>.
- [38] Alexander Rutherford et al. “JaxMARL: Multi-Agent RL Environments and Algorithms in JAX”. In: *Advances in Neural Information Processing Systems*. Ed. by A. Globerson et al. Vol. 37. Curran Associates, Inc., 2024, pp. 50925–50951. DOI: 10.52202/079017-1612. URL: https://proceedings.neurips.cc/paper_files/paper/2024/file/5aee125f052c90e326dcf6f380df94f6-Paper-Datasets_and_Benchmarks_Track.pdf.
- [39] Ray Documentation. *Deploying on Slurm*. Accessed 2026-05-28. 2026. URL: <https://docs.ray.io/en/latest/cluster/vms/user-guides/community/slurm.html>.
- [40] Ryan Lowe et al. *Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments*. Version 4, last revised 14 March 2020. 2017. DOI: 10.48550/arXiv.1706.02275. arXiv: 1706.02275 [cs.LG]. URL: <https://arxiv.org/abs/1706.02275>.
- [41] Chao Yu et al. “The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games”. In: *Advances in Neural Information Processing Systems*. Vol. 35. 2022. URL: <https://arxiv.org/abs/2103.01955>.
- [42] Eric Liang et al. “RLlib: Abstractions for Distributed Reinforcement Learning”. In: *Proceedings of the 35th International Conference on Machine Learning*. PMLR. 2018, pp. 3053–3062. URL: <https://arxiv.org/pdf/1712.09381>.

A Appendix

A.1 Notation

Symbol	Description
<i>Indices, Sets, and Regions</i>	
i	Mobile-agent index, $i = 1, \dots, N$
t	Time index, $t = 0, \dots, H$
r	Region index, $r \in \{\text{top}, \text{bottom}\}$
b	Tax-bracket index, $b = 1, \dots, B$
N	Number of mobile agents
H	Episode horizon / simulation length
B	Number of tax brackets
$\mathcal{I}_{r,t}$	Set of agents located in region r at time t
$r(i, t)$	Region in which agent i is located at tax collection time
<i>State, Actions, and Policies</i>	
s_t	Full environment state at time t
$\mathbf{a}_{i,t}$	Multi-discrete action vector chosen by agent i at time t
$\boldsymbol{\tau}_{r,t}$	Tax-schedule vector chosen by planner r at time t
$\tau_{r,t}^b$	Marginal tax rate chosen by planner r for bracket b
$\pi_{\boldsymbol{\theta}}$	Shared mobile-agent policy
$\boldsymbol{\theta}$	Shared neural-network parameter vector for all mobile agents
$\pi_{\boldsymbol{\phi}_r}$	Policy of regional planner r
$\boldsymbol{\phi}_r$	Neural-network parameter vector of planner r
$\mathbf{h}_{i,t}$	Recurrent hidden-state vector of mobile agent i
$\mathbf{h}_{r,t}^P$	Recurrent hidden-state vector of planner r
$\mathbf{m}_{i,t}$	Agent action-mask vector at time t
$\mathbf{m}_{r,t}$	Planner action-mask vector at time t
γ	Discount factor
<i>Agent Observations</i>	
$\mathbf{o}_{i,t}^{\text{world}}$	Agent i 's local spatial grid observation
$\mathbf{o}_{i,t}^{\text{idx}}$	Agent i 's local spatial index map
$\mathbf{o}_{i,t}^{\text{agent}}$	Agent-specific non-spatial state vector
$\mathbf{o}_{i,t}^{\text{market}}$	Agent i 's market observation vector
$\mathbf{o}_{i,t}^{\text{tax}}$	Agent i 's tax-policy observation vector
<i>Planner Observations</i>	
$\mathbf{o}_{r,t}^{\text{world}}$	Regional spatial map observed by planner r
$\mathbf{o}_{r,t}^{\text{idx}}$	Regional index map observed by planner r
$\mathbf{o}_{r,t}^{\text{tax}}$	Planner r 's regional tax-state observation vector
$\mathbf{o}_{r,t}^{\text{market}}$	Planner r 's local market-state observation vector
$\mathbf{o}_{r,t}^{\text{agents}}$	Planner-side per-agent entries for agents in region r

Table 1: Notation for indices, policies, actions, and observations.

Symbol	Description
<i>Coin, Income, and Labor</i>	
$x_{i,t}^c$	Coin endowment of agent i at time t
$\mathbf{x}_t^c$	Vector of coin endowments across all agents at time t
$\mathbf{x}_{r,t}^c$	Coin-endowment vector for agents located in region r at time t
$x_{i,t}^{c,\text{inv}}$	Available inventory coin of agent i at time t
$\bar{x}_{i,t}^c$	Coin snapshot of agent i from the start of the tax period
$z_{i,t}$	Period income of agent i before truncation at zero
$z_{i,t}^+$	Taxable income of agent i , defined as $\max\{z_{i,t}, 0\}$
$l_{i,t}$	Cumulative labor of agent i at time t
<i>Utility, Welfare, and Rewards</i>	
$\text{CRRA}(x_{i,t}^c)$	CRRA utility from coin consumption
η	Coefficient of relative risk aversion
κ	Labor-disutility parameter
$u_i(x_{i,t}^c, l_{i,t})$	Utility of agent i at time t
$\rho_{i,t}$	Reward of agent i at time t
$\rho_{r,t}^P$	Reward of planner r at time t
$\text{gini}(\mathbf{x}_{r,t}^c)$	Gini coefficient for agents in region r
$\text{eq}_r(\mathbf{x}_{r,t}^c)$	Regional equality measure, $1 - \text{gini}(\mathbf{x}_{r,t}^c)$
$\text{prod}_r(\mathbf{x}_{r,t}^c)$	Regional production measure
$\text{swf}_{r,t}(\mathbf{x}_{r,t}^c)$	Regional social welfare function for planner r
<i>Taxation and Redistribution</i>	
m_b	Lower cutoff of tax bracket b
$\mathbf{m}$	Vector of bracket cutoffs, $\mathbf{m} = (m_1, \dots, m_{B+1})$
$T_r(z_{i,t}^+)$	Tax liability under region r 's tax schedule
$\hat{T}_{i,t}$	Effective tax payment made by agent i
$R_{r,t}^{\text{tax}}$	Regional income-tax revenue in region r
$R_{r,t}^{\text{travel}}$	Regional revenue from travel payments
$R_{r,t}^{\text{tariff}}$	Regional revenue from import tariffs
$R_{r,t}$	Total regional revenue in region r
$g_{r,t}$	Lump-sum transfer per agent in region r

Table 2: Notation for economic variables, welfare, rewards, and taxation.

A.2 Plots

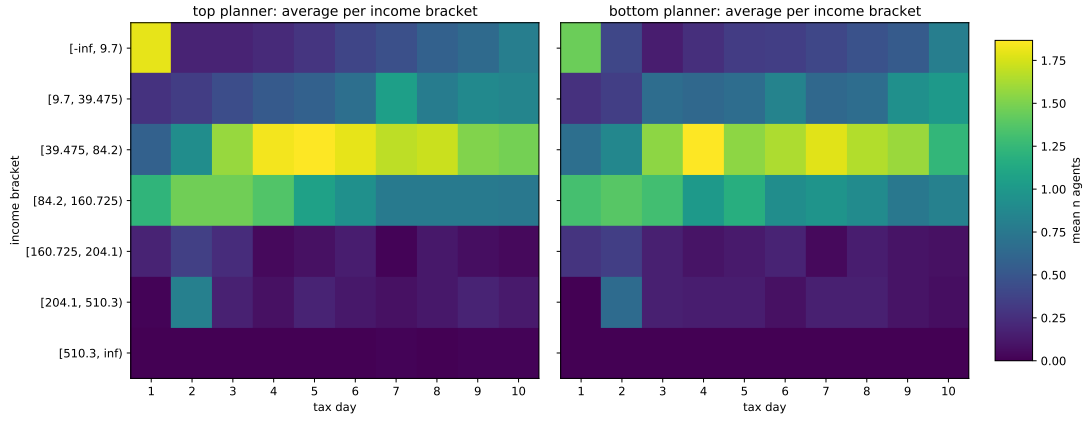
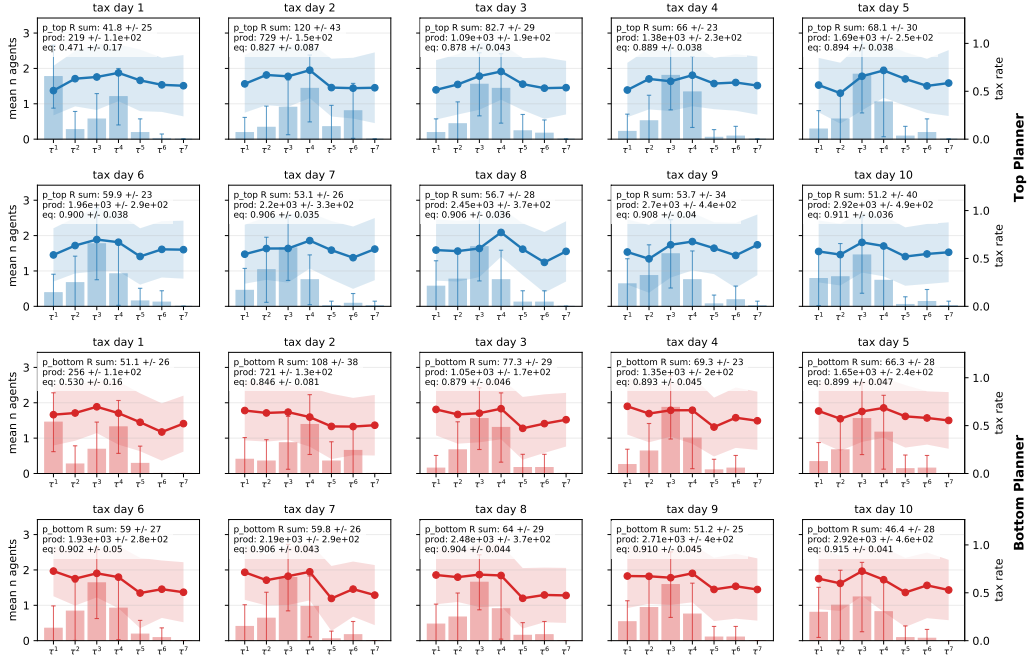
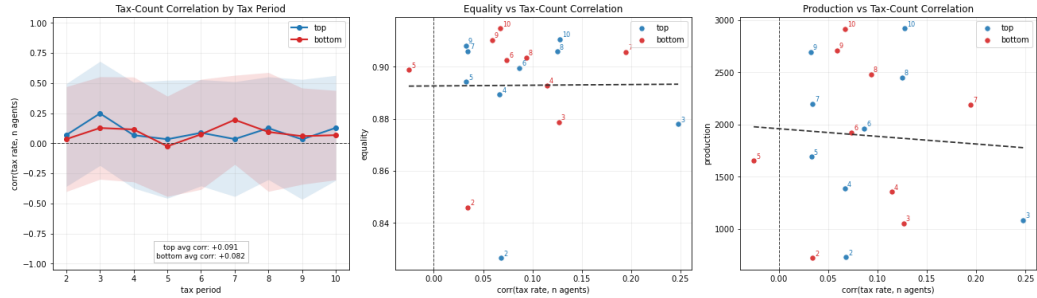


Figure 22: **Average income-bracket distribution in the baseline setting.** The heatmap shows the average distribution of agents across income brackets in the baseline experiment, averaged over rollout trajectories and training seeds. The figure provides a reference for interpreting planner tax schedules by showing which brackets are frequently populated and which brackets are rarely reached.



(a) **Average tax-day snapshots across dense logs and seeds.** Bars show the mean number of agents in each income bracket, while lines show the mean marginal tax rate selected for each bracket.



(b) **Tax-bracket alignment with equality and production.** The left panel shows the tax-count correlation across tax periods, while the middle and right panels compare this correlation with equality and production. The correlation is calculated over brackets τ_1 – τ_6 . The highest bracket is excluded since it almost never reached and might impact averages unnecessarily (as also shown in Appendix Figure 22). The first tax period is also excluded from the main figure because it reflects an early transient phase of the simulation. The corresponding plot including the first period is reported in Appendix Figure 37.

Figure 23: **Planner tax schedules and tax-count alignment.** Panel (a) compares the number of agents in each income bracket with the marginal tax rates chosen by each planner. Panel (b) summarizes this relationship using the correlation between bracket population and marginal tax rates, and compares the correlation with equality and production.

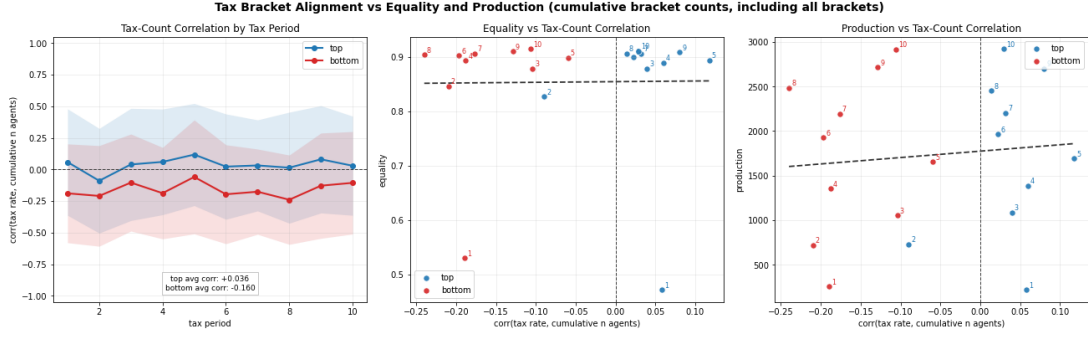


Figure 24: **Cumulative tax-count alignment in the baseline setting.** The figure shows the correlation between marginal tax rates and the cumulative number of agents reaching each income bracket. The cumulative measure is closer to the marginal tax mechanism, since it captures whether planners assign higher rates to brackets that agents actually reach. The systematic difference between planners appears in two out of three seeds and should therefore be interpreted as a descriptive training pattern rather than a robust planner asymmetry.

for bracket b , the cumulative count is

$$C_{r,t}^b = \sum_{j=1}^b n_{r,t}^j$$

where: $C_{r,t}^b$ is the cumulative number of agents up to bracket b . $n_{r,t}^j$ is the number of agents in bracket j . $r \in \{\text{top}, \text{bottom}\}$. t is the tax period. $b \in \{1, \dots, B\}$.

Giving the cumulative count vector:

$$C_{r,t} = (C_{r,t}^1, C_{r,t}^2, \dots, C_{r,t}^B)$$

and the correlation is:

$$\rho_{r,t} = \text{corr}(\tau_{r,t}, C_{r,t})$$

with:

$$\tau_{r,t} = (\tau_{r,t}^1, \tau_{r,t}^2, \dots, \tau_{r,t}^B)$$

The cumulative version is closer to the actual marginal tax mechanism: it asks whether the planner assigns higher marginal rates to brackets that agents actually reach.



Figure 25: **Market activity with and without cross-regional trade.** The figure compares market size, average transaction prices, and trade activity across settings with and without cross-regional trade. Market size is measured as the mean quantity of wood and stone held in escrow across runs. Large price movements in the earliest periods should be interpreted cautiously, since a small number of early market entries can strongly affect the average price when market activity is still limited.

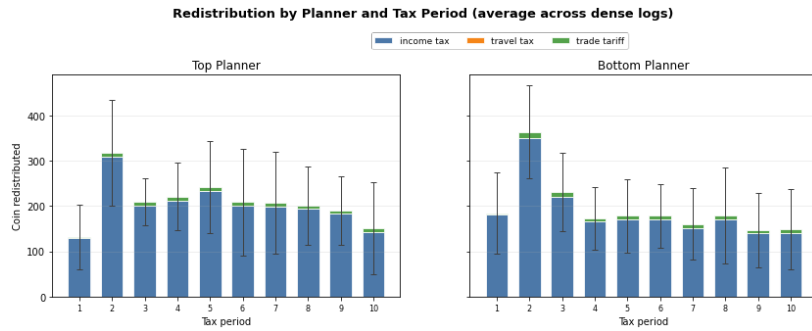


Figure 26: **Redistribution revenue in the trade-integrated setting.** The figure decomposes regional redistribution into income-tax revenue and import-tax revenue in the cross-regional trade experiment. The plot shows how market integration introduces an additional revenue channel through import taxation, although income taxation remains the main source of redistribution in most periods.

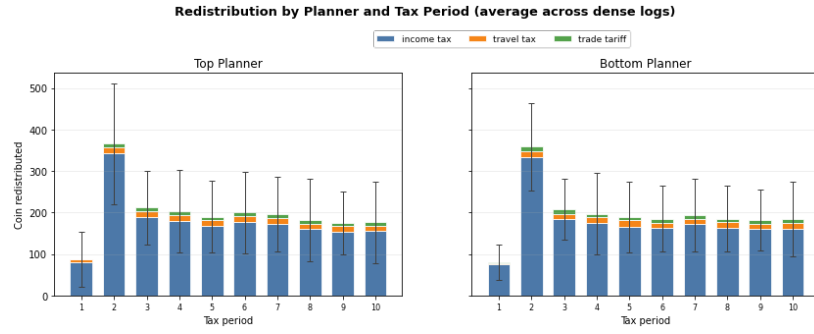


Figure 27: **Redistribution revenue in the travel enabled setting.** The figure decomposes regional redistribution into income-tax revenue, import-tax revenue, and travel revenue in the travel-enabled experiment. The plot shows how mobility adds a further revenue channel, while also changing the regional population over which redistribution is allocated.

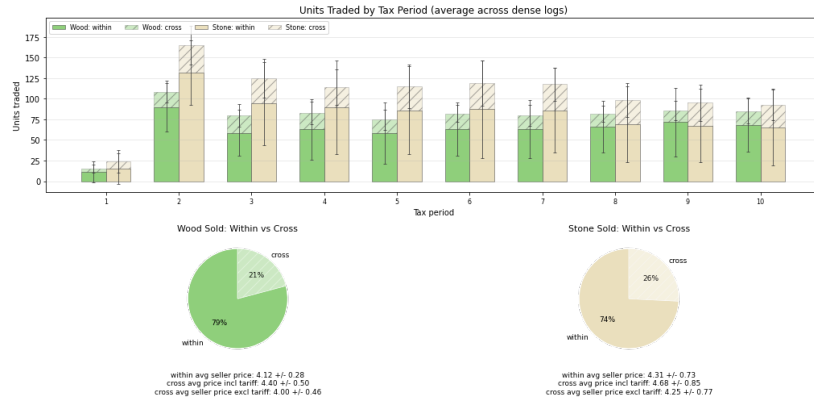


Figure 28: **Trade composition in the mobility setting.** The figure reports trade metrics for the travel-enabled experiment (comparable to the trade enabled setting shown in figure 12). Compared with the trade-integrated setting, stone accounts for a larger share of trading activity, although the standard-deviation bands indicate substantial variation across runs. This suggests that mobility changes trade composition, but that the pattern is not uniform across training seeds.

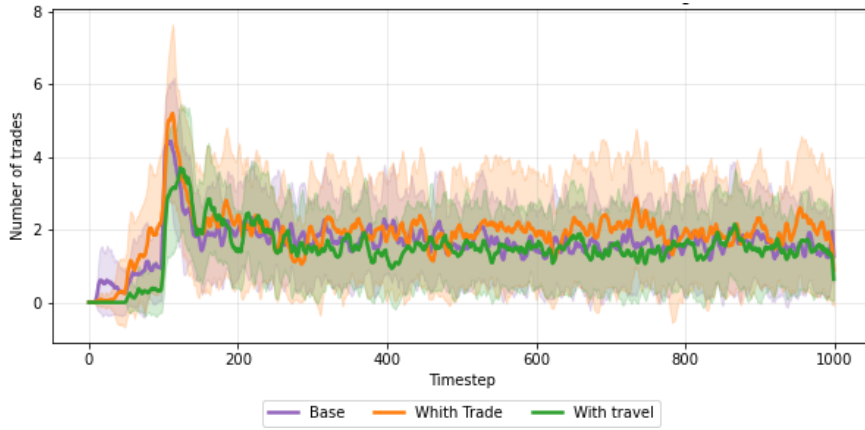


Figure 29: **Trade activity across experimental settings.** The figure compares trade counts over time in the baseline, trade-integrated, and mobility settings. Lines show mean trade counts across dense logs, while shaded areas indicate one standard deviation. The comparison shows how cross-regional trade and mobility change the level and variability of market activity relative to the closed-economy baseline.

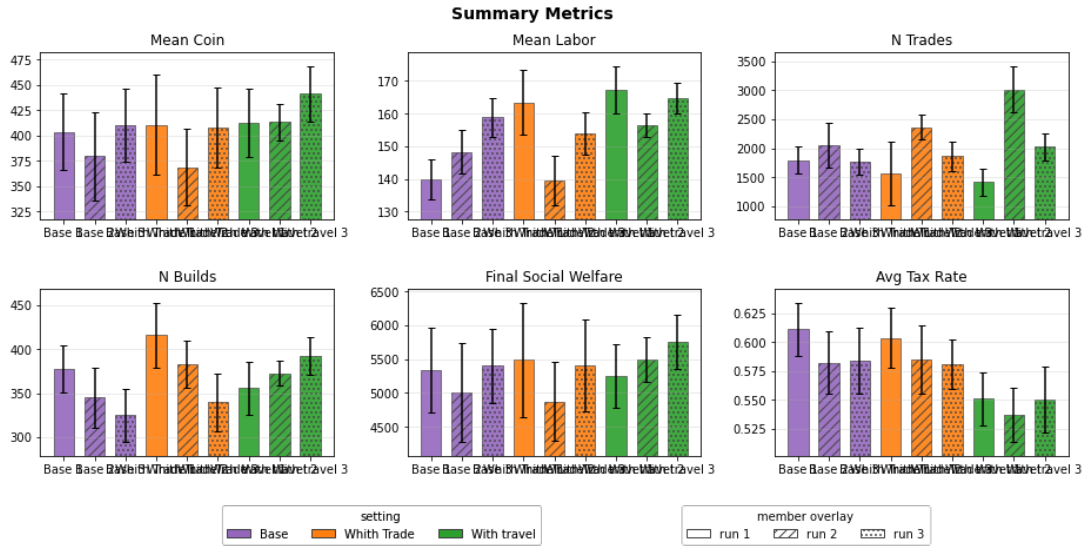


Figure 30: **Summary metrics across training seeds.** The figure reports key outcome metrics separately for each simulated training seed across the experimental settings. Showing the seed-level outcomes highlights the extent of variation across independently trained policies and supports the interpretation of the results as descriptive patterns rather than statistically definitive estimates.

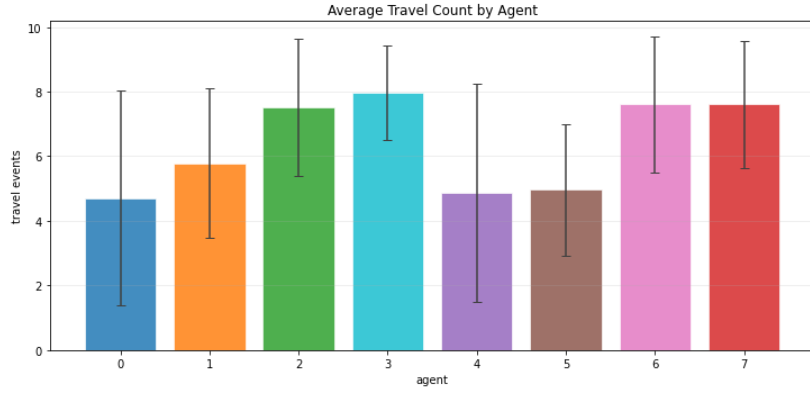


Figure 31: **Relocation events by agent.** The figure shows the average number of relocation events by agent index across the three training seeds. The distribution illustrates that mobility is not used uniformly across agents, suggesting that relocation behavior depends on agent-specific conditions such as skill, location, accumulated resources, and learned policy behavior.

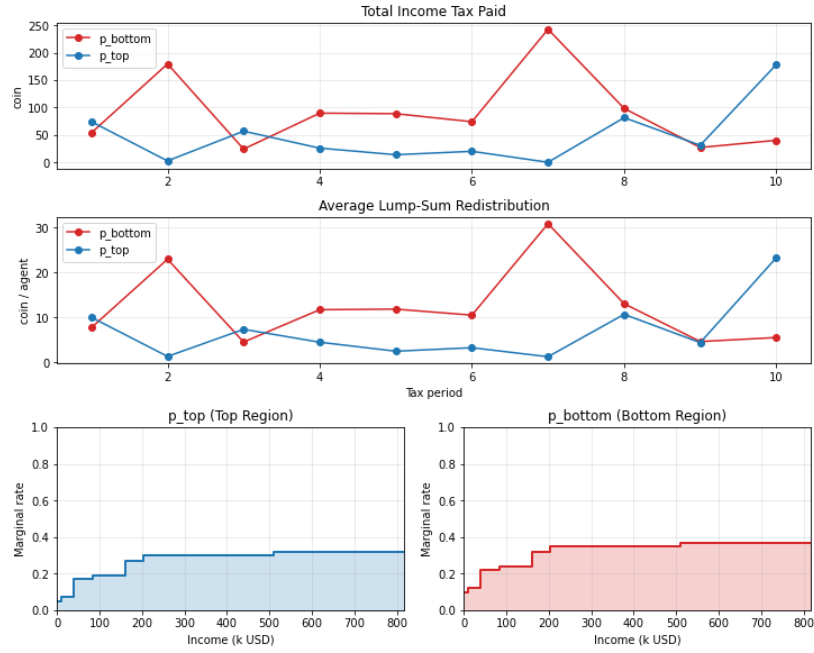


Figure 32: **Fixed-tax experiment outcomes.** The figure shows income and redistribution across tax periods and regions in the fixed-tax experiment. The lower panel reports the fixed tax schedules used in the experiment, where the top region has tax rates five percentage points lower than the bottom region in each bracket. The results provide a controlled comparison for examining whether a persistent tax advantage is sufficient to generate systematic relocation toward the lower-tax region.

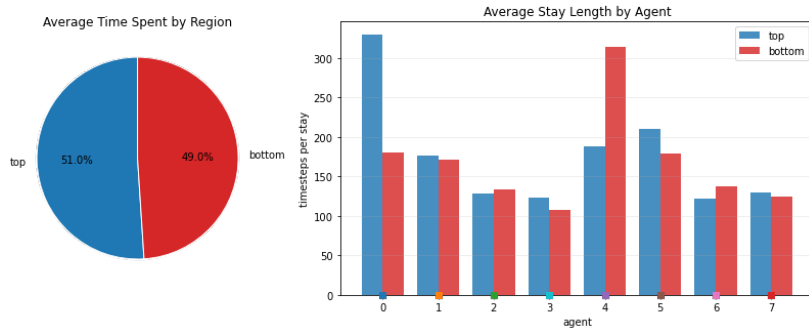


Figure 33: **Residence patterns in the regular mobility setting.** The figure reports the same residence measure as Figure 19, but for the regular travel-enabled setting without fixed tax schedules. The comparison provides a baseline for interpreting the fixed-tax experiment by showing how agents distribute across regions when tax schedules are learned rather than externally imposed.

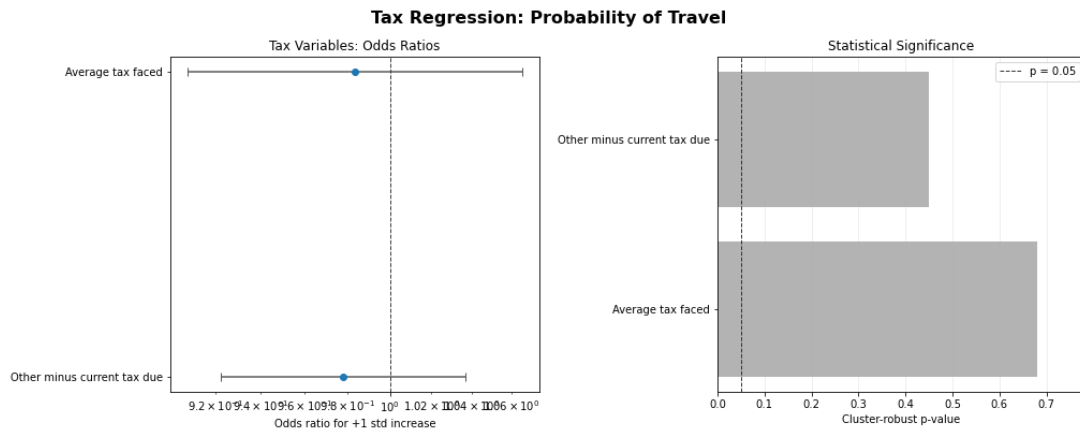


Figure 34: **Estimated tax effects in the relocation regression.** The figure reports odds ratios and p-values for the tax-related variables in the relocation regression. The estimates summarize whether current-region and alternative-region tax exposure are associated with relocation decisions, while the uncertainty measures indicate how cautiously these effects should be interpreted.

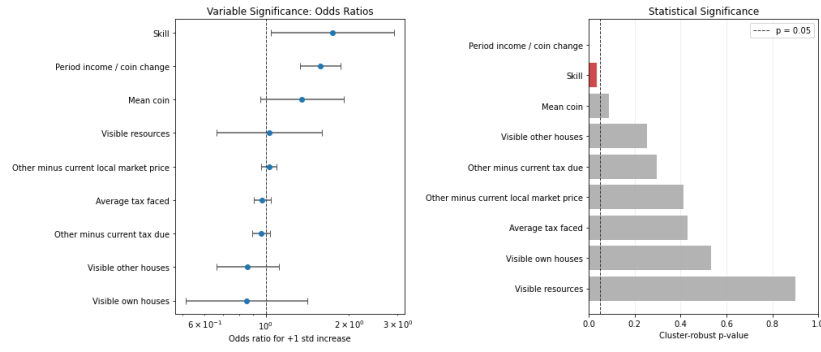


Figure 35: **Relocation regression without agent fixed effects.** The figure reports odds ratios and p-values from the pooled relocation regression, including tax-related variables but excluding agent fixed effects. The estimates are broadly consistent with the main regression results found in 17, suggesting that the relationship between relocation and observed economic conditions is not driven only by the inclusion of agent fixed effects.

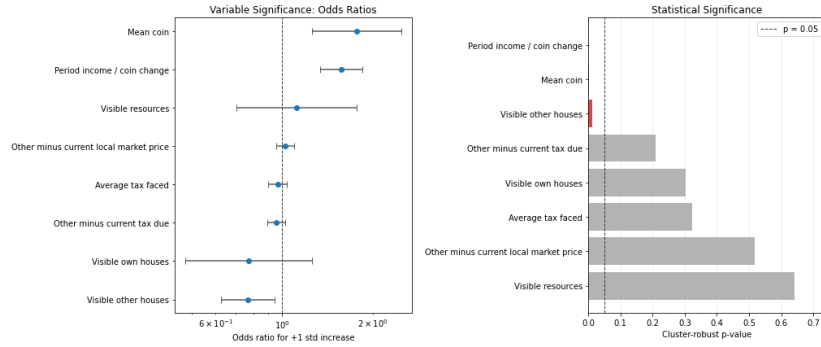


Figure 36: **Relocation regression with agent fixed effects.** The figure reports odds ratios and p-values from the relocation regression including agent fixed effects. This specification controls for persistent differences across agents, allowing the estimates to focus more directly on within-agent variation in tax exposure, income, resources, and spatial conditions. The results are consistent with the results found in 17.

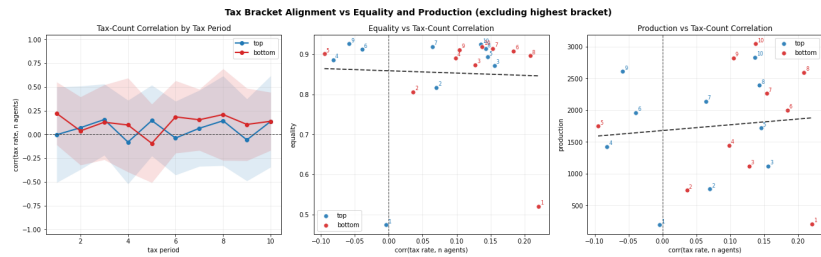


Figure 37: **Tax-count alignment including the first tax period.** The figure corresponds to the tax-count alignment analysis in panel (b) of figure 23, but includes the first tax period. The first period is visibly different from later periods, illustrating why it is treated as an early transient phase and excluded from the main tax-response analysis.

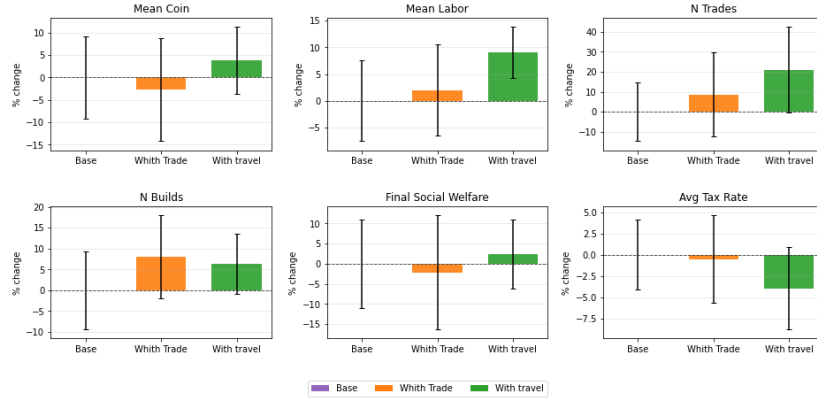


Figure 38: **Percentage differences in summary metrics.** The figure shows percentage differences in key summary metrics between the baseline and the trade-integrated and mobility settings. The comparison provides a normalized view of how the experimental extensions change aggregate outcomes relative to the closed-economy baseline.

A.3 Algorithm

Algorithm 1 summarizes the training procedure used in the simulations. The implementation follows RLlib’s multi-agent PPO setup. At each timestep, the shared mobile-agent policy chooses actions for all mobile agents, while each regional planner policy chooses a tax-action vector. Planner actions are sampled every timestep, but action masks restrict tax-rate changes to valid tax-decision times. The environment then advances one step, updates the economic state, and internally applies tax collection and redistribution whenever the tax-period condition is met.

Algorithm 1 Phased PPO training loop for mobile agents and regional planners

Input:

L (PPO rollout length),
 T (tax-period length),
 K (number of regions; $K = 2$ in the implemented simulations),
 $\mathcal{A}$ (PPO update rule),
 C (stopping criterion),
 $\mathcal{P}$ (training phase)

Output:

θ (shared mobile-agent policy parameters),
 $\{\phi_r\}_{r=1}^K$ (regional planner policy parameters)
Initialize environment state s_0 , agent observations $\{\mathbf{o}_{i,0}\}_{i=1}^N$, planner observations $\{\mathbf{o}_{r,0}^P\}_{r=1}^K$
Initialize recurrent states $\{\mathbf{h}_{i,0}\}_{i=1}^N$ and $\{\mathbf{h}_{r,0}^P\}_{r=1}^K$
Initialize policy parameters θ and $\{\phi_r\}_{r=1}^K$
Initialize rollout buffers $D^A \leftarrow \emptyset$ and $\{D_r^P\}_{r=1}^K \leftarrow \{\emptyset\}_{r=1}^K$
while training **do**
 for $t = 0, \dots, L - 1$ **do**
 for $i = 1, \dots, N$ **do**
 $\mathbf{a}_{i,t}, \mathbf{h}_{i,t+1} \sim \pi_\theta(\cdot \mid \mathbf{o}_{i,t}, \mathbf{h}_{i,t}, \mathbf{m}_{i,t})$
 end for
 for $r = 1, \dots, K$ **do**
 $\tau_{r,t}, \mathbf{h}_{r,t+1}^P \sim \pi_{\phi_r}(\cdot \mid \mathbf{o}_{r,t}^P, \mathbf{h}_{r,t}^P, \mathbf{m}_{r,t})$
 end for
 $s_{t+1}, \{\mathbf{o}_{i,t+1}\}_{i=1}^N, \{\mathbf{o}_{r,t+1}^P\}_{r=1}^K, \{\rho_{i,t}\}_{i=1}^N, \{\rho_{r,t}^P\}_{r=1}^K, d_t \leftarrow$
Env.step($s_t, \{\mathbf{a}_{i,t}\}_{i=1}^N, \{\tau_{r,t}\}_{r=1}^K$)
 Store agent transitions in D^A
 Store planner transitions in $\{D_r^P\}_{r=1}^K$
 Set s_t , observations, and recurrent states to their next values
 if d_t **then**
 Reset completed environment and recurrent states
 end if
 end for
 Estimate returns and advantages from D^A and $\{D_r^P\}_{r=1}^K$ using GAE
 if $\mathcal{P}$ trains mobile agents **then**
 Update θ using D^A and PPO update rule $\mathcal{A}$
 end if
 if $\mathcal{P}$ trains planners **then**
 for $r = 1, \dots, K$ **do**
 Update ϕ_r using D_r^P and PPO update rule $\mathcal{A}$
 end for
 end if
 Reset rollout buffers $D^A \leftarrow \emptyset$ and $\{D_r^P\}_{r=1}^K \leftarrow \{\emptyset\}_{r=1}^K$
 if criterion C is met **then**
 return $\theta, \{\phi_r\}_{r=1}^K$
 end if
end while

The phased PPO training loop alternates between rollout collection and policy optimization. During rollout collection, mobile agents and regional planners sample actions from their current policies, interact with the economy, and store the resulting observations, actions, rewards, value estimates, and recurrent states. After each rollout batch, returns and advantages are estimated using GAE. Depending on the active training phase, PPO updates the shared mobile-agent policy, the regional planner policies, or both. This structure allows agents and planners to co-adapt while also permitting selected phases in which one level of the system is held fixed.

In the implemented simulations, $K = 2$, corresponding to the top and bottom regions of the map. The eight mobile agents share the same policy parameters θ , while the two regional planners have separate policy parameters, ϕ_{top} and ϕ_{bottom} . Thus, experience from all mobile agents contributes to the update of the shared mobile-agent policy, whereas each planner is updated using its own planner-specific rollout data.

The notation $\mathbf{o}_{i,t}$ denotes the full observation available to mobile agent i , including its spatial map observation, index map, agent-specific state variables, market information, and tax information. Similarly, $\mathbf{o}_{r,t}^P$ denotes the full observation available to planner r , including its regional spatial observation, regional index map, tax state, market information, and planner-side per-agent entries. The masks $\mathbf{m}_{i,t}$ and $\mathbf{m}_{r,t}$ restrict the set of valid actions. For planners, these masks are especially important because they allow the planner policies to sample actions every timestep while only permitting tax-bracket updates at valid tax-decision times. At other timesteps, the valid planner action is effectively a no-op.

Tax collection and redistribution are not implemented as a separate call in the training loop. Instead, they are handled internally by `Env.step` through the regional tax components whenever the tax-period condition is met. Therefore, the rewards $\rho_{i,t}$ and $\rho_{r,t}^P$ already reflect the economic consequences of production, trade, travel, taxation, redistribution, and changes in utility or regional welfare.

The rollout buffers are written compactly in Algorithm 1. In practice, PPO stores additional quantities required for policy-gradient learning, including sampled actions, rewards, value-function predictions, action log-probabilities, done flags, action masks, and recurrent-state or sequence information. After each rollout batch is collected, returns and advantages are estimated using generalized advantage estimation. PPO then updates policy parameters by increasing the probability of actions with positive estimated advantage and decreasing the probability of actions with negative estimated advantage, while clipping the update to prevent the new policy from moving too far from the policy that generated the rollout data.

Which policies are updated depends on the training phase. Phase 1 trains the mobile-agent policy only. Phase 2 trains mobile agents and planners jointly. Phase 3A trains mobile agents while planner policy weights are held fixed; the planners still act in the environment, but their neural-network parameters are not updated. Phase 3B again trains mobile agents and planners jointly.

All trainable policies use the custom `keras_conv_lstm` architecture. The shared mobile-agent policy uses two convolutional layers, two fully connected layers with 128 hidden units, and an LSTM with 128 units. Each planner policy uses the same architecture but with two fully connected layers of 256 hidden units and an LSTM with 256 units. The actor and critic are implemented as separate policy and value branches.

A.4 Parameters

The key training hyperparameters used throughout the experiments are summarized in Table 3.

Table 3: Training hyperparameters for `simulation.run.py`

Parameter	Value
Number of rollout workers	4
Environments per worker	1
Parallel environment replicas	4
Sampling horizon per environment	200
Training batch size	800
SGD minibatch size	200
Number of SGD passes per batch	4
Minibatch gradient updates per iteration	16
CPUs per worker	1
GPUs	1
Framework	TensorFlow
Learning rate (agent)	0.0003
Learning rate (planner)	0.0001
Entropy regularization coefficient (planner)	0.02
Entropy regularization coefficient (agent, Phase 3A)	0.01
PPO clipping parameter ϵ	0.3
Episode length	1000
Tax period	100
SGD sequence length	25
Agent policy model	<code>keras_conv_lstm</code>
Planner policy model	<code>keras_conv_lstm</code>
Agent spatial input	11×11 local map
Planner spatial input	25×25 regional map
Agent convolutional layers	2 layers: 16, 32 filters
Planner convolutional layers	2 layers: 16, 32 filters
Convolutional kernel and stride	3×3 , stride 2
Index-map embedding dimension	4
Agent fully connected layers	2 layers, 128 hidden units
Planner fully connected layers	2 layers, 256 hidden units
Agent LSTM size	128 units
Planner LSTM size	256 units
Actor and critic structure	Separate policy and value branches
Agent action logits	57
Planner action logits	308 per planner
Phase 1 training duration	500 iterations, approx. 0.4M steps
Phase 2 training duration	1500 iterations, approx. 1.2M steps
Phase 3A training duration	500 iterations, approx. 0.4M steps
Phase 3B training duration	1500 iterations, approx. 1.2M steps
Phase 2 tax annealing schedule	(0, 0.0125)
Phase 2 tax annealing duration	400 iterations, approx. 320k aggregate steps
Travel enabled	No, in current <code>simulation.run.py</code> setup
Regional trade restriction	Yes
Fixed planner taxes	No

Table 4: Environment hyperparameters

Parameter	Value
World width	51
World height	25
Episode length	1000
Number of agents	8
Starting agent coin	0
Tax period	100
Planner subclasses	2
Build labor cost	2.1
Auction max bid/ask	10
Auction order labor	0.05
Auction max number of orders	10
Auction order duration	50
Gather move labor	0.21
Gather collect labor	0.21
Tax annealing schedule	$[-100, 0.0125]$

A.5 Tax table

Bracket	Income range	Planner action
1	$0 \leq z < 9.7$	τ^1
2	$9.7 \leq z < 39.475$	τ^2
3	$39.475 \leq z < 84.2$	τ^3
4	$84.2 \leq z < 160.725$	τ^4
5	$160.725 \leq z < 204.1$	τ^5
6	$204.1 \leq z < 510.3$	τ^6
7	$z \geq 510.3$	τ^7

Table 5: Income tax brackets corresponding to the US tax bracket system and used for most of the experiments. The bracket cutoffs are given by $m = [0, 9.7, 39.475, 84.2, 160.725, 204.1, 510.3, \infty]$.

A.6 Aneling Procedure

Training is organized into four phases. Phase 1 trains the mobile-agent policy while taxes and travel are disabled. This allows the agents to learn the basic gather, trade, and build dynamics before being exposed to regional mobility. Phase 2 trains mobile agents and planners jointly with regional taxation enabled. Phase 3A enables travel and trains the mobile agents while planner policy weights are held fixed. Phase 3B then trains both mobile agents and planners jointly with travel enabled.

The simulations use PPO with rollout fragments of length 200 and an episode length of 1000 timesteps. Thus, each rollout environment completes one full episode after approximately

$$\frac{1000}{200} = 5$$

PPO iterations. This conversion is useful for interpreting schedules that are defined in terms of completed episodes rather than PPO training iterations.

During Phase 2, the maximum tax rate available to the planners is gradually annealed. The implemented tax annealing schedule is

$$(0, 0.0125),$$

where the first element is the warmup period and the second element is the slope of the annealing schedule. The allowed maximum tax rate is computed as

$$\bar{\tau}_{\max} = 0.0125 \cdot (\text{completions} - 0),$$

bounded above by 1.0. Since the warmup period is zero, the tax cap begins increasing immediately. The cap reaches 100% when

$$1.0 = 0.0125 \cdot \text{completions}.$$

Solving for completions gives

$$\text{completions} = \frac{1.0}{0.0125} = 80.$$

Thus, the tax cap reaches 100% after approximately 80 completed episodes per rollout environment. Since one completed episode corresponds to approximately 5 PPO iterations, this is equivalent to

$$80 \times 5 = 400$$

PPO iterations. Therefore, under the implemented settings, the Phase 2 tax annealing schedule reaches the full tax range around Phase 2 iteration 400.

After the annealing period, planners can select from the full discrete tax-rate set. Before this point, action masking restricts the highest available tax rates, so planners gradually gain access to larger marginal tax rates over the course of training. This reduces the abruptness of early planner interventions and allows agents and planners to adapt gradually to the tax-policy action space.

A.6.1 Additional Implementation Specifications

The `keras_conv_lstm` model follows the architecture used in the AI Economist framework [12]. It combines convolutional layers for spatial observations, fully connected layers for the combined feature representation, and an LSTM for temporal information. The recurrent component allows each policy to maintain a hidden state over time, which is important in the partially observed multi-agent economy.

Dense logs saved after Phases 1, 2, and 3A are generated with deterministic action selection (`explore=False`), while the final Phase 3B dense logs used for analysis are generated with `explore=True`. The final reported rollouts therefore include stochastic policy sampling rather than purely deterministic greedy evaluation.

In the implemented configuration, planner policies use an entropy coefficient of 0.02. The shared agent policy uses RLlib’s default entropy coefficient of 0.0, except in Phase 3A, where the agent entropy coefficient is set to 0.01. This means that entropy regularization is applied asymmetrically across policies. Planner policies use a positive entropy coefficient to encourage exploration over the large discrete tax-policy space and reduce the risk of premature convergence to a narrow set of tax schedules. The shared agent policy uses no additional entropy bonus in most phases, relying instead on PPO’s stochastic action sampling. This avoids excessively noisy behavior in the agents’ already large multi-discrete action space. In Phase 3A, an agent entropy coefficient is introduced to encourage exploration of behavioral responses to fixed planner policies, including relocation, trade, gathering, and building decisions.

In this thesis, multi-agent PPO is implemented through RLlib’s `PPOTrainer` interface [42]. This corresponds to the older RLlib Trainer API used by the original AI Economist-style implementation; newer versions of RLlib organize PPO under the `ray.rllib.algorithms.ppo` API.

A.7 Action Space Specification

The implemented simulations use multi-discrete action spaces for both mobile agents and regional planners. In a multi-discrete action space, the policy selects one option from each action component at every timestep. Each component has its own no-op option, so the agent or planner can leave that component inactive while still selecting actions for other components.

For mobile agents, the action space consists of the following components:

Action component	Number of choices
Build	2
Buy Stone	12
Sell Stone	12
Buy Wood	12
Sell Wood	12
Gather	5
CrossWaterTravel	2
Total action logits	57

The buy and sell components have 12 choices because they include the 11 possible price points from 0 to 10, plus a no-op option. Similarly, the build and travel components each include an inactive/no-op choice. Thus, the mobile-agent policy outputs 57 action logits in total.

For planners, the action-mask/logit size is 308, corresponding to

$$14 \times 22 = 308.$$

Conceptually, each planner chooses marginal tax rates for seven income brackets. However, in the implementation the planner action space contains bracket-action keys from both regional tax components. Each bracket component has 22 choices: one no-op option plus 21 discrete tax-rate choices,

$$\{0.00, 0.05, 0.10, \dots, 1.00\}.$$

Action masks ensure that each planner can only apply the relevant valid tax-bracket actions at valid tax-decision times. At other timesteps, the valid planner action is effectively a no-op.

The spatial observations differ between mobile agents and planners. Mobile agents observe a local spatial window, while planners observe a regional map.

For mobile agents, the spatial observation is:

$$\text{agent world-map} : 7 \times 11 \times 11,$$

consisting of six environment/entity channels plus one padding or boundary channel used near map edges. The corresponding index map is:

$$\text{agent world-idx-map} : 2 \times 11 \times 11,$$

where the two index channels represent:

1. owner map, indicating ownership of private landmarks such as houses,
2. location map, indicating where agents are located.

For planners, the spatial observation is:

$$\text{planner world-map} : 6 \times 25 \times 25,$$

with the following channels:

1. Stone
2. Wood
3. House
4. Water
5. StoneSourceBlock
6. WoodSourceBlock

The planner index map is:

$$\text{planner world-idx-map} : 2 \times 25 \times 25,$$

with the same two index channels:

1. owner map, indicating ownership of private landmarks in the form of houses,
2. location map, indicating where agents are located.

Thus, mobile agents observe a local 11×11 view, while planners observe a regional 25×25 view. The mobile-agent map has seven map channels and two index channels, whereas the planner map has six map channels and two index channels.

Each planner also has fixed planner-side per-agent observation slots, denoted $p0, \dots, p7$. These slots exist for all eight mobile agents. For agents currently outside the planner's region, the corresponding slot is zero-masked. For agents currently inside the planner's region, the slot contains that agent's planner-side information. This keeps the planner observation structure fixed while ensuring that planner-side per-agent information is region-specific.

A.8 Binary Logistic Regression

The regression analysis is conducted at the agent-period level. The dependent variable is

$$Y_{i,p,r} = \begin{cases} 1 & \text{if agent } i \text{ travels at least once in tax period } p \text{ in rollout } r, \\ 0 & \text{otherwise.} \end{cases}$$

In the current simulations, the travel cooldown is approximately one tax period, so an agent can travel at most once within a tax period. More generally, however, the variable should be interpreted as an indicator for whether any travel occurs during the period.

The main regression specification includes the following predictors:

$$X_1 = \text{mean coin},$$

$$X_2 = \text{period coin change},$$

$$X_3 = \text{other minus current local market price},$$

$$X_4 = \text{mean visible resources},$$

$$X_5 = \text{mean visible other houses},$$

$$X_6 = \text{mean visible own houses},$$

$$X_7 = \text{skill}.$$

The variable period coin change is defined as

$$\Delta x_{i,p}^c = x_{i,\text{end of } p}^c - x_{i,\text{start of } p}^c.$$

It is therefore a net change in coin holdings over the tax period, not pure labor income, build income, or trade income. It reflects both coin earned and coin spent during the period.

The market-price variable is defined as

$$\Delta q_{i,p} = q_{i,p}^{\text{other}} - q_{i,p}^{\text{current}},$$

where $q_{i,p}^{\text{current}}$ is the average within-region market price in the agent's current region during period p , and $q_{i,p}^{\text{other}}$ is the corresponding within-region market price in the other region. A positive value means that the other region has a higher local market price than the agent's current region, while a negative value means that the other region has a lower local market price. If a region-period has no relevant trade observations, the plotting code uses period-level fallback averages where possible and otherwise sets the difference to zero.

Before fitting the model, each continuous predictor is standardized:

$$Z_j = \frac{X_j - \mu_j}{\sigma_j}.$$

Thus, the estimated coefficients describe the association between travel and a one-standard-deviation increase in the corresponding variable.

Without agent fixed effects, the model is

$$P(Y_{i,p,r} = 1) = \text{logistic} \left(\beta_0 + \beta_1 Z_{\text{coin}} + \beta_2 Z_{\text{coin change}} + \beta_3 Z_{\text{market price difference}} + \beta_4 Z_{\text{visible resources}} + \beta_5 Z_{\text{visible other houses}} + \beta_6 Z_{\text{visible own houses}} + \beta_7 Z_{\text{skill}} + \gamma_p \right),$$

where γ_p denotes tax-period fixed effects when these are included.

The logistic function is

$$\text{logistic}(x) = \frac{1}{1 + e^{-x}}.$$

Equivalently, the model is linear in log-odds:

$$\log \left(\frac{P(Y = 1)}{1 - P(Y = 1)} \right) = \beta_0 + \sum_j \beta_j Z_j + \gamma_p.$$

The agent fixed-effects specification adds an agent-specific intercept:

$$P(Y_{i,p,r} = 1) = \text{logistic} \left(\sum_j \beta_j Z_j + \gamma_p + \alpha_i \right),$$

where α_i denotes the fixed effect for agent i . In this specification, skill is omitted because it is constant within each agent and is therefore absorbed by the agent fixed effect.

Agent fixed effects control for stable differences between agents, such as persistent travel tendencies, skill level, starting position, long-run role in the economy, or systematic proximity to the regional border. The fixed-effects model therefore asks a within-agent question: when the same agent faces different conditions across tax periods, which variables are associated with travel?

Tax-period fixed effects control for period-wide differences shared by agents in the same period, such as changes in resource abundance, economy-wide wealth, tax schedules, or differences between early and late stages of an episode.

The model chooses the coefficients β that maximize the logistic likelihood. For observation n , let

$$p_n = P(Y_n = 1).$$

The likelihood contribution is

$$p_n^{Y_n} (1 - p_n)^{1 - Y_n},$$

and the full log-likelihood is

$$\ell(\beta) = \sum_n [Y_n \log(p_n) + (1 - Y_n) \log(1 - p_n)].$$

A coefficient β_j is the change in log-odds associated with a one-standard-deviation increase in predictor j , holding the other included variables fixed. To make this easier to interpret, the plots report odds ratios:

$$\text{odds ratio}_j = e^{\beta_j}.$$

An odds ratio above one indicates a positive association with travel, while an odds ratio below one indicates a negative association.

For example, if

$$\beta_{\text{resources}} = -0.40,$$

then

$$e^{-0.40} \approx 0.67.$$

This means that a one-standard-deviation increase in visible resources is associated with the odds of travel being multiplied by 0.67, holding the other included variables fixed.

For each coefficient, the null hypothesis is

$$H_0 : \beta_j = 0,$$

which is equivalent to

$$H_0 : e^{\beta_j} = 1.$$

A small p-value indicates stronger statistical evidence against the null hypothesis of no conditional association. The dashed reference line in the p-value plots marks

$$p = 0.05.$$

The reported p-values are cluster-robust. In the default specification, clustering is by agent, allowing repeated observations from the same agent across tax periods to be correlated. Cluster-robust p-values adjust the estimated uncertainty around the coefficients, but they do not change the coefficient estimates themselves.

This is distinct from including agent fixed effects. Cluster-robust standard errors adjust inference, while agent fixed effects change the regression specification by adding an agent-specific baseline. Thus, the pooled model with cluster-robust p-values still uses cross-agent variation, whereas the fixed-effects model focuses on within-agent variation over time.

Finally, the regression results should be interpreted as descriptive conditional associations rather than causal effects. The model shows how travel is associated with observed variables, holding the included controls fixed, but it does not identify the causal effect of changing those variables.

A.8.1 Fixed Effects: explanation

Tax-period fixed effects give each tax period its own intercept. Without tax-period fixed effects, the regression has one common baseline travel probability:

$$\text{logit}(P(Y = 1)) = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k$$

With tax-period fixed effects, it becomes:

$$\text{logit}(P(Y_{i,p,r} = 1)) = \beta_1 X_{1,i,p,r} + \cdots + \beta_k X_{k,i,p,r} + \gamma_p$$

where γ_p is a separate baseline effect for tax period p . This controls for things that affect all agents in the same period, such as: the economy becoming richer over time, resources becoming more or less available, agents being more likely to travel early or late, tax policies changing over the episode, market conditions changing over time. So tax-period fixed effects ask: Within the same tax period, which variables are associated with travel? Agent fixed effects give each agent its own baseline tendency to travel. With agent fixed effects, the model becomes:

$$\text{logit}(P(Y_{i,p,r} = 1)) = \beta_1 X_{1,i,p,r} + \cdots + \beta_k X_{k,i,p,r} + \gamma_p + \alpha_i$$

where α_i is a separate baseline effect for agent i . This controls for stable differences between agents, such as: skill level, typical location, persistent role in the economy, long-run wealth differences, learned behavioral tendencies, being naturally closer to the border. So agent fixed effects ask: When the same agent experiences different conditions over time, which variables are associated with travel? This is why skill drops out when agent fixed effects are included: skill is constant for each agent, so it is already captured by the agent's fixed effect. In Simple Terms: Tax-period fixed effects compare agents within the same period. Agent fixed effects compare an agent to itself over time. Together, they make the regression stricter: Is a variable associated with travel when comparing the same agent across different periods, while also accounting for period-wide changes in the economy? That is why the fixed-effects regression is often more cautious than the pooled regression.

A.8.2 Estimation Procedure for the Logistic Regression Models

The regression models used in the analysis are binary logistic regressions. The dependent variable is an indicator for whether agent i travels at least once during tax period p in rollout r :

$$Y_{i,p,r} = \begin{cases} 1 & \text{if agent } i \text{ travels during tax period } p \text{ in rollout } r, \\ 0 & \text{otherwise.} \end{cases}$$

Because the outcome variable is binary, the model is estimated using maximum likelihood estimation rather than ordinary least squares. The logistic model assumes that the conditional probability of travel is given by

$$P(Y_{i,p,r} = 1 \mid \mathbf{Z}_{i,p,r}) = \Lambda(\eta_{i,p,r}),$$

where

$$\Lambda(x) = \frac{1}{1 + \exp(-x)}$$

is the logistic function, and $\eta_{i,p,r}$ is the linear predictor.

For the pooled specification, the linear predictor is

$$\eta_{i,p,r} = \beta_0 + \mathbf{Z}'_{i,p,r}\boldsymbol{\beta} + \gamma_p,$$

where $\mathbf{Z}_{i,p,r}$ is the vector of standardized covariates, $\boldsymbol{\beta}$ is the corresponding coefficient vector, and γ_p denotes tax-period fixed effects when included. The standardized covariates include mean coin, period coin change, visible resources, visible houses, and the regional market-price difference. In specifications without agent fixed effects, building skill is also included.

For the agent fixed-effects specification, the linear predictor is

$$\eta_{i,p,r} = \mathbf{Z}'_{i,p,r}\boldsymbol{\beta} + \gamma_p + \alpha_i,$$

where α_i is an agent-specific fixed effect. The fixed effect gives each agent its own baseline tendency to travel, thereby controlling for stable differences between agents such as skill, starting position, persistent behavioral tendencies, or long-run differences in economic role. Since skill is constant within each agent, it is absorbed by α_i and cannot be separately estimated in the fixed-effects specification.

The likelihood contribution of each observation is

$$P(Y_{i,p,r} = y_{i,p,r}) = [\Lambda(\eta_{i,p,r})]^{y_{i,p,r}} [1 - \Lambda(\eta_{i,p,r})]^{1-y_{i,p,r}}.$$

The full likelihood is the product of these terms across all observations:

$$L(\boldsymbol{\beta}) = \prod_{i,p,r} [\Lambda(\eta_{i,p,r})]^{Y_{i,p,r}} [1 - \Lambda(\eta_{i,p,r})]^{1-Y_{i,p,r}}.$$

Equivalently, the log-likelihood is

$$\ell(\boldsymbol{\beta}) = \sum_{i,p,r} [Y_{i,p,r} \log \Lambda(\eta_{i,p,r}) + (1 - Y_{i,p,r}) \log(1 - \Lambda(\eta_{i,p,r}))].$$

The coefficient estimates are chosen to maximize this log-likelihood:

$$\hat{\boldsymbol{\beta}} = \arg \max_{\boldsymbol{\beta}} \ell(\boldsymbol{\beta}).$$

Thus, the estimator selects the parameters that make the observed pattern of travel and non-travel outcomes most likely under the logistic model.

The reported coefficients are transformed into odds ratios:

$$\text{odds ratio}_j = \exp(\hat{\beta}_j).$$

An odds ratio above one indicates that a one-standard-deviation increase in the corresponding covariate is associated with higher odds of travel, holding the other included variables fixed. An odds ratio below one indicates lower odds of travel.

Inference is based on cluster-robust standard errors. In the default specification, observations are clustered by agent, allowing repeated observations from the same agent across tax periods to be correlated. The cluster-robust correction changes the estimated uncertainty and p-values, but not the coefficient estimates themselves.

The resulting estimates should be interpreted as descriptive conditional associations rather than causal effects. The logistic model identifies how travel decisions are statistically associated with observed variables in the simulated data, conditional on the included controls and fixed effects. It does not identify the causal effect of changing those variables, since the regressors are themselves generated endogenously by the learned policies and the evolving simulated economy.

A.9 Shared Learning Parameters

Parameter sharing in mobile agents improves learning efficiency because all agents can learn from each other’s experiences. The main reason this is feasible, is because agents operate under identical rules and differ primarily in their states, locations, and skill levels. In this setting, a shared policy allows experience collected from all agents to contribute to a single learning process, significantly accelerating convergence compared to learning independent policies.

Beyond computational benefits, parameter sharing also serves as an inductive bias: it encourages agents to learn general behavioral patterns that are robust across different economic conditions. Because agents are exposed to diverse situations—such as varying resource availability, market states, and tax regimes—the shared policy learns to condition decisions on observations rather than relying on fixed, agent-specific behaviors. This is particularly important in a dynamic environment with mobility and cross-regional interaction, where agents may encounter a wide range of economic contexts over time that differ from their original learning environment.

However, this design requires the shared policy to represent multiple behavioral modes simultaneously, such as specialization in production or resource gathering. When agents face different constraints or roles, this can introduce gradient interference, as a single policy must approximate distinct strategies within one parameterization. Despite this limitation, the benefits of improved generalization, stability, and sample efficiency make parameter sharing well-suited for the present setting.

A.10 Parallel Training Configuration and Parameter Interpretation

This section explains the key training parameters used in the implementation and how they relate to data collection and PPO optimization. The values described here correspond to the current non-local `simulation_run.py` configuration.

A.10.1 Workers

The parameter `num_workers` specifies how many parallel rollout worker processes are used during training. Each worker runs an independent copy of the environment and collects experience by interacting with it using the current policies.

In the current non-local configuration,

`num_workers = 4.`

Increasing the number of workers increases the amount of experience that can be collected in parallel, but also increases CPU and memory requirements.

A.10.2 Environments per Worker

The parameter `num_envs_per_worker` determines how many environments are executed within each worker process. In the current configuration,

$$\text{num_envs_per_worker} = 1.$$

The total number of parallel environment replicas is therefore

$$\text{total environments} = \text{num_workers} \times \text{num_envs_per_worker} = 4 \times 1 = 4.$$

A.10.3 Rollout Fragment Length

The parameter `rollout_fragment_length` specifies how many timesteps each environment executes before returning collected experience to the trainer. Rather than waiting for a full episode to complete, RLlib collects shorter rollout fragments.

In the current configuration,

$$\text{rollout_fragment_length} = 200.$$

Thus, each environment contributes rollout fragments of approximately 200 timesteps. Since the episode length is 1000, one complete episode corresponds to approximately

$$\frac{1000}{200} = 5$$

rollout fragments per environment.

A.10.4 Training Batch Size

The parameter `train_batch_size` defines how many transitions are collected before one PPO training iteration is performed. In the current configuration,

$$\text{train_batch_size} = 800.$$

This is consistent with the rollout setup:

$$\text{train_batch_size} = \text{num_workers} \times \text{num_envs_per_worker} \times \text{rollout_fragment_length} = 4 \times 1 \times 200 = 800.$$

Thus, each PPO iteration uses approximately 800 newly collected transitions.

A.10.5 Mini-batch Size

The parameter `sgd_minibatch_size` determines how the training batch is split during optimization. Instead of updating the model once using the full training batch, PPO divides the batch into smaller mini-batches.

In the current configuration,

$$\text{sgd_minibatch_size} = 200.$$

Given a training batch size of 800, this gives

$$\frac{800}{200} = 4$$

mini-batches per pass through the training batch.

A.10.6 Number of SGD Iterations

The parameter `num_sgd_iter` specifies how many times PPO reuses the same collected batch for optimization. In the current configuration,

$$\text{num_sgd_iter} = 4.$$

The total number of mini-batch gradient updates per PPO training iteration is therefore

$$\frac{\text{train_batch_size}}{\text{sgd_minibatch_size}} \times \text{num_sgd_iter} = \frac{800}{200} \times 4 = 16.$$

This reuse of rollout data improves sample efficiency by extracting more learning signal from each collected batch.

A.10.7 Relation to Episode Length and Tax Period

The episode length is

$$1000$$

timesteps, while the tax period is

$$100$$

timesteps. Because the rollout fragment length is 200, each rollout fragment contains approximately two tax periods. A full episode contains ten tax periods and corresponds to approximately five PPO rollout fragments per environment.

A.10.8 Relation to the Original Training Procedure

The original AI Economist training setup also uses multiple parallel replicas of the environment. In the current implementation, the number of replicas is given by

$$\text{replicas} = \text{num_workers} \times \text{num_envs_per_worker}.$$

The number of steps collected per replica corresponds to `rollout_fragment_length`, and the total number of transitions collected before an update corresponds to `train_batch_size`. Mini-batch size and repeated optimization over the same rollout data are controlled by `sgd_minibatch_size` and `num_sgd_iter`. Thus, although the exact numerical values differ from the original AI Economist configuration, the overall structure remains the same: collect rollout data from parallel environments, concatenate it into a training batch, and perform multiple PPO gradient updates.

A.10.9 Summary

The training system consists of two main stages: parallel data collection and batched PPO optimization. The worker and environment parameters determine how much data is collected in parallel, while the batch size, mini-batch size, and number of SGD passes determine how that data is reused for learning.